Chapter 12: Central and South America

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Executive Summary

Extreme rainfall events (floods and droughts) have impacted many regions of Central America (CA) and South America (SA) in recent years and show a tendency to intensify due to climate change (*high confidence*). As a consequence, food production has been reduced with a subsequent impact on food security; transportation infrastructure has also been impacted as well as other economic activities in the area.

Amazon forests are vulnerable to drought (*high confidence*). The unprecedented drought of 2010 resulted in high tree mortality rates and variations in forest productivity. The combined effect of both impacts lead to a long-term decrease in a C stocks in forest biomass, compromising their role as a C sink (*high confidence*). This ecosystem, one of the richest biodiversity in the region, is threatened by the positive feedback between climate change and land use change (*high confidence*). This ecosystem regulates the regional climate, the local hydrological cycle (*high confidence*) and, through teleconnection, the global climate (*medium confidence*). Forest dieback is predicted under future climate change scenarios, particularly in the south and west regions (*medium confidence*).

A robust positive trend in precipitation has been detected since the beginning of the 20th century in Southeastern South America (SES), which is the most densely populated and agriculturally productive area of South America, with several large cities within its limits.

Increasing projected water scarcity, floods and competition over water, and more equal access to safe water services are important challenges for the region, particularly in large urban areas (*high confidence*). Adaptation experiences in the water sector have strongly increased over the past years in the region but are insufficiently documented in the scientific literature and analysed. Available evidence suggests that substantial co-benefits can be generated between improved water service, watershed conservation and flood mitigation when overcoming barriers related to institutional instability, fragmented services, inadequate governance structures or limited data availability, and adoption of more comprehensive water supply and demand focused approaches.

Small-scale farming systems with limited adaptive capacity are affected by climate change (*high confidence*). A focused intensification of agriculture appears as an adaptation option, especially for areas of rapid change. Heterogeneous impacts and adaptive capacity are observed among subregions, countries and productive systems. CA appears as one of the most affected areas. Advances and barriers for climate change adaptation are evidenced in the region.

Mountain communities are increasingly vulnerable (*high confidence*). Extremely poor indigenous and farmer communities are showing signs of climate stress, particularly in CA, Bolivia and Peru. Livelihoods, infrastructure and built environment in those regions are not prepared for flooding, droughts and extreme temperatures.

Climate change is amplifying the vulnerability of local and indigenous communities possibly promoting migration (*medium confidence*). These communities rely on fishing, forest products, and self-consumption farming and depend heavily on the services provided by the surrounding ecosystems. Migration events are regular but intensify with extreme climate events.

Inequality and its impact on the territory, as much as on the decision-making arena and the governance capacity, remains an important barrier for adaptation for all sectors in the region (*high confidence*). In spite of the increase in adaptation initiatives focusing on inclusion and social justice, these are still weakly attached to institutional initiatives undermining their results and continuity; they are barely financed and poorly reported therefore hindering feedback and evaluation for improvement.

The use of nature-based solutions has been growing in recent adaptation experiences with positive effects on reducing urban, ecological and social vulnerabilities in the cities (*medium confidence*). The capacity for cooperating with a variety of sectoral challenges, such as coastal protection, biodiversity preservation, drainage infrastructure, flooding reduction, carbon sequestration, water cleaning and supply, alleviation of urban heat bubbles, local food production and safety, among others, makes initiatives focused on green and blue infrastructure strategic for urban adaptation.

Public policies and non-governmental initiatives relevant to terrestrial and freshwater ecosystems adaptation have increased since AR5 both at national and sub national levels, most of them with an Ecosystem based Adaptation or Community based Adaptation approach (*high confidence*). Most reported limitations are financing (*high confidence*) and local biophysical and social conditions (*medium evidence, high agreement*). Those limitations can be addressed through technical support, cooperation and local empowerment.

12.1 Introduction

12.1.1 The Central and South America Region

Central and South America (CA and SA) is a highly diverse region of the world, both culturally and biologically, harbouring the highest biodiversity on the planet and a wealth of cultural diversity resulting from the many indigenous populations combined with the mestizo and European and African populations that migrated to the region starting in the XVI century. Moreover, it is one of the most urbanized regions in the world, with some of the most populated metropolitan areas. Several countries in the region have experienced a sustained economic growth in the last decade, making important advances in reducing poverty in the area. Yet, it is a region of substantial social inequality, where there still remains a large percentage of the population under the poverty line highly vulnerable to climate change and natural extreme events that constantly affect the region.

Land use changes in the region are large, mainly due to agricultural production for export purposes, one of the main sources of income for the area. These changes exacerbate the impacts of climate change and make the region play a key role in the future of the world economy and food production. Besides having the largest tropical forest on the planet, the region also has the largest potential for agricultural expansion and development, making it crucial to develop proper climate policies that guide this agricultural expansion without increasing substantially the emissions of greenhouse gases and the vulnerability of its population.

12.1.2 Approach and Storyline for the Chapter

The chapter follows an integrative approach in which the hazards, impacts, vulnerabilities and risks are discussed by geographical subregions following ATLAS WGI proposal to divide the region in eight climatically homogeneous subregions [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table]. Then, the discussion on adaptation practices, implemented and proposed, follows with a sectoral approach, connecting to the AR6-WGII cross-chapters themes. The storyline is then a description of the hazards, exposure, vulnerability and impacts providing as much detail as available in the literature at the subregional level, followed by the identification of risks as a result of the interaction of those aspects. This approach ensures a balance in the text, particularly for countries that are usually underrepresented in the literature, but that also show a high level of vulnerability and impacts, such as those situated in CA. The sectoral discussion on adaptation that follows will be more useful for policy makers and implementers in the field, who are usually focused and organized around specific sectors; for example, the different ministries in the government will be able to easily locate the adaptation information relevant to their own work. To ensure a coherence on the discussion, a section integrating impacts and adaptation is presented. The chapter closes with a presentation of case studies and a general discussion of the most relevant knowledge gaps for the Region.

12.1.3 Summary of the Fifth Assessment Report and Recent IPCC Special Reports

According to the WGII Fifth Assessment Report Chapter 27 (Magrin et al., 2014), increasing trends in precipitation have been observed in Southeastern South America (SESA) in contrast with decreasing trends in CA and central-southern Chile (*high confidence*). Frequency and intensity of droughts have increased in many parts of SA (IPCC, 2019c). Warming has been detected throughout CA and SA except for a cooling trend off the Chilean coast. Changes in climate variability and in extreme events have severely affected the region. Climate projections indicate increases in temperature for the entire region by 2100, but rainfall changes will vary geographically, with a notable reduction of -22% in northeast Brazil and an increase of +25% in SESA.

Deforestation and land degradation attributed mainly to increased agricultural production and cattle ranching are exacerbating the negative impacts of climate change in the region. This conversion of natural ecosystems is the main cause of biodiversity and ecosystem loss in the region and is an important source of greenhouse gas (GHG) emissions. The high and persistent level of poverty in most countries result in high vulnerability and increasing risk to climate variability and change. For example, increases in temperature and decreases in rainfall could decrease agricultural productivity by 2030 and threatening the food security of the poorest population. The Special Report by IPCC on Global Warming of 1.5ºC (IPCC, 2018b) concludes that limiting the warming to 1.5ºC is projected to result in smaller net reductions in yields of maize, rice, wheat and other cereal crops both for CA and SA. The heat stress is expected to reduce the suitability of Arabica coffee in Mesoamerica but it can improve in high latitude areas in SA (Olsson et al., 2019). There is limited evidence that these declines in crop yields may result in significant population displacement from the tropics to the subtropics (Hoegh-Guldberg et al., 2018). Undernutrition has worsened since 2014 in Latin America (Mbow et al., 2019). Evidence of climate change impacts on food security is emerging from indigenous and local knowledge studies in SA. Areas in CA with high proportion of subsistence crops tend to have less resources for adaptation (Mbow et al., 2019).

The expansion of sugarcane, soy, and oil palm used for food and biofuels will have an effect on land use, leading to deforestation in parts of the Amazon and CA, among other regions. Large conversions in tropical dry woodlands and savannas in SA (Brazilian Cerrado, Caatinga and Chaco) are observed (Arneth et al., 2019). The combination of deforestation with global warming may result in significant dieback of tropical rainforest occurring. Loses as high as 50% of biomass productivity are projected with a warming of 3ºC or 40% deforestation of the Amazon (Hoegh-Guldberg et al., 2018). Advances in second-generation bioethanol from sugarcane and other feedstock will be important for mitigation.

SA has been a significant contributor to the world's agricultural production growth in the last decades, but at the expense of seeing profound landscape transformations with high rates of native vegetation conversion. Recent initiatives by governments with the engagement of the private sector particularly from the food industry have reduced deforestation rates significantly in recent years. But deforestation control still requires stronger reinforcement mechanisms as indicated by recent increases in deforestation rates in Brazil. SA is one of the regions with the highest potential to increase food supplies particularly to more densely populated regions in Asia, Middle East and Europe but a great coordination effort by all countries and stakeholders involved is needed to guarantee a sustainable intensification process that ensures the conservation of the surrounding environment (Mbow et al., 2019).

Changes and weather and climatic patterns are negatively affecting human health in CA and SA, in part through the emergence of diseases in previously non-endemic areas. There is *high confidence* that constraining the warming to 1.5ºC would reduce risks for unique and threatened ecosystems safeguarding the services they provide for livelihoods and sustainable development (food, water) in CA and Amazon (Roy et al., 2018).

There is a *high confidence* that heat waves will increase in frequency, intensity and duration, becoming extremely long, over 60 days in duration. The risk of wildfires will increase significantly in SA (Jia et al., 2019). These processes are and will lead to increased desertification that cost between 8 and 14% of gross agricultural product in many CA and SA countries (Mirzabaev et al., 2019). Distinguishing climate induced changes from land use changes is challenging, but 5-6% of biomes in SA are expected to change by 2100 due to climate change (Olsson et al., 2019).

The first step toward adaptation to future climate changes is to reduce the vulnerability to present climate. Citizens and governments in the region need to address the challenge of building a new governance model focused on vulnerability reduction supported by adaptation strategies and inclusive development processes (Magrin et al., 2014).

12.2 Characterisation of the region with Subregions

Central America (CA) is a subregion that includes seven small countries in the middle of the continent, from Guatemala in the north to Panama in the south. For climate modelling purposes, the region is usually linked with Mexico, being referred as CAM - Central-America/Mexico in AR6-WGI. For geographical reasons, Mexico is included in North America region in AR6-WGII. Belize, in spite of historically and politically having more connections with the countries in the Caribbean, is included in AR6 WGII CA region for geographical reasons.

A narrow strip of land in between the Pacific and Atlantic oceans, Central America has a climate strongly influenced by these bodies of water which provide abundant rainfall (higher than 5 m yr^{-1} in some areas) which, together with the high temperatures, results in a typical tropical climate for most parts of the region. The topography generated by the Sierra Madre and the chain of volcanoes that crosses the entire isthmus results in a wide range of microclimates which in turn results in a region with very high biodiversity (both Guatemala and Costa Rica have been declared mega diverse countries), estimated to have 12% of the planet's biodiversity in 2% of its territory (Programa Estado de la Nación - Estado de la Región, 2016). A rain shadow effect by these high mountains result in a semi-dry region in the centre of the isthmus locally known as the "dry corridor" where the rainfall can drop to 400 mm yr^{-1} in some areas and with a dry season of up to 7 months (van der Zee et al., 2012).

The economy of the region relies heavily on export cash crops such as sugar, coffee and bananas with many small farmers in rural areas producing at the subsistence level. Manufacturing and tourism are also important; and in Guatemala, Honduras and El Salvador remittances from migrants in North America represent between 10 to 20 percent of the GDP (ECLAC, 2018). Costa Rica and Panama present a more diversified economy relying more on industry and commerce, which provide these countries with the largest GDP per capita in the region and the lowest poverty indices, ranked as countries with high development index. But still, the region presents higher poverty levels than its South America (SA) counterpart, at rates of 50% or higher. Inequality is high, particularly in Guatemala and Honduras, with poverty rates approaching 60% but still considered middle development countries.

The population in CA is more rural than in SA, with about 50% of the people living in cities in countries like Guatemala, Belize and Honduras, and 75% in Costa Rica and Panama. The capital cities in each country are by far the largest cities, with Guatemala City being the largest with an estimated 2 million people living in its metropolitan area. Guatemala is also the country with the largest population (1 of 3 Central Americans is Guatemalan), but El Salvador has the highest population density at 302 persons km^{-2} with Belize at the other end with 37 persons km^{-2}. Except for the wealthiest countries of Panama and Costa Rica, the rest of the countries show low socio-economic indicators particularly for rural areas and indigenous populations, highly represented in Guatemala with 42% indigenous population. In 2014, 59% of the population had at least one basic needs unsatisfied, usually related to the income or quality of the living quarters (Programa Estado de la Nación - Estado de la Región, 2016).

South America (SA) is divided in this report in seven climate subregions, following the assessment by WGI: Northwestern South America (NWS), Northern South America (NSA), Central South America (SAM), Northeastern South America (NES), Southeastern South America (SES), Western South America (SWS) and Southern South America (SSA).

The Northwestern South America (NWS) subregion comprises the coastal lowlands, the Andes and the upper Amazon basin from the Caribbean region in Colombia to southern Peru. This region is characterized by its high biological, agricultural, cultural and climatic diversity, favoured by a large altitudinal gradient and the cold and warm ocean currents that converge in the equator. The three countries included (Peru, Ecuador and Colombia), are considered megadiverse (Mittermeier, 1997), the region comprise two worldwide hotspots (Tropical Andes and Tumbes-Choco-Magdalena (Myers et al., 2000)), and host a wide diversity of terrestrial ecosystems, from deserts to tropical evergreen forests.

The agricultural frontier is in expansion, in regions such as the upper Amazonia and the Chocó. The rate of deforestation varies, being the highest in the forests of lowlands Ecuador (2.42 annual average, the biggest in Latin America). In Colombia, it ranges from 1-2 and in Peru is less than 1 (Armenteras et al., 2017).

A distinctive feature of this subregion are the indigenous populations, pertaining to more than 30 different indigenous peoples that amount more than 5 million people. The Quechua represent the most conspicuous group in this subregion (Sichra, 2009). There is a large population of small farmers and fishermen that rely on local production and is vulnerable to warming, changes in seasonality and total amount of rainfall, droughts, glacier receding, frosts, hails, landslides. The NWS includes global historical centres of domestication of plants such as cocoa, potatoes, and coca; up to the present there are more than 4000 varieties of potatoes (Centro Internacional de la Papa, 2014).

Three capital cities are part of this subregion (Bogota, Quito and Lima). Other big cities are Medellin, Cali and Guayaquil, and there are many intermediate and small cities. .The percentage of urban population is increasing: countries such as Ecuador will still have 33.5% of the population living in rural areas by 2021, while in Peru and Colombia that percentage will be less than 19% (CELADE, 2019).

The region's economy is highly dependent on the exportation of processed or not metals (mostly copper, gold and zinc), oil, coal and gas, fisheries, shrimp, banana, cocoa, coffee, flower crops, cattle and forest products (The World Bank, 2019). Food production for internal markets is highly important. Climate variability is critical for those countries, especially for fisheries and crops. Remittances from migrants are important for local economies. Industry is more developed and large in Colombia and Peru. Other important economic activities are tourism, and crops considered illicit such as coca and poppy, although their contribution to economy is hard to quantify (Vallejo, 2015).

The main sources of energy are hydropower and oil, with a growing tendency of the first. The region shows high poverty and inequalities indicators.

Northern South America (NSA) includes some of the Caribbean and the Atlantic Coast on the northern border down to Northern Amazonia (Brazil) in the southern border. Several countries share this region: Suriname, French Guyana, Guyana, Venezuela, eastern Colombia, and Brazil. It is structurally composed by the Andes to the west, the Brazilian Shield to the south and the Guyana Shield to the north and by the sedimentation basin to the centre, where the major rivers of the region are located (Amazon River, Orinoco, Rio Negro) (IPBES, 2018).

The mosaic of ecoregions in NSA, harbours high species richness and high degrees of endemism, with areas such as the Venezuelan tropical Andes considered as a "biodiversity hotspot" (Brooks et al., 2006; da C Jesus et al., 2009; Hoorn et al., 2010; Zador et al., 2015; Barlow et al., 2018).

The economy sectors within NSA subregion vary widely. Agriculture (overall beef and soy) and mining (gold, bauxite and aluminium) are important commodities explored in the region. The economy in Venezuela is strongly based on petroleum extraction (Minea, 2017). For Guyana, the agricultural sector is an important contributor to GDP (Government of Guyana, 2012). The bauxite extraction plays an important role in Suriname. However, activities such as wholesale and retail trade, hotels, restaurants, transport, communication, financial intermediation, real estate, renting and business activities, among others contribute with the 40% of the country's GDP (Berrenstein and Gompers-Small, 2016).

Brazil and Venezuela are the most populated countries of the subregion with 190 million and 27 million inhabitants respectively, most of them living in urban centres (84.4%). Even though, Brazil is a very populated country, the demographic density in the area covered within the NSA subregion is very low (Tritsch and Le Tourneau, 2016). In the Brazilian Amazon, areas of high deforestation are considered "deforested human deserts", which accounted, earlier this decade for almost one third of the total deforestation area (Tritsch and Le Tourneau, 2016), contrasting with very low densities forested areas (less than 1 inhab km^2) and big urban centres such as Manaus and Belem with 2-3 million inhabitants (IBGE, 2016). The NSA population includes an important percentage of indigenous peoples mostly settled in the forest (Kayapo, Warao, Yekwana, among others) (Minea, 2017).

According to the Gini coefficient, the inequality of countries in the NSA follow the order: Brazil (55) > Suriname (52.8) > Venezuela (43.4) = Guyana (43.2). Countries within the NSA subregion are below the global average value (65) (UNDP, 2010; Hellebrandt and Mauro, 2015).

The Central South America (SAM) subregion is characterized by tropical climate and vegetation from dense tropical humid forest, to flood lands, grasslands and savannah. Most of the area lies on Brazilian territory, including areas on Bolivia and part of the Peruvian Amazon. In Bolivia and Brazil, this area faces, in special, after the past 20 years, intense rates of deforestation and consequent increase of agricultural commodities production (soy, corn, sugar cane, meat and wood products). Global, including the SAM region, analysis of tropical land use dynamic related the increased rates of deforestation to the production of commodities (Strassburg et al., 2014; Henders et al., 2015).

The region has, therefore, important infrastructure for agriculture production, storage and transport, being an important supplier to the European and Asian markets (Henders et al., 2015). The region is home for more than 50 million people, but faces large inequalities in socio-economic conditions, especially in the areas of agriculture frontier (Strassburg et al., 2014; Moutinho et al., 2016; Martinelli et al., 2017). Issues related to widespread deforestation in the region (where the so-called 'Arc of Deforestation' is located), which includes a large portion in Bolivia and Peru (the southwestern portion of the Amazon Basin) go from particulate pollution resulting from biomass burnings affecting population health (Brondízio et al., 2016) to loss of biodiversity and ecosystem services and impact on indigenous population .

Changes in micrometeorological parameters associated to deforestation were identified in the Brazilian Amazon and Cerrado. Changes in surface cover (from native vegetation to pasture and crops) lead to changes in sensible heat fluxes, evapotranspiration and surface temperature, leading to temperature increase higher than the global average, up to 5°C (Dias et al., 2015; Panday et al., 2015; Silvério et al., 2015; Spera et al., 2016; Coe et al., 2017).

The Northeastern South America subregion (NES) is entirely within Brazil. It is a predominantly semi-arid region covered by seasonally dry tropical forest and woodlands (known as Caatinga) in its hinterland, with some localized, discontinuous strips of humid tropical forest on the coast (known as Atlantic Forest) (Silva et al., 2017). NES's socio-ecological systems are strongly controlled by rainfall seasonality, with an extended dry period and recurrent severe drought events (Marengo et al., 2017; Silva et al., 2017). Being an arid region, it is expected to have low biodiversity levels, but it is a particularly understudied region and recent surveys are showing that it is more biologically diverse than previously thought (Silva et al., 2017). The Caatinga has already lost about 45% of its original vegetation cover (Joly et al., 2019), with only about 1% of the remaining vegetation within fully protected areas (Jenkins et al., 2015).

NES is home to about 60 million people (estimate for 2019 from the Brazilian Institute of Geography and Statistics (IBGE), https://cidades.ibge.gov.br/), with >70% living in urban areas (data for 2010 from IBGE, https://cidades.ibge.gov.br/, (Silva et al., 2017)), and high poverty levels (> 50%, data for 2003 from IBGE, https://cidades.ibge.gov.br/). Despite significant improvements in the last decade, it is still the region with the lowest human development index in Brazil (Vieira et al., 2015; Silva et al., 2017). The region's formal economy is based mostly on public services (e.g., pensions and government expenditures), followed by a modest contribution of agriculture, which is mostly for subsistence, and the industry, scattered mostly in major urban centres along the coast (Silva et al., 2017).

The Southeastern South America subregion (SES) is a large territory that includes southern Brazil, Uruguay, Paraguay and most of Argentina. It has a heterogeneous landscape, going from a tropical rainforest along the northern coast to temperate grasslands in the south. The Brazilian Atlantic Forest, in northern SES, is one of the World's biodiversity hotspots (Myers et al., 2000). Biological diversity in SES decreases towards the south, with moderate levels in the Chaco and low diversity in Patagonia (Pimm et al., 2014). There is also a north-south gradient in economic development, from São Paulo State in Brazil, arguably the largest economic and industrial hub in South America, to Chaco and Patagonia in the south, recognized as wilderness areas (Mittermeier et al., 2003). SES has some of the most populous cities in South America, such as São Paulo and Rio de Janeiro in Brazil and Buenos Aires in Argentina (UNDESA, 2019).

Western South America subregion (SWS) comprises the south of Peru, Chile and the Andes up to the austral part of Chile and Argentina. This is a subtropical area that presents a variety of sub-climates: arid, semiarid, mediterranean and temperate. The climate and sub-climates present are influenced by the Pacific Ocean and the Andes mountain range. Water sources are mainly linked to the storage capacity of the mountain areas. The variety of climates is also reflected by the variety of ecosystems in the area which are determined by latitudinal gradient and topographic conditions.

The region includes the hyper arid area of the Atacama Desert with a precipitation of less than 2 mm yr^{-1} (Ritter et al., 2019); the semiarid area with sclerophyllous ecosystems; the mediterranean zone were large areas are allocated for agriculture; and the oceanic temperate area characterized by forest plantations and native forests.

The economy in the area is diverse in terms of productive sectors. While fisheries are present in all the sub areas; mining production is predominant in the hyper arid area; in the semiarid area presents mining and agriculture, and in the mediterranean area agriculture is the main economic activity. All activities are linked to natural resources. In this area, countries present an important proportion of fossil fuel energy consumption.

[PLACEHOLDER FOR SECOND ORDER DRAFT: Demography and urban settlements/cities, rural balance areas, socioeconomic indexes for poverty, inequality]

Southern South America (SSA) lies inside two countries: 10% in Chile and 90% in Argentina. It corresponds to the Patagonia, which basic geographic characteristics involve Andes Mountains, deserts, pampas, and grasslands to the east; it is a unique region surrounded by coasts with three oceans, its climate is mostly cool and dry with the presence of impressive ice fields and glaciers, the largest ice fields in the Southern Hemisphere outside of Antarctica with some volcanoes still being active.

Most of SSA is dominated by air masses coming from the Pacific Ocean. The region is located between the semi-permanent anticyclones of the Pacific and the Atlantic oceans. The strong, constant west winds are dominant across the region. The Andes play a crucial role in determining the climate of Patagonia. The north-south distribution of the mountains imposes an important barrier for humid air masses coming from the Pacific Ocean. Most of the central portion of Patagonia receives less than 200 mm yr^{-1}. Patagonia can be defined as a temperate or cool-temperate region.

The region of Patagonia has some of the most extensive wilderness areas in the planet. Natural grasslands comprise almost 30% of the Americas (White et al., 2000), dominating the landscape in a diversity of regions including the Patagonia steppe (Argentina) with the Colorado and Barrancas Rivers which run from the Andes to the Atlantic. Profound changes, affecting key ecosystem functions and ultimately human well-being, have occurred in South American temperate grasslands. Livestock grazing for over 400 years has reduced soil organic carbon stocks by an estimated 22% and net primary production by 24% (Piñeiro et al., 2006). South America cool temperate forests are found in Chile and Argentina. Over the past 20–30 years, the biodiversity of southern temperate forests has become widely recognized for its ecotourism and tourism values. High elevation areas have warmed in the southern northern Andes (Hofstede et al., 2014). Whether and the degree to which anthropogenic warming has affected tree growth and the position of the treeline along the Andes are still somewhat unclear. Anthropogenic warming seems to have affected tree growth and increased recruitment above treeline in some places, but not in others (Daniels and Veblen, 2004; Lutz et al., 2013). Some tree species have been moving upward below treeline (Feeley et al., 2011). Lack of or very slow upward movement of the treeline might reflect recruitment difficulties in high altitude grasslands (Rehm and Feeley, 2016) or under reduced precipitation in some parts of the Andes.

The main economic activities have been mining (coal, gold, silver), livestock (notably sheep), agriculture (crops and fruit production near the Andes and valleys), and oil and gas after its discovery near Comodoro Rivadavia in 1907. Traditionally, the region has exploited its hydroelectric potential by building big hydroelectric dams in several rivers with the tendency of incrementing socio environmental conflicts (Silva, 2016). Instead, wind is a significant renewable energy source that is increasing in Patagonia, which could generate a very large amount of electrical energy.

Patagonia is a sparsely populated region of about 3 million people in 1,043,076 km^2, that determines a density of 3 inhabitant km^{-2}. However, this region presents the lower proportion of inhabitants in rural zones (in Argentina only represents 9.9%) with some Mapuche indigenous communities. According to the Human Development Index (HDI) developed by the United Nations Development Program (UNDP), the Patagonia region in Argentina has a human development considered as "very high" (HDI= 0.861), but with a Gini index (inequality variable) of 0.379 in 2018.

12.3 Vulnerabilities and Impacts

12.3.1 Central America (CA) Subregion

12.3.1.1 Hazards

Significant trends in precipitation and temperature have been observed in Central America (CA). Observed warming (1970–1999) about 0.1ºC yr-1 in most of CA and cooling in parts of Honduras and northern Panama. Precipitations have increased 12 to 60 mm yr-1 between 1970 and 1999 in small regions of Guatemala, El Salvador and Panama (AR5 WGII Chapter 27 Executive Summary, (Magrin et al., 2014)). Several significant differences in regional mean temperature and range of temperature indices between the 1.5°C and 2°C global mean temperature targets are found (SR15 Fig 3.8). Nevertheless, non-significant differences in regional mean precipitation and range of precipitation indices between the 1.5°C and 2°C global mean temperature targets while where found (SR15 Fig 3.11). Related to extreme temperatures, annual maximum daytime temp (TXx) will change by +2°C for 1.5° scenario, and about +2.5°C for 2°C scenario, for 5-day extreme precipitation Rx5day: the projected changes are +10-20mm for 1.5 and 2°C scenarios and for (P-E, all RCP scenarios): (very) likely decrease (Hoegh-Guldberg et al., 2018).

Projected rainfall changes for the region range between -22% to +7% by 2100 ((AR5 WGII Chapter 27 ES), (Magrin et al., 2014)), with decrease in the northern part of CA and increase in Southern Panama consistent with the future south displacement of ITCZ (Hidalgo et al., 2017). The rainy season in Central America will likely experience more pronounced dry season by the end of this century, with a signal for reduced precipitation by the mid-century for the JJA and SON quarters, so mid-summer drought may intensify (Fuentes-Franco et al., 2014; Imbach et al., 2018; Naumann et al., 2018). This intensification has already been observed in the current decade. A report by FAO in June 2016 (FAO, 2016a) indicated that the dry corridor, particularly in Guatemala, Honduras, and El Salvador, was experiencing one of the worst droughts in 10 years. Son et al. (2018) analyzed NDVI from MODIS images for the period 2000-2016 and concluded that very dry moisture conditions occurred in 2002 and 2014 for the months of April-August.

The frequency of tropical cyclones (TC) is expected to decrease in the Atlantic and Pacific coasts of Central America, whereas the frequency of intense and long lasting tropical cyclones is projected to increase in the tropical Atlantic (Diro et al., 2014). When the statistical models are used to project future TC activity, some models suggest an increase in frequency by 2–8 TC yr^{-1} by the end of century. On the other hand, premised on the atmospheric circulation variables indicate either declines of approximately 1 TC yr^{-1} or no change by the end of century (Jones et al., 2016).

Recent increases in tropical cyclone intensification are detectable with a rate of anthropogenic forcing (Bhatia et al., 2019). [PLACEHOLDER FOR SECOND ORDER DRAFT: Sea level rise potential hazards to be discussed]. During the TC season more TC-driven events of extreme sea level exceed a 10-year return period (Muis et al., 2019).

The literature agrees on decreasing mean annual river runoff trends by 2050 for CA (Arnell and Gosling, 2013) and by the late 21st century (Hidalgo et al., 2013; Nakaegawa et al., 2013), possibly more pronounced in the northern than in the southern part of Central America. However, annual mean runoff patterns provide little indication for floods and landslide activity which in Central America are, among other, determined by the frequency and magnitude of tropical cyclones (IPCC, 2012). There is a scarcity of local or regional studies of future flood and landslide activity in the CA region, and an assessment therefore also needs to rely on global scale studies. Decreasing flood events are projected for CA (*low to medium confidence*) (Arnell and Gosling, 2013; Hirabayashi et al., 2013; Alfieri et al., 2017). Landslide activity is projected to increase over most of Central America as a result of climate change (mainly changes in precipitation patterns) (Gariano and Guzzetti, 2016), but there is a lack of studies and only *low confidence*. Furthermore, land-use practices, deforestation, overgrazing and soil degradation have an important effect on landslide activity (Seneviratne et al., 2012).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Sea level rise]

Figure 12.1: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections CA CMIP5/CMIP6]

12.3.1.2 Exposure, vulnerability and impacts

Of the 47 million Central Americans in 2015 (CELADE, 2019), 10.5 million lived in the dry corridor region, an area recently exposed to severe droughts that have resulted in 3.5 million people in need of humanitarian assistance (FAO, 2016a). About 44 percent of the CA population is poor or extremely poor). As of 2014, Honduras, Guatemala, Nicaragua and El Salvador had the highest poverty rates in CA and SA with 55.3, 50.5, 46.3 and 44.5 percent of their populations living in poverty. In 2011, rural poverty in Guatemala was 71%. In 2014, all CA countries except for El Salvador (excluding Belize in the analysis) had higher GINI coefficients (more inequality) than the average for Latin America (0.473), which in itself is the most unequal region in the world, with Guatemala (0.554) being the highest for the region (ECLAC, 2019).

Disasters from adverse natural events exacerbate Central America's economic vulnerability, accounting for substantial human and economic losses. Between 2005 and 2014, the region had a nominal combined cumulative loss of around US$ 5.8 billion, with more than 3410 deaths and hundreds of thousands of people displaced (Ishizawa and Miranda, 2016). Hydro-meteorological events are the most frequent and the ones with the largest impact in the region CA. In the period 1998-2012, a series of tropical storms and hurricanes hit the region causing thousands of deaths, millions of people affected and billions of dollars in damage (ECLAC, 2015). Ishizawa and Miranda (2016) show that one standard deviation in the intensity of a hurricane windstorm leads to a decrease in the growth of total per capita gross domestic product (0.9 to 1.6 percent), and a decrease in total income and labour income by 3% in CA. In turn, it increases moderate and extreme poverty by 1.5% points.

In 2013 a period of drought began that lasted until 2019. The most impacted group by the exposure to droughts and variable rainfall are small, subsistence farmers; a second target group is people in poor neighbourhoods. FAO (2016a) estimated 1.6 million people suffering from food insecurity in the dry corridor of CA. An evaluation in 2018 from a sample of 8,631 rural households in Guatemala showed a loss of 76% of the corn plantations and 74% of beans, the two staple foods for the region. 29.4% of the households had moderate or high food insecurity, and only 6.2% had received some aid by the government (Oxfam, 2019).

Rural livelihoods in Central America are highly threatened by Climate Change, particularly for small and medium-sized farmers; more impacts are expected in the future, particularly for indigenous communities in the mountains (*high confidence*) (Walshe and Argumedo, 2016; Bouroncle et al., 2017). Models suggest declines in suitability of crops, particularly coffee. Changes are likely at low and high elevation montane forest transitions and smallholders in the area have multiple vulnerability factors that put them at risk (Hannah et al., 2017). Unfortunately, most of those mountain areas have little or no climate information, making it hard to assess the expected changes or the possible adaptation measures (Pons et al., 2017).

Lowland forests have been better studied. A significant reduction in net primary productivity in tropical forests in CA is expected under both RCP4.5 and RCP8.5 scenarios as a result of temperature increase, precipitation reduction and droughts (Lyra et al., 2017). Wetlands are also expected to be highly affected by climate change in the region.

Between 2000 and 2010, 34% of the coffee departments or provinces of the region had high yields in coffee plantations (more than 0.8 T/ha), 40% had medium yields (0.3-0.8 T/ha) and 14% had a low yield (less than 0.3 T ha^{-1}) (CEPAL and CAC/SICA, 2014). However, under A2 scenario, it is expected that by 2100, only 12% of the departments or provinces will have a high yield, 40% will have a medium yield and 36% will have a low yield.

[PLACEHOLDER FOR SECOND ORDER DRAFT: other socioeconomic vulnerabilities - health, poverty, education, fragile groups, governance and political stability; industry, cities, key infrastructure]

In Central America, it is estimated that under the A2 climate change scenario, the economic losses in the agricultural sector could be equivalent to 5.4% and 19.1% of the Central American GDP for the years 2050 and 2100 respectively (CEPAL, 2010). At the country level, under the A2 climate change scenario, it is estimated that by 2050 the economic losses in the Costa Rican agriculture sector could represent between 1-2% of GDP. For Honduras, these losses could reach 5-8% of GDP and for Guatemala could be between 0.63% and 1.30% of GDP, and may be higher for farmers in subsistence conditions (Mora et al., 2010; Ordaz et al., 2010a; Ordaz et al., 2010b).

At the crop level, the impacts can be heterogeneous. For coffee crops, given that lower areas will be warmer and drier with climate change, coffee production in these areas will be less viable than at present (Bunn et al., 2015). Therefore, it is expected that coffee production will move towards higher lands, where climatic conditions are more favourable, mostly in areas above 1800 meters above sea level, but could also lose economic viability in some cases (Laderach et al., 2010). This will result in the redistribution of areas suitable for coffee production, adjusting to the range with conditions for growth and productivity. Under A2 climate change scenario, areas with excellent aptitude for Arabica coffee will decrease by 12% in Central America in the future. This change is expected to be greater in Honduras (24%), Nicaragua (21%) and Guatemala (13%). In addition, new areas with excellent aptitude are forecasted to rise in Guatemala (46% increase according to the current area with excellent aptitude), Costa Rica (12%) and Honduras (9%), and to a lesser extent in the other countries of Central America for the year 2050.

Regarding basic grains, climate change will affect the distribution of the areas suitable for their cultivation in the region. It is expected that the area suitable to produce maize will be reduced very slightly in Central America during the next 35 years. However, in some areas of Costa Rica and Guatemala, the land could be improved for the cultivation of maize by 2050. The area suitable for bean crops is expected to be reduced for the whole Central American region by the year 2050. Projections show that suitable areas with excellent aptitude under current conditions will decrease by 14%, mainly in Panama (41%) Costa Rica (21%) and El Salvador (20%). In addition, it is not expected that there will be new areas with excellent aptitude in the countries of Central America.

By the end of this century (A2 climate change scenario), the area cultivated with maize will be reduced by 35% in Central America. In addition, 62% of the areas that grow maize in the region will have yields below 1.5 T/ha. While it is expected that the regional production of beans will decrease by 43%, and 61% of the departments or cantons of the region will have yields lower than 0.55 T /ha. Panama could have the highest yield reduction in both crops (maize and beans). Regarding rice cultivation, under scenario A2 this would be the most affected crop with a 50% reduction in regional production, where Nicaragua and Belize would be the most affected countries, with reductions of over 69 and 57% respectively (CEPAL and CAC/SICA, 2013). Under a climate change scenario considering 3.5ºC temperature increase and a 30% reduction of rainfall, Vargas et al. (2018) estimated a reduction in production and export of crops and livestock affecting wages and dropping the GDP of Guatemala by 1.2% and increasing food insecurity.

The combination of high exposure to climate hazards with the exposure to earthquakes, together with a high vulnerability of the population and a low adaptive capacity, results in several of the CA countries repeatedly being ranked at the highest risk level worldwide. For the period 2012–2016 Guatemala, Costa Rica, El Salvador and Nicaragua were among the 15 riskiest countries in the world following the methodology by Kirch et al. (2017). Other authors also include CA as one of the world's hotspots for climate change (De Sherbinin, 2014).

[PLACEHOLDER FOR SECOND ORDER DRAFT: other losses already noticed or expected to happen - approaching losses in ecological, human and built systems, highlighting cities]

12.3.2 Northwestern South America (NWS) Subregion

12.3.2.1 Hazards

There is a *high confidence* of hazard increase in relation with warming, heat waves, sea level rise, coastal erosion and ocean and lake acidification. There is *medium confidence* in wet trend and coastal flood. There is a *low confidence* on frost, dry trend and drought. [PLACEHOLDER FOR SECOND ORDER DRAFT: still to complete with final WGI hazards table]

The projections suggest an increase in the occurrence of warmer days and nights (*medium confidence*), changes in the duration of dry spells (*medium confidence*), and decreases in water supply (*high confidence*). Warming has been observed in Peru for annual mean, maximum and minimum temperatures about 0.16 to 0.18ºC per decade (Vicente-Serrano et al., 2018). Warming and drier conditions are projected through the reduction of total annual precipitation, extreme precipitation and consecutive wet days, and increase in consecutive dry days (Chou et al., 2014) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table 11.6]. In the coastal area of Ecuador, an increasing trend of 0.12°C per decade in ambient temperatures has been observed in the period 1959–2011 (Contreras López et al., 2014) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 12.4.4.1]. In Quito, droughts have intensified in frequency and length since the middle of the 20th century, as well as precipitation and temperature in Ecuador (Cáceres et al., 1998; García-Garizábal et al., 2017; Domínguez-Castro et al., 2018). Heat waves will increase in frequency and severity in places close to the equator as Colombia (Guo et al., 2018) Positive increases in precipitation, in the range from 0.1–0.6 mm/day, with respect to the base climatology, are expected (Hidalgo et al., 2017) [PLACEHOLDER FOR SECOND ORDER DRAFT].

El Niño Southern Oscillation (ENSO) is the dominant phenomena affecting weather conditions along the Pacific Coast of NWS (Cai et al., 2015). The El Niño phase supposes the warming of the ocean, with local effects that can vary substantially, but with effects at a regional as heavy rains, storms, floods, landslides and heat and cold waves (Ashok et al., 2007; Wang et al., 2017; Rodríguez-Morata et al., 2018; Rodríguez-Morata et al., 2019). Results from five available CMIP6 models suggest that ENSO variability is likely to weaken under SSP1-2.6 and SSP2-4.5, beginning in the near-term (2021–2040), while there is no consensus on SSP3-7.0 and SSP5-8.5 [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Chapter 4 ES]. There is a *medium confidence* that extreme ENSOs will increase long after 1.5ºC warming stabilization (Wang et al., 2017). Extreme La Niña (which is a cooling of the ocean that can bring dryer conditions), are also expected to increase (Cai et al., 2015).

Other increasing hazards include air pollution, as seen in cities (Henríquez and Romero, 2019), and the increase of solar UV radiation (Cabrol et al., 2014), hails and snow storms. The projections suggest a sea level rise ranging between 0.26 – 0.77 m by 2100 with a 1.5°C rise in temperature (*medium confidence*) (Reguero et al., 2015; Reyer et al., 2017; Hoegh-Guldberg et al., 2018; Muis et al., 2018).

Figure 12.2: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections NWS CMIP5/CMIP6]

12.3.2.2 Exposure, vulnerability and impacts

ENSO-triggered floods altered completely the annual discharge of many watersheds. Anomalous years as 2015–16 (among others) signified enormous fluvial discharges draining towards the Pacific Ocean, but also to the Atlantic. These floods affected large cities built on medium-latitudinal Andes like Lima and Quito (Isla, 2018). In the Andean countries from Venezuela to Chile, floods make up about 50% of all recorded disasters of the past decades, followed by landslides (29%). In the Andes, Peru and Bolivia have the highest record of climatically related disasters, followed by Colombia and Ecuador. In Peru alone, the ENSO of 2017 left in 6–9 billion US dollars on monetary losses, 114 deaths, more than a million inhabitants affected, 6614 km of roads damaged, 326 bridges destroyed, 41,632 homes destroyed or uninhabitable, and 242,433 homes, 2,150 schools and 726 health posts damaged (French and Mechler, 2017). Sediment transport as well as river erosion are derived processes (Morera et al., 2017). A moderate increase in the frequency of disasters in the NWS has been reported, yet without clear trends (Huggel et al., 2015a; Stäubli et al., 2018). Scale studies indicate an increase of flood risk for the subregion during the 21st century, consistent with more frequent floods (*medium to high confidence*), being worst in higher emission scenarios (Arnell and Gosling, 2013; Hirabayashi et al., 2013; Alfieri et al., 2017).

Exposure and vulnerabilities for precipitation, and emergencies and disasters related, have increased in the cities of Ecuador for the period of 1980–2009 (Briones-Estébanez and Ebecken, 2017). Those living on river banks and slums built on steep slopes are among the most affected by floods of all kinds, but also livelihoods dependant on activities as agriculture or tourism. Irrigation, potable water, roads, bridges, health, education and housing buildings, are frequently damaged or destroyed by extreme ENSO events (Lowe et al., 2017; Martínez et al., 2017; Rosales-Rueda, 2018). The absence of proper drainage systems inside buildings and cities increases the vulnerability to floods, to cold and heat waves, and wind storms. Informal housing and settlements, usually located in the highest risk land, undermines the adaptive capacity (Miranda Sara and Baud, 2014; Cuvi, 2015; Miranda Sara and Jameson, 2016; Lampis, 2017), while systems of draining such as that of Guayaquil increases adaptation. The highest vulnerability and exposure of informal settlements occur in the highlands, in cities such as Bogota, Quito or Cusco. In Quito, a bigger exposure to air pollutants as particles is also related with socioeconomic conditions (Rodríguez-Guerra and Cuvi, 2019).

Coastal lowlands are exposed to sea level rise in the form of coastal flooding and erosion, and subsidence and saltwater intrusion. The percentage of the population living at the low elevation coastal zone, according to Neumann et al. (2015), is 2% for Peru, 3% for Colombia, and 14% for Ecuador, which means around 15 million people. Cities as Lima and Guayaquil host up to 10 and 3 millions of people, respectively.

Sea temperature warming will affect marine life and fisheries in the three countries, being Peru the most sensitive because the proportional contribution of fisheries to GPD (Magrin et al., 2014; Cai et al., 2015; Ding et al., 2017; Gutiérrez et al., 2017). In Ecuador it is projected a loss of more than 50% of the current maximum catch potential (Cheung et al., 2018).

Glaciers have been in constant retreat as an impact of climate change. The tropical Andes have lost about 30% of their area since the mid–1980s, a bigger percentage than non-tropical regions of the Andes (Reinthaler et al., 2019). In Ecuador, the glacier of the Antisana has lost mass, related to a 15% decrease in surface area and a 235 m retreat of the glacier snout (Basantes-Serrano et al., 2016). In Colombia, glaciers lost 52% of their area between 1985 and 2016 (Rabatel et al., 2018). In Peru the loss was of 29% for the period 2000–2016 (Seehaus et al., 2019), with mass changes of about 0.4 m w.e. yr^{-1} for the period 2001–2017 (Dussaillant et al., 2019). In a low emissions scenario, by the end of the 21st century, Peru will lose about 50% of present glacier surface, while in a high-emission scenarios there will remain very small areas of only about 3–5% on the highest peaks, about 3–5% (Schauwecker et al., 2017). For Antizana, Ecuador, a loss of area of 72% and 98% is projected for RCP2.6 and RCP8.5 scenarios, respectively (Cuesta et al., 2019). Overall, based on climate-glacier model intercomparison runs an average glacier mass loss of about 60% to 90% is projected for the tropical regions by the end of the century (Hock et al., 2019a; Hock et al., 2019b).

Changing glaciers, snow and permafrost, have implications for the occurrence, frequency and magnitude of derived hazards (*high confidence*). That process also leads to landscape transformations through the formation of new lakes and drying of existing ones, alteration of hydrological dynamics with impacts on water for human consumption, agriculture, industry and hydroelectric generation, modification of biodiversity, floods, slope instability, landslides (Michelutti et al., 2015; Carrivick and Tweed, 2016; Emmer, 2017; Mark et al., 2017; Milner et al., 2017; Reyer et al., 2017; Vuille et al., 2018; Cuesta et al., 2019; Drenkhan et al., 2019; IPCC, 2019b). These changes will have diverse outcomes: new lakes represent both a source of future hazards and water scarcity, as well as an opportunity for water resource management (Colonia et al., 2017; Drenkhan et al., 2019).

Glacier shrinkage typically implies an initial increase of runoff, followed by a decline when ‘peak water’ is passed (Hock et al., 2019b). The timing and extent of peak water is spatially highly variable. Peak water passed for a large number of tropical Andes glaciers (*medium confidence*) (Baraer et al., 2012; Huss and Hock, 2018). The impact of changing runoff on people and economy depends, among other, on the glacier melt contribution to total runoff of the basin, and the type and extent of people depending on it. While the impact during the wet season may be small, it can be substantial during seasonal dry or drought periods (Buytaert et al., 2017; Schoolmeester et al., 2018). In Peru, a 50% reduction in glacial runoff would result in a decrease in annual hydro electrically-generated power output of around 10% (Vergara et al., 2007); a similar process could occur in Ecuador and Colombia, where the proportional participation of hydroelectric sources is increasing. The endangerment of freshwater supplies and ecosystem services is leading to costly extensions of large water infrastructure (Miranda Sara and Baud, 2014; Miranda Sara et al., 2017), leading at the same time to processes of conservation and restoration, as in Quito and Bogotá.

Slope stability is expected to decrease, but the magnitude and timing of this process are poorly defined (Haeberli et al., 2017). In the 20^{th} and early 21st century, many large glacier related disasters killed hundreds to thousands of people in single events of floods or landslides (Huggel et al., 2007; Iribarren Anacona et al., 2015; Emmer, 2017). Glacier lake outburst floods, ice and rock avalanches, debris flows, including lahars from ice-capped volcanoes are among the most severe threats. Observed trends are difficult to determine due to the rare occurrence of the hazards but there is typically a delay in response of hazard occurrence to cryosphere change (decades or longer) (Harrison et al., 2018). Consequently, projections for the future are uncertain.

The altitudinal distribution of ecosystems is changing, and causing extinctions and habitat reduction for some species, and increase for others (*very robust evidence, very high agreement*) (Ramirez-Villegas et al., 2014; Duque et al., 2015; Morueta-Holme et al., 2015; Andrango et al., 2016; Báez et al., 2016; Aguirre et al., 2017; Cuesta et al., 2017; Cuesta et al., 2019), including vectors of diseases, at times related with events such as ENSO (Lowe et al., 2017). Extinctions in amphibians have been related with temperature raises and acting in synergy with diseases (*very low confidence, low agreement*) (Catenazzi et al., 2014). Ecosystems such as high-elevation wetlands could degrade or disappear because of the disruption in water flow due to glacier shrinkage (Cuesta et al., 2019). As water availability has an important role in shaping spatial patterns of carbon stocks in NWS, water deficits can enhance tree mortality and favour small trees (Álvarez-Dávila et al., 2017). The ecotones are migrating at slower rates, so it is expected to be a hard barrier to migration to several species, leading to drastic population and biodiversity losses (Lutz et al., 2013). The agricultural frontier is also going upwards as it becomes suitable crops as potatoes or maize (*very robust evidence, very high agreement*) (Morueta-Holme et al., 2015; Skarbø and VanderMolen, 2016), and the freezing level height (FLH) has risen over the last decades (Schauwecker et al., 2017).

Farming systems are vulnerable, particularly those small, as they are highly dependent on water and biodiversity. Climate change related hazards could foster rural poverty. It has led to the modification of agriculture calendars and irrigation adjustments (Postigo, 2014). Impacts in society includes the loss of cultural values as those associated with glaciers or watersheds, opportunities for recreation, food security, spirituality, demography and migration (Carey et al., 2017). Many people choose to migrate as adaptation, with some good consequences in economic terms, but also others as the loss of local and indigenous knowledge. Migration to cities can mean a good adaptation, but also can worsen the problems as urban poor people can become even more exposed and vulnerable. In Ecuador, environmental variables are most likely to enhance international than internal migration (Gray and Bilsborrow, 2013). In the central highlands of Peru, there are different patterns, including daily circular migration to combine the scarce income from agricultural production with urban income, rather than abandoning the farming land (Milan and Ho, 2014; Zimmerer, 2014).

Vector-borne diseases such as dengue, malaria, zica, chikungunya, among others, can be related with the increase of floods in lowlands, and the expansion of the diseases' altitudinal distribution ranges. The predicted number of dengue cases between the 3.7ºC and the 1.5ºC warming scenarios by 2050 and 2100, range from 28,900 to 88,800 in Peru, 34,600 to 110,000 in Equator, and 97,400 to 317.000 in Colombia, although this does not consider uncertainties such as dengue vaccines evolution or socioeconomic trajectories (Colón-González et al., 2018). The increase of health problems is also expected in relation with the increasing of UV and solar radiation. In Quito, in July of 2017 solar radiation was significatively stronger than the previous six years (Environmental Secretariat of Quito, 2018).

A changing in fire regimes and fire risk is expected to occur in highland ecosystems as paramo and cloud forest (Anderson et al., 2011), although it is still difficult to determine the influence of anthropogenic and climate change influence on fire patterns in all the subregions (Armenteras et al., 2020).

12.3.3 Northern South America (NSA) Subregion

12.3.3.1 Hazards

The observed temperature extremes presented with *medium confidence* increases in minimum and maximum temperatures (Donat et al., 2016; Almeida et al., 2017). Precipitation showed increasing trends in the annual and wet season over the eastern part and decreasing trends of the dry season (Almeida et al., 2017).

Regional changes, such as increases in rainfall amounts in western NSA, are expected. Projections also suggest increases in the occurrence of warm days and nights (*medium confidence*), changes in the duration of dry spells (*medium confidence*), and decreases in water supply (*high confidence*) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Chapter 12 ES].

An overall increase in temperature by the end of century is projected for all the seasons. A decrease in precipitation over the tropical region (Teichmann et al., 2013; Sánchez et al., 2015) is also expected. Changes in the dry season in the central part of South America due to the late onset and late retreat of monsoon, decreases in precipitation over the Amazon and central Brazil are expected (Coppola et al., 2014; Giorgi et al., 2014; Llopart et al., 2014).

Projected warming (Chou et al., 2014) and decreasing of total annual precipitation, heavy precipitation (R95p) and consecutive wet days (CWD) (Seiler et al., 2013; Chou et al., 2014) while (Giorgi et al., 2014) shows an increase of R95p. Increase in dryness is shown in (Marengo and Espinoza, 2016) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table 11.6]. The projected extreme maximum temperatures (TXx) are +2.5°C for 1.5°C scenario, and about +3.2°C for 2°C scenario. For precipitation SR1.5 presents an increment of +10–30mm for Rx5day in both 1.5 and 2°C scenarios. In all RCP scenarios (P-E) will (very) likely decrease (Hoegh-Guldberg et al., 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: sea level rise, river floods, landslide and pluvial flood hazards]

Figure 12.3: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections NSA CMIP5/CMIP6]

12.3.3.2 Exposure, vulnerability and impacts

A significant increasing trend on temperature and precipitation, increments in extreme flood and drought events as well as unusual dry periods over the past two decades have been identified in the NSA subregion (Marengo and Espinoza, 2016; Marengo et al., 2018). Studies show great uncertainty regarding future projections of climate change for the NSA, suggesting however that droughts can intensify and more intense and recurrent extreme events throughout the 21st century will take place (*high agreement, medium evidence*) (Marengo and Espinoza, 2016; Liu et al., 2018). Anomalies in climate variability, as well as unprecedented extreme events reported for the Amazon, have led to severe negative impacts on human, natural and managed systems.

The assessment of climate change exposure, vulnerability and impact in NSA subregion is particularly complex due to the variety of biomes encompassed; from the coastal zones of Venezuela, Guyana and Suriname to the Amazonian forests, including part of the Andes and the vast savannahs of Venezuela. The scientific research in NSA is significantly biased towards the Amazon biome, the information gap for the rest of the subregion needs to be bridged for a more comprehensive assessment in the future. Low elevation coastal zones (LECZ) of Venezuela, Guyana and Suriname are increasingly exposed to sea level rise (Reguero et al., 2015). These countries have been classified as highly vulnerable to climate change (CAF, 2014; Villamizar et al., 2017; Nagy et al., 2019). The percentage of the national population living in LECZ areas is 56% for Guyana, 68% Suriname and 6% Venezuela (Nagy et al., 2019). Under the different climate change scenarios, 85% of the Venezuelan coastline (5600 km^2 of wetlands, mainly mangroves, with 19 protected areas) are at risk of being impacted by sea elevation (Nagy et al., 2019). According to CAF (2014), Suriname has experienced coastal erosion and flooding, causing damage to infrastructure, agriculture and ecosystems while Georgetown, the capital of Guyana, has suffered a significant number of floods.

The unprecedented extreme events of floods (2009, 2012 and 2014) and drought (2010) in the Amazon basin led to increased societies vulnerability (Mansur et al., 2016; Debortoli et al., 2017; Menezes et al., 2018). Predisposition of societies to be adversely affected by climate change varies according to poverty levels, ethnic composition, age structure, governance, access to natural resources, adaptive capacity and spatial accessibility (Pinho et al., 2015; Menezes et al., 2018; Parry et al., 2018). The disruption of the region natural hydrology dynamics, as a consequence of consecutive years of extreme events can increase the sensitivity of the food and transport systems of the indigenous and rural resource-dependent communities and reduce the ability of these communities to cope with climate change induced alterations (e.g., Pinho et al. (2015). According to Pinho et al. (2015), the sensitivity increase of social determinants, as well as the decrease in social resilience, exacerbates communities socio-environmental vulnerabilities (*medium confidence*). Nevertheless, the dynamics of the adaptive capacity of the indigenous and rural resource-dependent communities is a complex issue. There is a robust and increasing literature showing that resource-dependent communities located in remote areas, address climate anomalies by reducing the vulnerability of socio-ecological systems through indigenous and local knowledge (Mistry et al., 2016; Vogt et al., 2016).

The unprecedented extreme drought and flood events of the last decade have impacted indigenous and rural resource-dependent communities in various dimensions of their livelihoods. Food security has been strongly impacted since it is based on two sectors highly vulnerable to climate change such as fishing and small scale agriculture. During extreme droughts and floods, there is a decrease in fishing due to limited access to fishing grounds; during drought there is also a reduction in fish size and species availability (Pinho et al., 2015; Camacho Guerreiro et al., 2016). Small scale agriculture practices in this area (e.g., floodplain agriculture and slash and burn) are highly coupled with natural hydrological cycles and therefore severely affected by climate change induced alterations such as extreme floods and droughts (Cochran et al., 2016). Livelihoods are also impacted by disruptions in land and river transport, restrictions in drinking water access, increased incidence of forest fires and disease outbreaks caused by extreme droughts and floods (Marengo et al., 2013; Pinho et al., 2015; Marengo and Espinoza, 2016; Marengo et al., 2018).

In NSA subregion, urbanised areas are highly exposed to extreme flood events. In the Amazon Delta and Estuaries (ADE) 41% of the total population of urban centres, are exposed to flooding (Mansur et al., 2016), while in Santarem, a Brazilian Amazon city, population and infrastructure are highly exposed to floods and flash floods (Andrade and Szlafsztein, 2018). In urbanised areas, poor planning and high population densities increase exposure levels (Mansur et al., 2016). Based on conceptual models and multicriteria indexes, Mansur et al. (2016) showed that 60–90% of the population (ca 1.6 million people) in the urban centres of ADE live in conditions of moderate to high degree of vulnerability. In Santarem, where 80% of the area is highly or moderately vulnerable to flooding and flash floods, the vulnerability of extreme events is determined by the level of exposure and adaptive capacity (Andrade and Szlafsztein, 2018). Inequalities in the spatial accessibility in cities are another subjacent driver of social vulnerability in urban centres. According to Parry et al. (2018), Amazon populations located in remote urban centres with limited or non-existing roads in relation to more connected urban centres, are more vulnerable to extreme events (drought/flood).

In the Amazon basin, approximately 80% of the population is concentrated in cities due to migrations in search of improvements in education, health and goods and services (Pinho et al., 2015). These populations are settling in areas prone to flooding combined with various levels of unhealthiness (Pinho et al., 2015; Mansur et al., 2016; Andrade and Szlafsztein, 2018; Parry et al., 2018). According to Parry et al. (2018), extreme events impact these populations in their health through outbreaks of waterborne and metaxenic diseases. Food security is also impacted due to cities dependence on imported products whose transportation (road/river) is affected by floods and droughts. In addition, flood events have caused losses of homes and disruption of public and commercial services.

Research on the impacts of climate change in population health in the Amazon basin is scarce, particularly in relation to metaxenic diseases and microbial proliferation (Brondízio et al., 2016). Several diseases such as malaria and leishmaniasis are endemic to the region, however socio-environmental changes (e.g., deforestation and climate change) are altering natural dynamics such as geographical variation of the vector niche, forms of transmission and drug resistance (Confalonieri et al., 2014b). An important relationship between the outbreak of infectious diseases and changes in climatic events (e.g., droughts, floods, heat waves, ENSO, etc.) or environmental events (e.g., deforestation, dam construction and habitat fragmentation) have been found for the Brazilian Amazon (Filho et al., 2016; Nava et al., 2017). These impacts are more severe in poor populations with limited access to health services (Pan et al., 2014).

Early studies show that Amazonian forests constitute one of the major carbon (C) sinks on Earth (Pan et al., 2011) and therefore they play an essential role in precipitation and C balance at a regional scale (*high confidence*) (Marengo et al., 2018). For these reasons, any alteration affecting the functioning of these forests as a C sink would have an impact on atmospheric CO2 and hence on global climate change. There is robust evidence of the vulnerability of the Amazonian forests to drought (*high confidence*) (Brienen et al., 2015; Olivares et al., 2015; Feldpausch et al., 2016; Zhao et al., 2017; Anjos and De Toledo, 2018; Sampaio et al., 2018; Yang et al., 2018). The 2010 drought increased tree mortality rate in these forests (Doughty et al., 2015; Feldpausch et al., 2016), while productivity didn’t show a consistent change; some authors report a drop in productivity (Feldpausch et al., 2016) and others found no significant changes (Brienen et al., 2015; Doughty et al., 2015). Nevertheless the combined effect of increasing tree mortality with variations in growth, results in a long-term decrease in C stocks in forest biomass compromising their role as C sink (*high confidence*) (Brienen et al., 2015).

Under the RCP 8.5 scenario for 2070, drought will increase the conversion of rainforest to savannah in Eastern Amazonia (Anadón et al., 2014; Olivares et al., 2015; Sampaio et al., 2018). Prolonged droughts will also promote the expansion of savannahs in Guyana and Suriname for 2081–2100 (Olivares et al., 2015). The transformation of rainforest into savannahs brings forth biodiversity loss and alterations in ecosystem functions and services such as food production, climate regulation, C sink and hydroelectric capacity, among others (Anadón et al., 2014; Olivares et al., 2015; Sampaio et al., 2018). In this region the synergic effects of deforestation, fire, climate change and extreme events will exacerbate the risk of savannisation (Sampaio et al., 2018).

12.3.4 Central South America (SAM) Subregion

12.3.4.1 Hazards

The Amazon has been identified as one of the areas of persistent and emergent regional climate change hotspots in response to various representative concentration pathways (Diffenbaugh and Giorgi, 2012). In a vast transition zone between the Amazon and the Cerrado Biomes within the region, analysis of seasonal precipitation trends suggested that almost 90% of the observational sites showed reduced in the length of the rainy season in the region (Debortoli et al., 2015), on a period from 1971 to 2010. Changes in the hydrological and precipitation regimes, which characterize by reduction in rainfall in Southern Amazonia, contrasting to an increase in the northwest Amazonia, and overall an increase of climate extreme event, is being reported by several authors (Fu et al., 2013; Almeida et al., 2017; Marengo et al., 2018; Espinoza et al., 2019).

[PLACEHOLDER FOR SECOND ORDER DRAFT: additional results related to observed changes]

Brazil has experienced extreme climatic events recently (Marengo and Bernasconi, 2015) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Chapter 12]. In Bolivia, CMIP3/5 models projected an increase in temperature (2.5º–5.9ºC), with seasonal and regional differences. In the lowlands, both ensembles agreed on less rainfall (-19%) during drier months (June–August and September–November), with significant changes in inter annual rainfall variability, but disagreed on changes during wetter months (January–March) (Seiler et al., 2013). As a consequence of higher temperatures and reduced rainfall, an increased water deficit would be expected in the Brazilian Pantanal (Marengo and Espinoza, 2016). The largest increases in warmer days and nights are expected in the Central South America area (SAM) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Chapter 12].

Figure 12.4: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections SAM CMIP5/CMIP6]

12.3.4.2 Exposure, vulnerability and impacts

[PLACEHOLDER FOR SECOND ORDER DRAFT: Explore how large-scale drivers are interacting to local processes (e.g., climate change and land use change) leading to increased exposure and impact]

The SAM region is where the largest expansion of croplands (soybean, corn and sugarcane) has been observed in the past two decades, partly in response to an increased demand for biofuels to abate GHG emissions and partly to attend external demand for such commodities (Lapola et al., 2014; Cohn et al., 2016). Feedbacks to the climate system resulting from such land-use changes are intricate. On the one hand the clear-cutting of Amazon forest and Cerrado savannah in the SAM region lead to a local warming due to an increase in the Bowen ratio (Malhado et al., 2010). On the other hand, the implementation of sugarcane (and presumably soybean) fields on land previously occupied by pastures in the Cerrado biome lead to a local near-surface cooling of ~0.9ºC owing to the resulting change in albedo and evapotranspiration. Deforestation of the Amazon for pastures and soybean fields have decreased evapotranspiration during drought months and caused a localized lengthening of the dry season in Northwest SAM by 6.5 (± 2.5) days since 1979 (Fu et al., 2013).

It is not surprising therefore that while SAM is the region in Central and South America that experienced the highest temperature increase in the last century, it is also where most of the fire spots in sub-continent are located, owing also to the prevalent use of fires in pasturelands (Bowman et al., 2009).

Although SAM is pointed out as a region where agricultural production will be especially impacted by climate change , there is considerable uncertainty regarding the effects of CO_2 fertilization and level of investments needed for adaptation (Lapola et al., 2014) on the geography of land use in the region.

[PLACEHOLDER FOR SECOND ORDER DRAFT: Ecological vulnerabilities (biodiversity sensitivity, water, specific biomes, oceans)]

Historic land cover change and concurrent climate change in the SAM region has created an extinction debt of 657 plant species for the Cerrado, which is more than four-fold the global recorded plant extinctions (Strassburg et al., 2017). The largest expanses of remaining Cerrado-biome vegetation are located in SAM, but the region is depleted in biodiversity-protected areas (only 7.5% inside protected areas), which will leave fauna and flora with little room for moving across the landscape in face of climate change. Protected areas—indigenous lands included—have markedly refrained forest clear-cutting in the Amazon deforestation arc (most of which is inside SAM) (Nolte et al., 2013). However nearly one hundred protected areas in the Amazon, Cerrado and Pantanal biomes inside SAM have been identified as highly or moderately vulnerable to future climate change and demand deep adaptation interventions (Feeley and Silman, 2016; Lapola et al., 2019).

Yet, the maintenance of these protected areas or even the halting of deforestation may do little to impede a large-scale tipping point of the Amazon forest or even more subtle changes caused by climate change in the region (Lapola et al., 2018; Lovejoy and Nobre, 2018). Considering that SAM hosts the ecotone between the Amazon forest and the South American Cerrado is it expected that the first impacts of a drier climate over the forest will be seen there. In fact an increasing dominance of drought-affiliated genera of tree species has been reported in the southern part of the Amazon forest in the last 30 years (Esquivel-Muelbert et al., 2019).

SAM hosts the headwaters of important South American rivers such as the Paraguay, Madeira, Tocantins-Araguaia and Xingu. Modeling studies project decreases in river discharge rate in the order of 27% for the Tapajós basin and 53% for the Tocantins-Araguaia basin for the end of the century, which may affect freshwater biodiversity, navigation and generation of hydroelectric power (Marcovitch et al., 2010; Mohor et al., 2015).

The rise of the large-scale soybean agroindustry in the early 2000's has led to a faster increase in human development indicators in Mato Grosso compared to other neighbouring regions, with the development of industrial facilities and infrastructure (e.g., major ports in the Madeira Waterway) and the generation of job posts that are tightly linked to the soybean production chain (Richards et al., 2015). Such a development also came at a considerable cost for the environment (e.g., Neill et al. (2013)) and the regional climate, even though a moratorium implemented in 2006 to refrain new soy plantations on deforested areas reduced deforestation by a factor of five (Macedo et al., 2012; Kastens et al., 2017). The same sort of supply chain interventions along with incentive-based public policies (rather than punitive measures) applied for the beef supply chain could minimize the need for agricultural expansion in the SAM deforestation frontier (Nepstad et al., 2014).

Projections point out soybean yields will suffer one of the strongest negative impacts due to climate change and will demand strong investments for adaptation should it continue to be cultivated in the same localities as today (Oliveira et al., 2013). As such the future socio-economic vigour of the region will be, to a large extent, connected to an unlikely stability of the regional climate and eventual fluctuations of global markets affecting the agricultural supply chain (Nepstad et al., 2014).

SAM has a low population density, and the majority of population is located in cities. The population of some of these cities are indicated as highly vulnerable to the direct effects of future climate change considering the enormous social inequalities embedded in these cities (Darela-Filho et al., 2016). Cuiabá and Brasília urban areas for example are projected to have an increase of more than 600% in deaths related to heatwaves from 2031 to 2080 compared to the period of 1971–2020 (Guo et al., 2018). Observations from recent past droughts in SAM indicates how the incidence of respiratory diseases may worsen under a drier and warmer climate. Northwest SAM had a ~54% increase in the incidence of respiratory diseases associated with forest fires during the 2005 drought compared to a no-drought 10-year mean (Ignotti et al., 2010; Pereira et al., 2011; Smith et al., 2014). On the other hand, the high risk of floods (high-frequency and high-incurred damage) is found in the Brazilian states of Acre, Rondônia, and Southern Amazonas and Pártá (Andrade et al., 2017).

The agriculture in the SAM region is highly depended on the climate, which is related basically to ¾ of the agricultural yields' variability in the region. The remaining fluctuation on yields refers to issues related to infrastructure, economic, policy and social. Good infrastructure, on transport logistics, quality of roads and storage, strongly influences the vulnerability of the agriculture sector.

Global-scale studies indicate an increase of flood risk for the SAM region during the 21st century (consistent with more frequent floods) (*medium to high confidence*) (Hirabayashi et al., 2013; Arnell et al., 2016; Alfieri et al., 2017). Higher emission scenarios result in substantially higher flood risks than low emission scenarios (Alfieri et al., 2017).

12.3.5 Northeastern South America (NES) Subregion

12.3.5.1 Hazards

There is *medium to high confidence* that extremes derived using daily minimum temperatures (TN) are increasing faster than those derived using daily maximum temperatures (TX) (Donat et al., 2016), with the largest warming rates observed over NES for cold nights [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 11.9.5]. In this region, all temperature extremes like annual maximum temperature have increased, with some non-significant trends in 1957/1962–2011 TN (Lacerda et al., 2015), and with more recent (1970-2012) significant warming trends in MATOPIBA region (a region intercepting the boundaries Maranhão, Tocantins, Piauí and Bahia states in Brazil) (Salvador and de Brito, 2018) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 12.4.4.1]. The projected warming for the extreme annual maximum temperatures over NES is TXx: +2°C for the 1.5° scenario and about +2.5°C for 2°C scenario (Hoegh-Guldberg et al., 2018).

Orlowsky and Seneviratne (2012) projected an increased number of tropical nights with minimum temperatures exceeding the 20°C-threshold, particularly in NES [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 12.4.4.1].

Non-significant decreases in total annual precipitation was found in NES [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 12.4.4.3]. Most precipitation extremes indices in São Francisco River Basin decreased from 1947 to 2012, although without statistical significance (Bezerra et al., 2018). The northern part of NES shows positive precipitation change with respect to the 1961–1990 period in three simulations, but negative change in the RCP 8.5 Eta-HadGEM scenario (Chou et al., 2014; Giorgi et al., 2014). An increase in dryness is projected for the subregion due to the combination of increased temperatures, less rainfall, and lower atmospheric humidity (5 to 15% relative humidity reduction). These conditions create water deficits, projected for the entire region after 2041 (3–4 mm/day reduction), particularly over western NES and over the semiarid region (Marengo and Bernasconi, 2015; Marengo et al., 2017).

Figure 12.5: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections for NES CMIP5/CMIP6]

12.3.5.2 Exposure, vulnerability and impacts

NES is a semi-arid region that suffers from periodic and sometime extreme drought events. The region is already experiencing an increase in temperature, and is projected to endure further warming and increased dryness. People and assets in the region are exposed to these projected climatic hazards, with an increased vulnerability due to the region's high poverty levels. The human population in NES is highly vulnerable to droughts, which have well-documented impacts on water and food security, human health and well-being in the region (e.g., Confalonieri et al. (2014a); Marengo et al. (2017); Bedran-Martins et al. (2018)). The majority of the population lives in urban areas (about 40 million people, data for 2010 from the Brazilian Institute of Geography and Statistics, https://cidades.ibge.gov.br/). In capital cities about 45% of the population live in poverty (data for 2003 from IBGE, https://cidades.ibge.gov.br/), often in slums with already deficient water supply and sewage systems and poor access to health and education, increasing their vulnerability to droughts. The rural population (about 15 million people, data for 2010 from IBGE, https://cidades.ibge.gov.br/) already suffers from natural scarcity of water in the countryside. A predicted increase in drought, coupled with inadequate soil management practices by small farmers increases the susceptibility to desertification (Spinoni et al., 2015; Vieira et al., 2015; Tomasella et al., 2018), threatening livelihoods in rural areas.

Recent studies predict strong negative impact of climate change on NES' production of rice, corn, tobacco, citric fruits, coffee, live animals, milk (Ferreira Filho and Moraes, 2015), mangaba fruit (*Hancornia speciosa*) (Nabout et al., 2016), and cocoa (Gateau-Rey et al., 2018). A projection of climate change impacts on Brazil's agriculture identifies NES as the most susceptible region in the country (Ferreira Filho and Moraes, 2015), concentrating the bulk of the predicted negative effects of the loss of regional gross domestic product (GDP). Although agriculture gives a modest contribution to the regions' economy, the study predicts that the drop in agriculture activity in NES would have a severe impact on the poorest workers and households in rural areas, by shrinking the agricultural labour market and increasing the price of consumption bundles (Ferreira Filho and Moraes, 2015). Finally, the expected increase in dryness is predicted to impact hydroelectric power generation in NES, due to an expected reduction in river flow (Marengo et al., 2017; de Jong et al., 2018).

NES has about 3800 km of coastline, where people, infrastructure and economic activities are exposed to sea level rise. The high concentration of cities on the coast increases the region's vulnerability (Martins et al., 2017). The capital city of all NES' state except for one (Piauí) are on the coast, totalling almost 12 million people exposed to sea level rise (estimate for 2019 from Brazilian Institute of Geography and Statistics (IBGE) available at https://cidades.ibge.gov.br/). The ports of São Luís, Recife and Salvador are important exporters of Brazilian commodities and also exposed to sea level rise. NES' beaches are an international touristic destination (Pegas et al., 2015), bringing considerable revenues to the region (Ribeiro et al., 2017). Another important economic activity on NES' coast is aquaculture for shrimp production. The region is responsible for 99% of the Brazilian shrimp production, with projected impacts on productivity from sea level rise as well as increased ocean temperature and acidification (Gasalla et al., 2017). The construction of infrastructure to contain the impacts of sea level rise (e.g., dicks) has not been a practice in the region.

NES' natural systems are also vulnerable to climate change. The 913,000 km2 of NES' Caatinga vegetation (Silva et al., 2017) are exposed to predicted increase in dryness. The Caatinga is particularly sensitive to variations in water availability (Seddon et al., 2016; Rito et al., 2017), and a global scale analysis placed it among the most sensitive vegetations to climate change worldwide (Seddon et al., 2016). The vulnerability of the Caatinga vegetation to climate change is further increased by the extensive conversion of native vegetation into human dominated landscapes (Lapola et al., 2014; Silva et al., 2017; Joly et al., 2019) with a documented interacting effect between precipitation and human disturbance on plant diversity in Caatinga dry forests (Rito et al., 2017). Some ecosystem services provided by nature in NES are predicted to be impacted by climate change, including carbon storage and provision of firewood (Althoff et al., 2016), prevention of soil erosion (Martins et al., 2017) and pollination (Giannini et al., 2017).

There are few studies projecting the likely impact of climate change on NES' biodiversity as compared to forested systems in South America, such as the Amazon and the Atlantic Forest. Studies with terrestrial animals predict that most groups would be negatively impacted by climate change (endemic vertebrates, bats and ants), especially in wetter environments within NES, while some species might benefit from it (some endemic vertebrates) (de Oliveira et al., 2012; Arnan et al., 2018; da Silva et al., 2018), but all studies agree that habitat loss increases vulnerability to climate change. There is an even larger knowledge gap on the likely impacts of climate change on NES's aquatic biodiversity. Still, NES has a surprisingly high rate of endemism of freshwater fish. Fifty-two percent of the fish in the region are found nowhere else, representing 203 endemic fishes (Lima et al., 2017) exposed to predicted reduction in river flow due to climate change in the region (Marengo et al., 2017; de Jong et al., 2018).

Due to the biogeographic uniqueness of NES' coastal waters it is classified as a separate marine ecoregion (the Tropical Southwestern Atlantic) (Spalding et al., 2007). The 685 Km^2 of coral reefs along NES's coast (estimated from UNEP 2018, but likely underestimated (Moura et al., 2013) are exposed to increased sea temperatures. They represent most coral reefs in the Southern Atlantic Ocean (Leão et al., 2016), increasing NES coast's conservation and touristic value. Coral reefs are well known for its high sensitivity to climate change, particularly rising sea temperatures. A number of coral bleaching events associated with abnormal increase in sea temperatures have occurred in NES (Krug et al., 2013; Leão et al., 2016), but thus far corals have been able to recover after sea water temperature returned to normal values (Leão et al., 2016). The vulnerability of NES' coral reefs to climate change, however, is increased by the synergism between increased sea level temperature and other well-documented anthropogenic disturbances in the region, such as coastal runoff, urban development, marine tourism, overexploitation of reef organisms, and oil extraction (Leão et al., 2016).

12.3.6 Southeastern South America (SES) Subregion

12.3.6.1 Hazards

Temperature extremes have shown a decrease in maximum temperatures TX extremes but there is *medium confidence* that warm maximum temperature extremes (TXx and TX90p) have decreased in the last decades over most of SES during austral summer (Rusticucci et al., 2017; Wu and Polvani, 2017). With *high confidence* TX extremes showed decrease over south western SES [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table 11.7].

There is *low confidence* that the decrease in hot extremes over SES is related with an increase of extreme precipitation, produced by more intense cyclones and anomalous easterly flow related with a southward shift of the tropospheric jet due to ozone depletion in summer months (Wu and Polvani, 2017) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 11.9.5].

Projected trends in minimum and maximum temperature (TN and TX) extreme indices show a general warming over SES during the austral summer. The increase in the frequency of warm nights (minimum temperature over the 90th percentile, TN90p) is larger than that projected for warm days (maximum temperature over the 90th percentile TX90p) consistent with observed past changes. The larger increases in TN compared to TX have been related with changes in cloud cover that affect differently daytime temperatures (TX) as compared to night time temperatures (TN) (López-Franca et al., 2016; Menéndez et al., 2016) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 11.9.5];

Over SES most of the stations have registered an increase in annual rainfall, largely attributable to changes in the warm season and it is one of few regions where a robust positive trend in precipitation [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 10.19] has been detected since the beginning of the 20th century (Vera and Díaz, 2015; Saurral et al., 2017; Lovino et al., 2018) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI 11.9.5]. The projected warming from annual maximum temperature over SES is TXx: +2.3°C for the 1.5°C scenario and about +3°C for 2°C scenario.

Increases in rainfall amounts in SES are projected and also suggest increases in the occurrence of warm days and nights (*medium confidence*), changes in the duration of dry spells (*medium confidence*), and decreases in water supply (*high confidence*) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Chapter 12] In general, long-term and short-term droughts in the La Plata Basin will be more frequent in the medium-term future (2011–2040 with respect to the 1979–2008 period), but also shorter and more severe. Similar changes are also projected for the distant future (2071–2100) and for a more extreme emission scenario (RCP8.5) (Carril et al., 2016)

A higher observed frequency of extratropical cyclones in the region has been detected, with three cyclogenetic foci: South-southeast Brazil, extreme south of Brazil and Uruguay, and southeast of Argentina (Reboita et al., 2018). On the other hand, these authors have detected a negative trend in the annual number of cyclone events in the long-term future of 3.6 to 6.5% (2070–2098), consistent with Grieger et al. (2014), that showed a decrease of 3 to 11% (2080–2100 for the A1B scenario). Although there is a general negative trend in cyclone events, there seems to be no significant change in their main characteristics (Reboita et al., 2018). In Montevideo, mean sea-levels increased over the past 20 years, reaching 11cm from 1902 to 2016, and a recent accelerating trend has been observed (Gutiérrez et al., 2016b).

Figure 12.6: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5 projections for SES subregion CMIP5/CMIP6]

12.3.6.2 Exposure, vulnerability and impacts

SES has undergone a significant increase in rainfall amounts, and is projected to undergo a general warming and endure more extreme rainfall events, with increase in rainfall amounts together with an increase in frequency and severity of droughts, and associated decrease in water supply. These hazards will expose a large number of people in the region, particularly in cities.

The observed and projected increase in rainfall exposes people and infrastructure to landslides and flash floods. In the Plata basin, urban floods have become more frequent, causing infrastructure damage and sometimes substantial mortality (Barros et al., 2015). Studies show high exposure and risk to extreme precipitation in SES, mostly in Buenos Aires, Rio de Janeiro and São Paulo (Vörösmarty et al., 2013; Nehren et al., 2019). A global-scale study projects an increase in flood frequency in the Parana river, but a decrease in the southernmost part of SES following the most severe climate scenario (Hirabayashi et al., 2013). Other studies show a similar trend, where floods are expected to be more frequent and lasting in Parana and Uruguay rivers in the La Plata river basin (Camilloni et al., 2013; Nagy et al., 2019). A large increase in landslides and flash floods is also predicted for the Brazilian portion of SES, where they are responsible for the majority of the deaths related to natural disasters in the country (Debortoli et al., 2017), exposing around 84 million people (data for 2010 from the Brazilian Institute of Geography and Statistics, https://cidades.ibge.gov.br/). The vulnerability these events in SES is increased in the poor population, which often lives in marginal housing in risk areas (slopes or close to rivers) and have limited access to public services and goods, education, health care, financial resources and face exclusion (Debortoli et al., 2017).

Despite the observed increase in rainfall amount in the region, between 2014 and 2016 Brazil endured a water crisis that affected the population and economy of major capital cities in the SES region Brazil (Nobre et al., 2016a). Due to uncontrolled urban growth, 21.5 million people living in the large Brazilian cities of São Paulo, Rio de Janeiro and Belo Horizonte (estimate for 2019 from Brazilian Institute of Geography and Statistics (IBGE) available at https://cidades.ibge.gov.br/) are expected to be exposed to water scarcity, despite great water availability in the region (Marengo et al., 2017). Extremely long dry spells have become more frequent in southeast of Brazil, affecting 40 million people and the economies in cities such as Rio de Janeiro, São Paulo and Belo Horizonte, which are the industrial pole of the country (PBMC, 2014; Nobre et al., 2016a; Cunningham et al., 2017). Recently, the Cantareira System reached a near collapse state, affecting water supply of 8.8 million people in the region of São Paulo (Cunningham et al., 2017). Severe dry spells have also been documented for Paraguay (Benitez and Domecq, 2014) and Argentina (Barros et al., 2015).

The expected increase in temperature also exposes the population in large cities to extreme heat. Urban heat islands are already a reality in large cities in the region, such as Buenos Aires (Camilloni and Barrucand, 2012), Rio de Janeiro (Peres et al., 2018) and São Paulo (Vemado and Pereira Filho, 2016), with reported impact on human health in the latter (e.g., Araujo et al. (2015); Son et al. (2016)). These cities alone represent 22 million people exposed to increased heat (estimate for 2019 from Brazilian Institute of

Geography and Statistics (IBGE) available at https://cidades.ibge.gov.br/ and from INDEC. Proyecciones elaboradas en base a resultados del Censo Nacional de Población, Hogares y Viviendas 2010).

Many large cities in SES are located on the coast, increasing the region's vulnerability to sea-level rise. Buenos Aires, Porto Alegre, Rio de Janeiro, and Montevideo are at risk of flooding from sea-level rise (Nagy et al., 2019). Around 3.3 million people are affected by high waves in Argentina, Uruguay and Brazil, and 3.4 million people are affected by weak storm surges in Brazil and Argentina (Calil et al., 2017). Amongst the impacts related to sea-level rise are habitat destruction and the invasion of exotic species, affecting biodiversity and the provision of ecosystem services (Nagy et al., 2019). Impacts are expected to worsen, and ENSO events will become more frequent and intense due to climate change (Nagy et al., 2019). People and infrastructure of large coastal cities in SES such as Rio de Janeiro, Vitória, Joinville and Florianópolis in Brazil (Zanetti et al., 2016), Montevideo in Uruguay, and Buenos Aires, La Plata, Mar del Plata, Bahía Blanca and Comodoro Rivadavia in Argentina, house together more than 27 million people exposed to sea-level rise (estimate for 2019 from Brazilian Institute of Geography and Statistics (IBGE) available at https://cidades.ibge.gov.br/; from the INDEC. Proyecciones elaboradas en base a resultados del Censo Nacional de Población, Hogares y Viviendas 2010; and from the Instituto Nacional de Estadística (INE) – Estimaciones y proyecciones de población (revision 2013)).

The SES region concentrates 41.9% of the population of Brazil as well 55.3% of the national GDP (Cunningham et al., 2017). Extremely long dry in the Cantareira system in São Paulo impacted industry and agriculture, and the operation of hospitals and schools (Cunningham et al., 2017). Agricultural prices increased by 30% in some cases and harvest yields of sugar cane, coffee and fruits suffered a reduction of 15–40% in the region. The number of fires increased by 150%, and energy prices increased by 20–25%, as most electricity from hydroelectric power (Nobre et al., 2016a). Damages on crop production caused by severe droughts driven by climate change were also reported for Paraguay, which is extremely vulnerable as its economy is mostly based on agriculture. Records show a decrease of 4.2% in GDP in Southern Paraguay during the most severe drought episodes (Benitez and Domecq, 2014). In Argentina, agricultural losses of 29% for soybean, 20% for wheat and 12% of maize were detected, while flooding events affected wheat and maize yields and milk production (Barros et al., 2015). In Argentina, projected changes in hydrology of Andean rivers associated to glacier retreat are predicted to have negative impacts on the region's fruit production (Barros et al., 2015).

Sea-level rise and intense storms surges in the large coast of SES is exposed exposes the tourism industry in general (Nagy et al., 2019). Farming of mussels and oysters in the region is predicted to be negatively impacted by climate change, particularly sea-level rise, and ocean warming and acidification (Gasalla et al., 2017). Sea-level rise also exposes nuclear power plants along the coast of Brazil (Angra nuclear plant), although no studies have assessed its risk.

The region's natural systems are also exposed to climate change. SES region is houses two important biodiversity hotspots: the Cerrado and the Atlantic Forest, where about 72% of Brazil's threatened species can be found (PBMC, 2014). The Atlantic Forest is predicted to be one of the most threatened terrestrial biodiversity hotspot in the world, due to the combined effect of climate change, invasive species and land use change (Bellard et al., 2014). The combination of high levels of endemism and > 75% loss of original forest cover characteristic of the Atlantic Forest and the Cerrado (Myers et al., 2000) reduces species' ability to cope with climate change. An increasing number of studies show that the Atlantic Forest and Cerrado are risk of biodiversity loss [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGII Cross-Chapter Paper 1 Biodiversity Hotspots], largely due to projected reduction of species' geographic distributions in many different taxa (e.g., Loyola et al. (2012); Ferro et al. (2014); Loyola et al. (2014); Hoffmann et al. (2015); Martins et al. (2015); Aguiar et al. (2016); Vale et al. (2018); Borges et al. (2019); Braz et al. (2019), including wild edible plants species of high local importance (de Oliveira et al., 2015). Reductions in species' distribution are also projected in the La Plata Basin for subtropical amphibians (Schivo et al., 2019) and the river tiger (*Salminus brasiliensis*), a keystone fish of economic value (Ruaro et al., 2019). Some more localized habitats are also at risk of losing area due to climate change, such as the meadows of northwest Patagonia (Crego et al., 2014) and mangroves of southern Brazil (Godoy and Lacerda, 2015).

12.3.7 Western South America (SWS) Subregion

12.3.7.1 Hazards

Warming has been observed in the region (*high confidence*) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table 11.7, AR6 ATLAS 5.5.2.1], except for a cooling off the Chilean coast (*high confidence*). Andes temperature shows a significant warming trend during austral spring, summer and autumn seasons (Burger et al., 2018; Vicente-Serrano et al., 2018). From 1961 to 2016, Chile has shown a significant increment of 23% in heatwaves (duration and frequency) (*medium confidence*). The most important increments have been observed for Atacama Desert (northern Chile) and Metropolitan Region (Central Chile) (Campos-Caba, 2016; Ceccherini et al., 2016; Piticar, 2018). During January 2017, an extraordinary heat wave affected central Chile (Antofagasta and Curicó recorded the most prolonged warm periods with extreme high temperatures for 14 and 17 days, respectively. In 2016–2017, Chile had its second warmest summer since 1964, a winter several cold spells and heavy snowfalls, and a Pacific Ocean cooling (Blunden et al., 2018).

Robust drying trend in Chile (30ºS–48°S) (*medium confidence*) (Saurral et al., 2017; Boisier et al., 2018) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGI Table 11.7, AR6 ATLAS 5.5.2.1]. Warming and decrease in precipitation in the northern Chile (Saurral et al., 2017; Meseguer-Ruiz et al., 2018) and southern portion of SWS (Chou et al., 2014). An increase in the precipitation intensity (heavy daily rainfall) over the Titicaca basin and in the southern Pacific (Heidinger et al., 2018) have been observed.

Figure 12.7: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5. Projections SWS CMIP5/CMIP6]

Four sets of downscaling simulations based on the Eta Regional Climate Model forced by two global climate models (Chou et al., 2014) projected warmer conditions for all the region [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS Table 17] and drier conditions in total annual and extreme precipitation for Southern Chile. Projected extreme 5-day precipitation (Rx5day) for 1.5ºC and 2°C scenarios are about +10 to 20mm (Hoegh-Guldberg et al., 2018).

From December 2016 to February 2017, as a consequence of 'coastal El Niño', southern Peru showed a high variability in precipitation (from 0 mm to 3281 mm), where some Peruvian areas (Arequipa) received precipitation amounts that exceeded the historical records from 1982 (Rodríguez-Morata et al., 2019). In the Peruvian coast, 'Coastal El Niño' was associated with an increase of +5ºC, and coastal precipitations with a maximum of 8 mm day^{-1} (monthly average) which produced a decrease in the southern winds and decrease in the ocean cooling (*high confidence*) (Echevin et al., 2018; Garreaud, 2018; Hu et al., 2019).

Shifts of the Pacific anticyclone (Schneider et al., 2017) have been associated to changes in wave direction (from WSW to SW) in Central coast of Chile (Martínez et al., 2018). These anomalies have also presented a high correlation with ENSO warm events (Molina et al., 2011; Martínez et al., 2018). For Central Chile, a significant increase in wave heights between 1957 and 2016 has been observed, mainly caused by an increase in the extreme coastal events (Martínez et al., 2018). These extreme events have increased from 5% to 20% in the last 60 years. From 1982 to 2016, sea level at central Chile have increased 5 mm by year, where El Niño events of 1982–1983 and 1997–1998 caused an extreme increase of 15 to 20 cm in the mean sea level (Martínez et al., 2018).

As a consequence of the Pacific anticyclone poleward migration, Humboldt Current System (HCS) shows an intensification of wind stress and upwelling events (*high confidence*) (Sydeman et al., 2014; Aguirre et al., 2018; Jacob et al., 2018; Bindoff et al., 2019). Current upwelling intensification has increased the number of cold events (more than 14 days per decade) at the West South-America, resulting in a decline in the Surface Sea Temperature (SST) of almost –0.4ºC per decade (Lima and Wethey, 2012). Peruvian coast already shows a consistent cooling trend of –0.36ºC per decade (*high confidence*) (Gutiérrez et al., 2011). Between 2003 to 2013, the poleward displacement of the South Pacific Anticyclone in central Chile resulted in a cooler and saltier water column as a consequence of the intensification of upwelling events (Jacob et al., 2018).

CMIP3 and CMIP5 global climate models suggest an upwelling intensification in HCS (Rykaczewski et al., 2015; Wang et al., 2015; Garreaud, 2018; Oyarzún and Brierley, 2019), mainly in the southern upwelling zone at surface depths (i.e., Peru and Chile) (Oyarzún and Brierley, 2019). As a consequence, Humboldt Current is projected to experience widespread calcium carbonate undersaturation within a few decades (Bindoff et al., 2019).

Glaciers in the southern regions of SWS (and the SSA region) have lost more thickness (i.e., higher negative mass balances) than in the central Andes but due to their large size the relative area change rate over the past 3 decades is often lower (about 0.3 to 1% yr^{-1} while up to >3%yr^{-1} for smaller glaciers) (Braun et al., 2019; Reinthaler et al., 2019).

The area loss since 1985 in the region is in a range of 20 up to 60% (Reinthaler et al., 2019). The current glacier area in Chile has been reported as 23,700 ±1,185 km^2 (Barcaza et al., 2017). Peak water is thought to have passed, or to be reached in the next 10–20 years for most glaciers in the SWS (and SSA) region, and summer runoff is expected to significantly decline in glaciated catchments during this century (*medium confidence*) (Huss and Hock, 2018; Schoolmeester et al., 2018). In terms of hazards in relation with the changing cryosphere the SWS region faces similar processes and challenges as the NWS region, including glacier lake outburst floods (GLOF), ice and rock avalanches, debris flows, and lahars from ice-capped volcanoes (Iribarren Anacona et al., 2015; Jacquet et al., 2017; Reinthaler et al., 2019).

Table 12.1: [PLACEHOLDER FOR SECOND ORDER DRAFT: Glacier area and mass changes across the Andes, 1960s to present and under future scenarios]

Table 12.2: [PLACEHOLDER FOR SECOND ORDER DRAFT: Snow cover changes, observed and projected, for the southern Andes]

The glaciers of the Southern Andes (including the SWS and SSA regions) show the highest glacier mass loss rates worldwide with more than 40 m water equivalent (w.e.) over the period 1961–2016 (-1.18 ±0.38 m w.e. yr^{-1}), contributing to about 3 mm sea level rise of this period (Dussaillant et al., 2018; Zemp et al., 2019). For the observational period of the 21st century, the contribution of Andean glaciers to sea level rise is reported as about 0.05 to 0.08 mm yr^{-1} (Jacob et al., 2012; Gardner et al., 2013; Braun et al., 2019).

Regional sea-level change for Chile predicted until the end of the 21st century for RCP4.5 and RCP8.5 scenarios, show the total mean sea-level rise along the coast will lie between 34 cm and 52 cm for the RCP4.5 scenario and between 46 cm and 74 cm for the RCP8.5 scenario (Albrecht and Shaffer, 2016).

Floods and landslides in the SWS region are typically strongly correlated with ENSO activity and events. Warm ENSO phases are linked with more landslides (Moreiras, 2015; Fustos et al., 2017). In areas influenced by the cryosphere flood and landslide activity is additionally influenced by changing snow patterns, glacier retreat and permafrost degradation. For instance, in the Aconcagua region landslide activity is more related to snow melt than rainfall (Moreiras et al., 2012). For the central Andes trends towards earlier snowmelt and earlier high river discharges in the year are projected (Reyer et al., 2017). Degrading permafrost generally decreases slope stability and may be a contributing factor to high mountain landslides (Krautblatter and Leith, 2015). While there exists information on permafrost landforms (rock glaciers) in the dry subtropical and central Andes (Brenning, 2005; Rangecroft et al., 2014); very little is known about the effects of warming and degrading permafrost on slope instability and landslides in these regions (Iribarren Anacona et al., 2015).Extreme floods in the SWS region which strongly exceed peak annual river discharge have repeatedly occurred in relation with glacier lake outburst floods. Flood events with peak discharge of up to several thousand cubic meters per second and a reach of several tens of kilometres have been recorded at several locations (Worni et al., 2012; Iribarren Anacona et al., 2015; Jacquet et al., 2017). Limited information is available concerning future flood hazards and risks.

Global-scale studies indicate decreasing flood risks in the 21st century for the SWS region (*low to medium confidence*) (Arnell and Gosling, 2013; Hirabayashi et al., 2013; Alfieri et al., 2017). In Andean regions affected by de-glaciation and cryosphere change slope stability is generally expected to decrease but magnitude and timing of this process are poorly defined (Stoffel and Huggel, 2012; Haeberli et al., 2017).

Climate drivers strongly influence global wildfire activity as related to the increasing length and frequency of fire weather seasons (Jolly et al., 2015). In Chile since 1946 to 2017, the number of fires and the areas burned have increased significantly (*high confidence*) (González et al., 2011; Úbeda and Sarricolea, 2016; de la Barrera et al., 2018; Urrutia-Jalabert et al., 2018). From 1976 to 2013, the annual average area affected by wildfires was approximately 7,000 ha with more than 5,100 fire events (Urrutia-Jalabert et al., 2018). In Chile, wildfire activity and its increasing trend have been closely related to changes in temperatures (González et al., 2011; de la Barrera et al., 2018; Gómez-González et al., 2018), land uses (Gómez-González et al., 2018; McWethy et al., 2018), changes in precipitation patterns (Gómez-González et al., 2018; Urrutia-Jalabert et al., 2018), topography (Gómez-González et al., 2018), and human density (McWethy et al., 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Trends and projections in ocean warming, ocean acidification and ocean deoxygenation]

12.3.7.2 Exposure, vulnerability and impacts

The southern Andes of Peru and northern Bolivia have been suffering for the last decade cold waves with increasingly lower temperatures. There are indications that observed migration trends inside Peru and Bolivia has impacted negatively local and indigenous climate knowledge together with climate change, and market integration (Valdivia et al., 2010). Migration is co-driven by such climatic variability which exacerbates already vulnerable and extremely poor rural population and indigenous communities (mainly Quechua and Aymara speakers) who recurrently lose their livelihoods (both crops and animals), and get affected by pneumonia and other related illnesses.

The most recent extreme rainfall caused by El Niño recorded in Chile during 2015 and 2017 caused more than 30 casualties and 16,000 people affected (Barrett et al., 2016). During the 2016–2017, coastal El Niño, Peru suffered the most severe disasters since the 1997–1998 El Niño event. Heavy rainfalls and severe flooding led to catastrophic damages to houses and infrastructure, with more than 600,000 people affected, 100 deaths and 6000 dengue cases (Rodríguez-Morata et al., 2018). Central Chile is most strongly affected by floods where more than 270,000 people were affected and 26 people were killed by floods in 2000, 2002 and 2006 (Fustos et al., 2017).

Heat-related mortality in Santiago de Chile from 1998 to 2002 strongly affected female population >65 years old, mainly associated to cardiovascular problems (Bell et al., 2008). Mega-fires that occurred in Chile during the austral summer December 2016–February 2017 caused at least 11 fatalities, affecting thousands of people, and destroying some towns. As a reference, the mortality rate in Chile attributed to air pollution is 64 per 100,000 inhabitants.

During 2017 in Chile, wildfires in the central-southern Chile experienced 91 wildfires (mostly mega fires, >40,000 ha) that burned more than 500,000 ha in total, consuming mainly forest plantations, native shrub lands, and native forests (de la Barrera et al., 2018). These events resemble the mega-fires that occurred in Chile during the last austral summer December 2016 to February 2017 that caused at least 11 fatalities, affecting thousands of people, and destroying some towns.

The dependence of people and economy on glacier melt water varies within the SWS region. In Chile and Argentina together, the respective area is about 1000 hectares of land irrigated with glacier melt water. Similarly, for domestic use, in Chile about a quarter million people depend on glacier melt water and in Argentina about 100,000 people (Buytaert et al., 2017; Schoolmeester et al., 2018). While the absolute numbers of percent glacier melt contribution, corresponding irrigated area and people depending on it, are uncertain, the picture that emerges is relatively robust.

The built environment (infrastructure, housing and urban facilities), are also quite vulnerable not being prepared for such temperatures and events, such as hailstones, snow storms and icy temperatures below –20°C unable to protect their inhabitants. In urban settings such as Santiago high levels of spatial asymmetry to heat stress (measured as heat vulnerability index) are present and the poorest urban areas highly affected (Inostroza et al., 2016; Romero, 2019).

Mountain communities are increasingly vulnerable to lose their sources of water for irrigation and human consumption, due to snow melt and glacier retreat (Moreiras et al., 2012), being Bolivia´s communities the most vulnerable ones (Buytaert et al., 2017; Schoolmeester et al., 2018), loss of houses and infrastructure due to landslides, glacier lake outburst floods, ice and rock avalanches, debris flows, and lahars from ice-capped volcanoes is a widespread threat in the region (Iribarren Anacona et al., 2015; Jacquet et al., 2017; Reinthaler et al., 2019).

This region has a long coastal line with several and populated coastal cities and settlements vulnerable to drought, sea level rise, floods (Fustos et al., 2017), heat waves and epidemics. Aspects such as the level of urban poverty, quality of housing, infrastructure, services and proximity to the coastline define the level of vulnerability. Cities of the southern Peruvian coast and the northern Chilean coast are among the most exposed and vulnerable to flooding and drought events associated to ENSO, both El Niño and La Niña (Vargas et al., 2015; Barrett et al., 2016; Rodríguez-Morata et al., 2019) whose populations are already suffering water stress. Elder persons and children are the most vulnerable urban population to heat waves. Santiago, for example, shows high levels of spatial asymmetry to heat stress (measured as heat vulnerability index) highly affecting the poorest urban areas (Romero, 2019).

Fisher towns, artisanal ports, fisheries and aquaculture mostly from Peru and Chile are vulnerable to ocean warming and acidification, sea level rise and upwelling intensification, also affecting their food security (Manríquez et al., 2013; Vargas et al., 2015; Aguirre-Velarde et al., 2016; Ramajo et al., 2016).

Increasing temperatures have important and significant impacts on marine species subject to fisheries and aquaculture. ENSO and warming are related with harmful algal blooms (HABs) in the SWS region which have increased in frequency, intensify and geographic extent (Díaz et al., 2019). In Chile, 39 HABs have been recorded between 1972 and 2018 (Cuellar-Martinez et al., 2018) provoking high mortalities of marine organisms such as fish and molluscs (Díaz et al., 2019). Additionally, HABs events have generated important impacts on human health: in Chile, since 1970 to 2010 more than 1,000 human intoxications and 31 deaths were registered. In 2016, HAB events in Chiloé Island (Chile) affected salmon farms with important economic losses (500 million dollars) caused by the 39,000 tonnes of fish death (Clément et al., 2007) and 1700 people unemployed (Díaz et al., 2019).

Ocean warming and acidification impact the physiology, reproduction, development and survivorship of multiple species of Peru and Chile such as mussels, scallops, gastropods and seaweeds (Manríquez et al., 2013; Vargas et al., 2015; Aguirre-Velarde et al., 2016; Ramajo et al., 2016). These biological impacts are exacerbated by other stressors such as toxins (i.e., algal blooms) (Mellado et al., 2019), changes in salinity (Duarte et al., 2018), decreasing oxygen levels, and increasing temperatures on mussels and scallops. Additionally, higher biological impacts are expected as variations in upwelling intensity occur on fisheries (*high confidence*) and aquaculture (*medium confidence*) (Sydeman et al., 2014; Rykaczewski et al., 2015; Ramajo et al., 2019).

In Chile and Peru, fisheries and aquaculture are one of the most important economic and socially productive activities. In Chile, from 1994 to 2014, there has been a clear reduction, from 8 million to 3.8 million tones in landings reflecting a deterioration of the stocks with important socio-economic problems associated (lower incomes and reduced food security) (Yáñez et al., 2017).

Observed and projected climate change impacts are highly diverse along South America. High latitude regions, such as Chile are projected to increase their fish catch potential (Cheung et al., 2018), while tropical South Pacific countries such as Ecuador are projected to lose more than 50% of the current maximum catch potential (Cheung et al., 2018). Also, climate change will have dissimilar impacts at long and short-term and at different spatial scales on the aquaculture and fisheries in CA and SA as a consequence of the dissimilar sensibilities, exposure, vulnerability and adaptive capacity to climate change (Ding et al., 2017). All this will have significant consequences over the entire value chain (i.e., prices, trade and consumption) and the competitiveness of the region which is highly dependent on fisheries and aquaculture products (Ding et al., 2017; Soto et al., 2018).

Harmful impacts of phytoplankton blooms in SWS region have been associated with decreasing levels of oxygen, high levels of nutrients, increasing temperatures by the El Niño events (Garreaud, 2018; León-Muñoz et al., 2018).

Coastal 'El Niño' have severe impacts on the infrastructure damaged, crops and people. This extreme event also has important impacts on marine resources such as anchovies with a consequent decrease of 1.3% in the Peruvian gross domestic product (Takahashi et al., 2018).

Extreme storm events in the Pacific Ocean related with high sea wave heights (up to 6.4 m) have caused in several regions of Chile a significant loss of sediment, infrastructure damages, local transit impacts (Campos-Caba, 2016; Martínez et al., 2018), and higher economical costs (more than 1000 million Chilean pesos) (Campos-Caba, 2016).

There exists reports on a declination of food security, this situation is critical for some areas of Peru (Porkka et al., 2013). Chile's agricultural sector remains responsible for a substantial share of the country's overall GHG emissions, no sector-specific target or plan has been set that could help in achieving emission goals (OECD, 2018).

12.3.8 Southern South America (SSA) Subregion

12.3.8.1 Hazards

Mean temperatures in the SSA subregion are projected to continue to rise up to +2.5°C in 2080 with respect to the present climatology (Kreps et al., 2012). Extreme maximum annual temperature (TXx) will increase +2.3°C for 1.5°C scenario and +3°C for 2°C scenario (Hoegh-Guldberg et al., 2018). Changes projected by regional multi-model ensemble (MME) RCM-ENS (from 2079 to 2098 minus from 1991 to 2010) in summer suggest cooling on minimum temperatures over the region (López-Franca et al., 2016).

Analysing centennial precipitation series, (Saurral et al., 2017) observed an increase in precipitation in Trelew but no change for Comodoro Rivadavia, both stations located at Eastern Patagonia, while negative trends in austral summer rainfall in southern Andes were observed (Vera and Díaz, 2015). Changes in rainfall intensities do not show the same patterns of change spatially all over the subregion. Extreme 5 days precipitation (Rx5day) will increase 40–50 mm for 1.5 and 2°C scenarios (Hoegh-Guldberg et al., 2018).

It is expected that droughts will intensify through the 21st century in SSA. Kitoh et al. (2011) suggested a decrease of up to 1 mm/day in the southern Andes and Southern South America (SSA). The probability of having extended droughts, such as the recently experienced mega-drought (2010–2015), increases to up to 5 events/100 years (Bozkurt et al., 2017).

Wildfire spread is controlled by complex system-specific interactions of fuel quantity and quality modified by environmental heterogeneity, climate and the occurrence of both natural and anthropogenic ignitions. Chile's wildfires in Patagonia (fire frequency and intensity) have grown at an alarming rate (Úbeda and Sarricolea, 2016).

A rise in temperature means that the isotherm of 0ºC will move up the mountains leaving less surface for accumulation of snow (Barros et al., 2015). Rasmussen et al. (2007) reported that a warming in an altitude of 850 hPa at 50°S, the latitude of the icefields, of about 0.5°C both winter and summer, would have decreased the fraction of precipitation falling as snow. The calculated 850-hPa precipitation flux at that latitude lacks noticeable trend over 1960–1999, indicating that the decline of snowfall has been due to warming rather than drying. This is consistent with Malmros et al. (2018) that indicated an increase in air temperature causes 0ºC isotherm to ascend to higher elevations resulting in a larger proportion of precipitation falling as rain as opposed to snow. In the first assessment on snow cover over the Aysén basin in Patagonia-Chile by using MODIS data from the period 2000–2016, Pérez et al. (2018) reported a mean annual decrease in snow cover of −20.01 $km^2 yr^{-1}$.

Figure 12.8: [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 ATLAS 5.5. Projections SSA CMIP5/CMIP6]

12.3.8.2 Exposure, vulnerability and impacts

Grasslands make a significant contribution to food security in Patagonia through providing part of the feed requirements of ruminants used for meat, wool and milk production. There is a lack of information regarding the combined effect of climate change and overgrazing and the consequences for pastoral livelihoods that depend on rangelands. Gaitán et al. (2014) found that temperature and the amount and seasonal distribution of precipitation were important controls of vegetation structure in Patagonian rangelands. They found that over two-thirds of the total effect of precipitation on ANPP was direct, and the other third was indirect (via the effects of precipitation on vegetation structure). Similarly, Crego et al. (2014) reported that suitable areas for meadows (very productive areas for livestock production) will decrease by 7.85% by 2050 given predicted changes in climate. Thus, if evapotranspiration and drought stress increase as temperature increases and rainfall decrease in water-limited ecosystems, it would be expected a greater exposure of ranchers due to a reduction of stocking rate and therefore families' income. The number of farmers (mainly family enterprises) exposed to climatic hazards (drought) is approximately 70–80 thousand that have 14–15 million sheep in Argentina.

A major drought from 1998 to 1999 coincident with a very hot summer led to extensive dieback in a Nothofagus species (Suarez et al., 2004). In another dominant Nothofagus species, several periodic droughts have triggered forest decline as of the 1940s (Rodríguez-Catón et al., 2016). These exposures enterprise related to logging (sawmills and timber production) and tourist activities. According to Gonzales-Reyes et al., 2017 decreasing rainfall pattern in Punta Arenas is closely associated with the SAM variability at inter-annual to inter-decadal time scales.

There are reports related to a decrease in survival, growth and vulnerability to drought and fire-severity species of native forest due to climate change and wildfire (Mundo et al., 2010; Landesmann et al., 2015; Whitlock et al., 2015; Jump et al., 2017; Camarero et al., 2018; Venegas-González et al., 2018). Rodríguez-Catón et al. (2016) reported a coincidence between major changes in regional decline growth of forests with severe droughts due to climatic variations over northern Patagonia. Once the forest decline begins, other contributing factors such as insects (e.g., defoliator outbreaks) and diseases increase the forest vulnerability or accelerate the loss of forest health of previously stressed trees (Piper et al., 2015). This region hosts unique temperate rainforests and it is particularly rich in endemic and long-lived conifer species (e.g., *Fitzroya cupressoides*), which may be vulnerable to declines in soil moisture availability (Camarero and Fajardo, 2017). Also, Crego et al. (2014) reported that meadows in Patagonia will likely be vulnerable by a decrease in precipitation regimes due to climate change, probably, and consequently many species that rely on meadows in an arid environment will also be impacted.

Fire has been found to promote or halt biological invasions. For example, an analysis of Pinus spreading after wildfires in Patagonia reveals that there is a high risk of pines becoming invasive if ignition frequency increases as a result of climate change (Raffaele et al., 2016).
According to Inostroza et al. (2016), the Magellan Region is one of the most fragile regions in Patagonia and despite its low population densities, it is under a silent process of anthropogenic alteration where between 53.1% and 68.1% of the area needs to be considered as influenced by human activity whom are occupying pristine ecosystems even extensive conservation designations (Inostroza, et al. 2016).

Patagonia main cities have developed thanks to oil and gas extraction which demand massive quantities of water due to fracking and drilling techniques. Vaca Muerta is the major region in South America where those techniques are used to extract oil and gas, and this will lead to an exacerbation of current water scarcity and to competition with irrigated agriculture (Rosa, et al., 2018) which in the context of drought may exacerbate socio environmental conflicts.

The development of various human activities and water infrastructure are decreasing water sources, changing river basins from exoreic to endoreic and the disappearance of one lake in 2016 (Scordo et al., 2017). Numerous dams for irrigation, some also used for hydropower have been and are planned to be built despite wind power generation potential (Silva, 2016). Oil and gas have played an important role in the rise of

Neuquén-Cipolleti as Patagonia's most populous urban area, and in the growth of Comodoro Rivadavia, Punta Arenas, and Rio Grande, as well.

Patagonian icefields in South America are the largest bodies of ice outside of Antarctica in the southern hemisphere. They are losing volume due partly to rapid changes in their outlet glaciers which end up in lakes or the oceans, becoming the largest contributors to eustatic sea level rise (SLR) in the world, per unit area (Foresta, et al. 2018)(Moragues et al., 2019). All 28 major calving glaciers in the Southern Patagonia icefield retreat during the last century. Upsala Glacier retreat generated slope instability and a landslide movement destroyed the western edge in 2013. The Upsala Argentina Lake has become potentially unstable and may generate new landslides (Moragues et al., 2019).

The potential impact of climate change is of special concern in arid and semi-arid Patagonia, a >700 000 km^2 region of steppe-like plains in Argentina. Thus, melting snow and ice in the glaciers of Patagonia and the Andes will alter surface runoff into interior wetlands, sea level rise of between 20 and 60 cm will destroy coastal marshes, and an increase in extreme events, such as storms, floods, and droughts, will affect biodiversity in wet grasslands (after Junk et al. (2013); Joyce et al. (2016)). Kubisch et al. (2016) reported the extinction risk of three species of lizard from Patagonia as a result of global warming.

Meier et al. (2018), covering the area between 41 and 56ºS, reported an absolute glacierized area loss of 5,455 ± 1,269 km^2 (19.4%) in the last ~150 years, where the annual area reduction increased by 0.25% a-1 for the periods 2005–2016 (in small glaciers in the north of the Northern Patagonian Icefield had over all periods the highest rates of 0.92% a-1).

12.4 Key Impacts and Risks

This section provides a synthesis on key risks across the Central and South America region; it follows the definition and concept of risk provided in AR5, distinguishing the risk components climatic hazard, exposure and vulnerability of people and assets. This concept is refined in AR6 and described in more detail in chapter 16, according to which key risks are potentially severe risks. Key risks may refer to present conditions or to the future, with a focus on the 21st century. Both mitigation and adaptation can moderate the extent or severity of risks. The identification and evaluation of risks implies values which may vary across individuals, communities or cultures.

In line with chapter 16 of this report a risk outcome perspective is adopted here, i.e., the consequences related to risks are in the focus which potentially can be a result of different combinations of hazards, exposure and vulnerabilities. There is limited literature with a focus on severe risks in the Central and South America region, and limited specific and explicit consideration of risk drivers such as level of warming, level of exposure, vulnerability and adaptation. In addition to evidence from the literature the assessment here therefore relies on experts' understanding of biophysical and social systems.

Criteria for identification of key risks for this chapter include the magnitude of the consequences, in particular the number of people potentially affected; the severity of the negative effects of the risk (e.g., life threatening, major negative effect on livelihoods, well-being, or economy; the importance of the affected system (e.g., for vital ecosystem services, for large population groups), and the irreversibility of either the process leading to the risk or of the consequences; and the potential to reduce the risk.

Several of the key risks identified for the Central and South America region align well with the overarching key risks assessed in AR5 (Oppenheimer et al., 2014) and later in (O'Neill et al., 2017). The identified key risks include 1) risk of death, injury or severe ill-health due to extreme weather events such as storms, floods, landslides; 2) risk of food insecurity due to repeated and/or extreme drought conditions; 3) risk of severe health effects due to increasing epidemics (in particular vector-borne diseases); 4) risk of water insecurity due to glacier shrinkage and snow reduction in the Andean region and failed water management in other regions; 5) systemic risks due to cascading impacts of extreme weather events and/or epidemics and public service system failures; 6) risk of loss of terrestrial ecosystems and biodiversity, in particular large-scale ecological transformation of the Amazon forest; 7) risk of loss of coastal infrastructure and marine ecosystems, in particular coral reef bleaching, and risks related to intensification of ocean upwelling and ocean acidification (Table 12.4).

The assessment of key observed and projected impacts and risks shows that in the Central and South America region several systems are already approaching critical conditions under current warming levels and are likely to cross these thresholds, partly in an irreversible way, for different degrees of future warming (glaciers in the Andes, coral reefs in Central America, loss of the Amazon forests). Most key risks and their severity and extent are strongly driven and determined by exposure, vulnerability and adaptation. In particular, the high vulnerability of large populations groups, infrastructure and service systems, high inequality and poor governance are a critical factor in increasing and becoming key risks.

[PLACEHOLDER FOR SECOND ORDER DRAFT: assessment of key risks against different levels of warming and different SSP's and adaptation measures]

Table 12.3: [PLACEHOLDER FOR SECOND ORDER DRAFT: Glacier changes, associated impacts and adaptation measures across the Andean region]

Table 12.4: Key risks and impacts identified and assessed for the Central and South America region.

Nature of key risk and impact	Geographic region	Consequences considered severe	Driving hazard conditions	Driving exposure conditions	Driving Vulnerability conditions	Top adaptation options	Confidence in key risk identification	References
Food and nutrition insecurity due to frequent / extreme droughts	Central and South America, in particular CA, NES, SWS, SSA	More frequent/longer dry/hot periods, increasing desertification, together with high vulnerabilities, disruption of food provision chains/inability to acquire food for large parts of the population, reduced capacity or production of goods (food, fiber, fuel, such as for northeast Brazil)	More frequent and/or longer drought and very hot periods. High variability in the yearly rainfall patterns, particularly a severe decrease in rainfall at the onset of the rainy season. Decrease amount of rainfall overall	More people exposed to food and nutrition insecurity due to spatially more extensive drought, high population growth rate (including in rural areas) and more population dependent on goods (failure on food sovereignty)	Reduced capacity of farmers, especially small farmers living in poverty, to adapt to changing climatic conditions. Soil degradation. Limited institutional and governance related capacities, limited water storage capacity and irrigation systems	Improved water governance (storage, irrigation), improved water eco-efficiency (recycling and reuse), reducing grazing pressure on vulnerable lands, long-term planning on biomass production, institutional support to small farmers (funding, information), reducing inequalities and power relations in food production systems, and more efficient food provision chains	*Medium confidence (medium evidence, high agreement)*	(Rosenzweig et al., 2014; Greve and Seneviratne, 2015; Marengo and Bernasconi, 2015; Vieira et al., 2015; Arnell et al., 2016; Kummu et al., 2016; Kuzdas et al., 2016; Veldkamp et al., 2017; Tomasella et al., 2018)

Risk to life and infrastructure due to floods and landslides	Central and South America	more frequent and stronger storms/hurricanes and heavy precipitation events, with risk to death and severe health effects, disruption of critical infrastructure and basic service provision systems	more frequent and stronger storms/hurricanes and heavy precipitation events. For some regions possibly higher rainfall variabilities combined with higher extreme rainfall	More people exposed to floods and landslides due to changing hazards, land-use and increased population. More people in poverty living in high-risk areas on steep slopes in urban areas or flood plains in urban and rural areas	vulnerable populations are usually low income. Low resilience in infrastructure and critical service systems after an extreme event.	Generally high potential for adaptation, with future risks strongly depending on adaptation and development pathways. Integrated disaster risk management, increasing resilience of infrastructure and systems. Relocation of people in high-risk areas. Better land-use planning and urban development planning. Early warning systems.	*Medium evidence, medium agreement*	(Hirabayashi et al., 2013; Kundzewicz et al., 2014; Jongman et al., 2015; Miranda Sara and Jameson, 2016; Alfieri et al., 2017; Rodríguez-Morata et al., 2018; Aristizábal and Sánchez, 2019)

Water insecurity	Andes, Pacific Coast and Central America	Seasonal water scarcity due to glacier shrinkage and snow cover change, more pronounced dry periods, failed water management, unequal water distribution and governance	glacier/snow depletion, prolonged dry periods/precip/circulation changes	increase in population and water use/demand dependent on glacier/snow melt.	decreasing water management and governance capacities, economic sectors' dependence on melt or rain water contribution, low water infrastructure efficiency and investment, vulnerable and growing urban areas with poor sewage system,, unequal distribution	improved and integrated water governance with higher levels of investment,, increase efficiency of water supply services, diversification of water sources, improved eco-efficiency of water use	*Medium to high confidence*	(Drenkhan et al., 2015; Barcaza et al., 2017; Buytaert et al., 2017; Miranda Sara et al., 2017; Braun et al., 2019; Drenkhan et al., 2019)
Cascading impacts and risks of storms, floods, epidemics surpassing infrastructure and public service system capacities	Central and South America	Break-down of public service systems, including infrastructure and health services affecting large population groups	Higher frequency and magnitude of climatic shocks/stresses (storms, floods)	More people and infrastructure exposed to climate/weather events	Increasing vulnerability of public service and infrastructure systems	Increase of systems' resilience, based on identification of thresholds in the system and where impacts cascades can be broken	*Medium confidence (medium evidence, medium agreement)*	(Brondízio et al., 2016; French and Mechler, 2017)

Increase in epidemics, particularly vector-borne diseases	Central and South America	HIgher epidemics of diseases like Malaria, Dengue fever and the such.	Higher temperatures increase the geographical range of vectors	Density of population increased by increased urbanization, resulting in higher transmission rates	Lower income housing with no screens. Low sanitation conditions, particularly in low income neighborhood.	Vector control measures like fumigation. Screens in houses. Better public health services	*Medium to high confidence (medium evidence, high agreement)*	(French and Mechler, 2017)
Risk of large-scale ecological transformation of Amazon forest	South America	More frequent, stronger and persistent dry conditions. Land-use and deforestation practices.	More frequent, stronger and persistent dry conditions.	Reduced availability of natural sources (food, fiber), for local people.	Strong dependence on non-climatic drivers (land-use change, deforestation)	Policies for reduced rates of deforestation, and reforestation; landscape planning and increase of conservation areas	*Low confidence* (large scale ecosystem changes) *Low to medium* (small scale ecosystem changes)	(Nobre et al., 2016b; Lapola et al., 2018)
Coral reef bleaching	Central and South America (in particular NES),	Degradation and possible death of coral reef in northern Central America, the second largest reef in the world, and in Northeastern Brazil, with the largest coral reef in the Southern Atlantic Ocean	Higher temperatures, higher carbon dioxide leading to ocean acidification		Ecosystem highly sensitive to water temperature and pH	Limit global warming and carbon dioxide emissions	*High confidence (high evidence, high agreement)*	

Sea level rise, intensification of upwelling and ocean acidification	Central and South America	More than 0.70 m of sea level rise under 1.5° C, loss of fisheries	More intense and persistent coastal flooding, salt water intrusion, coastal erosion and subsidence	More people, infrastructure (ports) and services (coastal tourism) exposed, increase need of relocation of millions of people	water scarcity, water stress, loss of fresh water from underground sources, lower fish food availability	Coastal protection, planned settlements relocation, coastal integrated management strategies	*Medium to high confidence*	(Villamizar et al., 2017; IPCC, 2018a; IPCC, 2019b)

Climate change is critically important for the current and future status of mining activity and its impacts on surrounding communities and environments (Odell et al., 2018). While mining operations are vital to provide the materials needed for the modern way of life, they are often located in areas with limited water supply, a limitation exacerbated by climate change (Northey et al., 2017). Mining industries operate on a large variety of geographical conditions originating diverse levels of exposure and vulnerability to climate change. According to Loechel et al. (2013) effective climate change adaptation in mining communities is reliant on action by both local authorities and mining operations.

There appears to have been little research investigating climate adaptation in these sectors. Literature linking climate change and mining is scanty, with some studies carried out mainly in other parts of the world and only a few in our region in countries with a long mining tradition, Chile (Odell et al., 2018) and Peru (Corzo and Gamboa, 2018).

The energy and water intensity of large-scale mining may make this sector particularly vulnerable to climate change; additionally, the relative lack of power of resource-poor communities living in areas where such mining is making claims on water and energy resources renders these communities even more vulnerable (Odell et al., 2018). Given new conditions generated by changes in a growing demand and climate change, mining industries will need to increase resilience to extreme events; additionally, the declining concentrations of mineral of interest in the raw material require greater energy input for extraction and processing and new methods to avoid associated emissions are required (Hodgkinson and Smith, 2018).

12.5 Adaptation

A revision of the most recently submitted National Communications, Nationally Determined Contributions and National Adaptation Plans (https://unfccc.int) shows that public policies and non-governmental initiatives for climate change adaptation in the region have increased since AR5, both at national and sub national levels (*high confidence*). This section assesses adaptation goals, trends, barriers and initiatives in the region, through a sectoral approach.

12.5.1 Terrestrial and freshwater ecosystems and their services

Climate change adaptation strategies for natural systems must be conceived within a socioecosystem framework, which recognizes the important and complex interactions between the biological, ecological, cultural, ethical, economic and political dimensions of vulnerability (Ostrom, 2009; Joaqui Daza and Figueroa Casas, 2014; Colding and Barthel, 2019). Central and South America conform a culturally and biologically diverse region with unique socioecosystems that are and will be strongly impacted by climate change in the coming decades, with significant consequences for the people that depend on their integrity (*high confidence*) (CAF, 2014; Camacho Guerreiro et al., 2016; Li et al., 2018). Warming has generated extreme heat events in many parts of South America, with increasing frequency, intensity and duration (IPCC, 2019a). Droughts and floods will seriously affect the integrity of terrestrial and freshwater ecosystems in the entire region (CAF, 2014). Water availability for irrigation is decreasing due to population growth and increasing urbanization, along with multiple stressors on socioecosystems resulting from land use changes induced by climate change and variability (Jat et al., 2016). Human migration, between as well as within countries, constitute a climate change impact on human livelihoods as well as a stressor for socioecosystems in the region, frequently resulting in increased vulnerability (Chisari and Miller, 2016).

A widely discussed strategy for increasing ecosystems' resilience to climate change is to reduce, at the local scale, other global change stressors, such as land cover change, pollution, overexploitation, and invasive species. This strategy works on the premise that eliminating current stressors will increase ecosystem's ability to resist or recover from future disturbances. Common management approaches include, therefore, protecting relatively pristine areas and restoring degraded ones. A review of the most recently submitted National Communications, Nationally Determined Contributions and National Adaptation Plans (https://unfccc.int) in South and Central America reveals that all countries in the region have formulated policies to cope with climate change that include either strategies or measures relevant for socioecosystem adaptation. They include policies that are relevant to terrestrial and freshwater ecosystems, applying the protection and restoration approach, with an emphasis on forests and water basins (*high confidence*).

There are also policies for terrestrial and freshwater ecosystems conservation based on protected areas and land management. There is a wide variety of ecosystem based adaptation (EbA) measures proposed or implemented in the region, including ecosystem restoration, communitarian management of natural resources, natural reserve conservation, forestation and reforestation, ecological corridors, seed conservation and germplasms banks, adaptive territorial planning, protection of local and indigenous agricultural systems, genetic diversity protection, integrated management of hydric resources and payment for ecosystem services (PES). Despite the fact that all countries in the region are designing or implementing some kind of adaptation plan, strategy, guideline or priorities, only three have formally submitted National Adaptation Plans to the United Nations Framework Convention on Climate Change (Brazil, Colombia and Chile).

Implementation of regional adaptation in terrestrial and freshwater ecosystems has included conservation, restoration and management measures and actions. Although conservation and restoration has been reported as effective to reduce ecosystems vulnerabilities, especially in protected areas (*low evidence, high agreement*) (Anderson et al., 2010; Borsdorf et al., 2013; Keenan, 2015; Pires et al., 2017), their effectiveness depends on the integration of conservation actions with enhancement of local socioeconomic conditions (*medium evidence, high agreement*) (Scarano and Ceotto, 2015; Pires et al., 2017; Kasecker et al., 2018). Since AR5, there has been an increase in measures and actions reported for adaptation through natural resources and ecosystem services management. The main approaches are EbA and community based adaptation (CBA) (*high confidence*). In both cases, adaptation relevant to terrestrial and freshwater socioecosystems basically consists in addressing the provision of ecosystem services for water, agriculture and forestry (*high confidence*), especially at forests, plantations, grasslands, mountains, savannahs, and wetlands (Doswald et al., 2014).

Integrity and adaptive capacity of socioecosystems is highly affected by human activities like farming. Common EbA practices like dispersed trees, home gardens and live fences are implemented by smallholder basic grain farmers in Guatemala and Honduras based on their experience, knowledge and perception of social and biophysical changes (Chain-Guadarrama et al., 2018). Harvey et al. (2017) found a mean of 3.8 EbA practices per farm in a survey of 300 smallholder coffee and maize farmers in six landscapes in Central America, a number that could be improved by facilitating knowledge exchange among farmers.

Research, monitoring systems and other initiatives for knowledge production and information storage, systematization and dissemination are promoted in most countries in the region in the context of terrestrial and freshwater socioecosystem adaptation, from dedicated programs to the identification of priority research topics (*high confidence*) (https://unfccc.int). In Chile, for example, the Eco-social Observatory of Climate Change Effects for High Altitude Wetlands of Tarapacá has been collecting information on physical, biological and social variables, integrating knowledge from natural and social sciences since 2013 (Uribe Rivera et al., 2017). Research programs for adaptation tend to be focused on assessing impacts and vulnerability, but better impact assessments do not necessary mean better decision making. Emerging research approaches integrating traditional forest and ecosystem sciences with social, economic and behavioral disciplines may greatly improve decision making (*high confidence*) (Keenan, 2015). Recognition of local management needs and integration of indigenous knowledge systems, promoting a shared understanding among governmental institutions, communities and other stakeholders. can greatly improve design and implementation of adaptation strategies (*high confidence)* (Keenan, 2015). Cuesta et al. (2019) advocates for reinforcing interdisciplinary research agendas with active participation of government agencies and social organizations. They should have a cross-sectorial focus and include knowledge sharing and capacity building for all relevant stakeholders (Cuesta et al., 2019). More interdisciplinary and transdisciplinary research is needed in the region in order to better understand and successfully manage the relationship between governance, implementation, knowledge, wealth distribution and trade-offs between adaptation, mitigation and the implementation of the Sustainable Development Goals (SDG) (*high confidence*) (Keenan, 2015; Hurlbert and Gupta, 2016; Roco et al., 2016; Kasecker et al., 2018).

Some innovative approaches that integrate different dimensions of vulnerability and adaptation potential are worth mentioning. Kasecker et al. (2018) identified "EbA hotspots" in Brazil on the premises that poverty is a driver of climate change vulnerability while conservation and sustainable use of ecosystems foster adaptation to climate change. Using an indicator with three parameters: poverty, proportion of natural-vegetation cover, and climate hazard, they identified 398 EbA hotspots among the 5.565 Brazilian municipalities. Most of them were in the Amazon, Cerrado and Caatinga which are recognized as some of the most vulnerable ecosystems to climate change in the world (Kasecker et al., 2018).

Effective adaptation to climate change requires the integration of scientific knowledge and local and indigenous knowledge; the latter can be very detailed and usually relates to actual people priorities. In Manaus, fishers perceived increase in fish mortality, low capture level, reductions in fish size and disappearance of some species as consequences of droughts. Floods, on the other hand, increase difficulties to access fishing grounds (Keenan, 2015; Camacho Guerreiro et al., 2016). There are important opportunities to avoid maladaptation or improve adaptation in terrestrial and freshwater socioecosystems by the integration of EbA and CBA approaches with scientific and local and indigenous knowledge, together with the engagement of communities, policy makers and other stakeholders from early stages of vulnerability assessments (*medium evidence, high agreement*) (Valdivia et al., 2010; Borsdorf et al., 2013; Rangecroft et al., 2013; Doswald et al., 2014; Metternicht et al., 2014; Keenan, 2015; Lasage et al., 2015; Camacho Guerreiro et al., 2016; Hurlbert and Gupta, 2016; Roco et al., 2016; Santos et al., 2016; Walshe and Argumedo, 2016; Uribe Rivera et al., 2017). There is a significant gap on identifying limits to adaptation in the region, despite its critical role for the design and implementation of adaptation strategies and estimation of potential risks of climate change driven loss and damage (Dow et al., 2013).

Most countries identified inadequate financing and access to technology as barriers for adaptation relevant to terrestrial and freshwater socioecosystems (*high confidence*), and insufficient institutional coordination is also frequently mentioned (Rangecroft et al., 2013; Cameron et al., 2015). These limitations could be at least partially addressed through technical support, multilateral cooperation and local empowerment (Harvey et al., 2017; Calispa, 2018; Chain-Guadarrama et al., 2018). CBA tends to empower people, increasing their adaptive capacity and resilience to climate change and water scarcity (Rangecroft et al., 2013).

Lack of coordination between policy and implementation has also been recognized as a problem. According to Calispa (2018) Ecuadorian climate change public policy points towards a CBA approach to adaptation, but in the implementation phase it is often downscaled. Analyzing national reports from Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, El Salvador, Paraguay, Peru and Uruguay, Ryan (2012) found that important steps have been taken in recent years toward adaptation, both in climate change policymaking and in institutional arrangements, although climate policies were often weakly integrated with other sectorial policies. The analysis revealed a widespread recognition of the importance of citizen participation in adaptation planning and implementation. Some type of social participation mechanisms is present in most countries, but their characteristics and levels of implementation vary considerably (Ryan, 2012). Other examples of ineffective or incomplete implementation of policies or legal frameworks for adaptation in terrestrial and freshwater ecosystems include the Brazilian Native Vegetation Protection Law (NVPL) whose proper implementation in Rio Doce watershed, known for centuries of land degradation and the protagonist of in major environmental disasters in Brazilian history, could reduce deforestation and soil erosion and improve water quality, in addition to having important climate change mitigation benefits (Pires et al., 2017).

Territorial biased has been reported in adaptation priorities for research and implementation. According to Chazarin et al. (2014) most scientific research on adaptation in Peru focuses on the highlands and coastal regions while mitigation research is focuses on the forest. This hinders the ability to design and implement comprehensive adaptation programs. Combined adaptation and mitigation strategies can produce positive results, but tend to be disconnected (Locatelli et al., 2015). Most reviewed cases in agriculture and forestry in Latin America (84% of 274 cases) reported positive synergies between adaptation and mitigation outcomes. Nevertheless, research on Latin American forests mainly focus on mitigation, while studies on agriculture tend to be oriented towards adaptation (Locatelli et al., 2017).

With one of the highest rates of biodiversity and ecosystem services loss in the world, Latin America also shows a substantial asymmetry in the access to their benefits (Laterra et al., 2019). To overcome inequality in the burdens of vulnerability and the benefits of adaptation, along the lines of gender, age, socioeconomic conditions and ethnicity, is a big challenge for the region. In the design of adaptation strategies it is important to consider asymmetries in needs, vulnerabilities and capacities, and consider power relations and priority differences among stakeholders (*high confidence*) (Tengö et al., 2014). Asymmetries need to be acknowledged and addressed in order to avoid unintentionally generating or increasing social disruption and inequality. Rural communities in the Cusco Region in Peru ground their ability to successfully respond and adapt to climate change on four cultural values, known in Quechua as *ayni* (reciprocity), *ayllu* (collectiveness), *yanantin* (equilibrium) and *chanincha* (solidarity) (Walshe and Argumedo, 2016). Adaptation strategies could benefit from integrating these and other insights from traditional cultures.

Despite the relative progress in adaptation in the region, many gaps remain. Socioecosystems from the Puna ecoregion are considered some of the most vulnerable to climate change, but there is no clear understanding about the ways in which climate change will affect them at a regional scale (Uribe Rivera et al., 2017). According to Hurlbert and Gupta (2016) in Chile and Argentina successful policy response to climate change could improve by using adaptive management that can test hypothesis to build adaptive capacity, engaging with people to develop adaptation solutions and employing adaptive co-management to negotiate cost issues. Adequate information transfer, a proactive role of the State, and appropriate mechanisms for cost estimation are identified as beneficial to increase adaptive capacity (Santos et al., 2016). There is a pressing and increasing need for implementation of successful (effective, just and equitable) climate change adaptation strategies in Latin America and other developing regions, and it is critical and indispensable to find the right balance between assessment and action (Metternicht et al., 2014).

12.5.2 Ocean and coastal ecosystems and their services

12.5.2.1 Past and current situation

Population in coastal zones of CA and SA continue increasing, and it is expected to rise to 38 million people by 2060 (Neumann et al., 2015). Coastal zones show high levels of poverty, exposition to multiple climate and non-climate hazards, and high vulnerability to be vastly dependent on the natural resources and touristic activities (Reguero et al., 2015). AR5 pointed that coastal hazards would impact the 30% of the total population living in CA and SA region, multiple ecosystems (i.e., sandy beaches, coral reefs, mangroves), and coastal infrastructure (i.e., ports) putting on risk a significant number of ecosystem services such as food provision (fish stocks), recreation and tourism (Magrin et al., 2014). To AR5, adaptation strategies in the ocean and coastal ecosystems at CA and SA were mainly based on the conservation of key ecosystems by creating marine protected areas (MPAs) with significant fails during their implementation (i.e., no considering connectivity among populations), while adaptation measures to respond to sea level rise (SLR) were better implemented and mainly based on the protection of some ecosystems and the promotion of investments in new infrastructures. According to AR5, adaptation to climate change in CA and SA region was importantly constrained by the low adaptive capacity, the high vulnerability (inequality), the lack of a political agenda and the financial and technological resources (Magrin et al., 2014).

Since AR5, only a few governments (i.e., Chile, Guyana and Nicaragua) have specific focus on ocean and coastal ecosystems and their services, but adaptation planning is still deficient. To date, adaptation strategies in CA and SA are mainly focused on the implementation of measures to cope with SLR in urban areas and touristic settlements (Calil et al., 2017; Villamizar et al., 2017). Currently, poverty, weak governance, top-down strategies, low flexibility, lack of awareness of climate risks, economic hardships, lack of stakeholder's engagement, the depletion of ecosystems and natural resources are important barriers and limits that CA and SA countries met to implement adaptation plans to climate change on ocean and coastal areas (Leal Filho, 2018; Nagy et al., 2019). However, recent progresses (increasing adaptation efforts on coastal management strategies, ecosystems protection and warning systems) in addition with potential positive changes may be expected in the near future (Leal Filho, 2018; Nagy et al., 2019).

12.5.2.2 Ocean and coastal ecosystem services

In CA and SA, kelp forests, seagrass beds, mangroves, coral reefs, beaches and coastal dunes formed-habitats have an essential role in the coastal dynamic by reducing the movement of water created by currents and waves, storms and hurricanes, preventing the beach erosion, sequestering carbon, increasing the concentration of dissolved oxygen, and rising pH values (Inniss and Simcock, 2017). Additionally, they provide a suitable habitat to a high biodiversity of fish and invertebrates supporting many important regional fisheries, tourism, and other economic activities (CEPAL, 2018) [PLACEHOLDER FOR SECOND ORDER DRAFT: AR6 WGII Chapter 3].

Figure 12.9: [PLACEHOLDER FOR SECOND ORDER DRAFT: Regional map showing the type and area of the different marine and coastal ecosystems in CA and SA]

12.5.2.2.1 Fisheries, aquaculture and food security

Fisheries and aquaculture contribute significantly to food security and livelihoods in CA and SA by creating employment, food, income and economic growth (FAO, 2018). With the exception of Honduras, more than 45% of the total fisheries in CA and SA are based on marine products (fish) (CEPALSTAT, 2019). Four SA countries are among the 15 counties with the major marine capture production worldwide (Peru, Chile, Argentina and Ecuador) (FAO, 2018), highlighting Chile and Peru as the most productive worldwide in terms of fish caught (Gutiérrez et al., 2016a; Vannuccini et al., 2018). Aquaculture is another important activity in CA and SA. More than 90% of resources produced by aquaculture activities are of marine origin in Ecuador, Panama, Chile and Nicaragua. On the other hand, aquaculture is much less developed in countries such as Uruguay, Argentina and Colombia, Brazil, Costa Rica and El Salvador (CEPALSTAT, 2019). During 2016, marine aquaculture in Chile produced 1.3% of the world total of food fish, while the rest of the countries of Latin America (including Mexico) contributed with the 2.1%. To date, Chile highlights as the third major producer of marine and coastal finfishes and fourth of molluscs worldwide, while Brazil is among the 15 countries with major production of marine crustaceans (FAO, 2018).

Fisheries in Brazil, French Guyana, Suriname, Guyana and Venezuela is mainly export-oriented shrimp fishery being critical for Guyana economy (6% GDP) (Oxenford and Monnereau, 2018). Ecuador, Peru and Chile are top exporter countries of fish and fish products (over 2 million tonnes in 2016) (Vannuccini et al., 2018).

In CA and SA countries, fisheries and aquaculture employ more than two million people (FAO, 2018).

The contribution by women to fishery economies is overlooked, largely due to “fishing” is mainly defined as catching fish at sea from a vessel by using specialized gears (Harper et al., 2017). In CA and SA, direct fishery activities (catch) are controlled by men, while the women participation is mainly associated to indirect fisheries activities (i.e., processing).

To date there is an important lack of data and information reflecting the real role and magnitude of women participation in fishery and aquaculture activities in CA and SA (FAO, 2016b; Bruguere and Williams, 2017), however in some SA countries such as Peru, women participation (considering direct and indirect fishery activities) reaches the 33% (Harper et al., 2017), while in Chile the number of women fishers have shown a trend to increase significantly (7% to 23% from 2004 to 2014) and today the artisanal fishers are the almost the 25% of the total (Gallardo-Fernández and Saunders, 2018).

Table 12.5: [PLACEHOLDER FOR SECOND ORDER DRAFT: Update information (from 2015 to 2019) for fishery and Aquaculture GDP by CA and SA country to add in Table 12.5 or figure suggested and a summary paragraph here; Update information for fishery and Aquaculture employment (average from 2015 to 2019) by CA and SA country to add in Table 12.5 and a summary paragraph here; Fishery and Aquaculture women employment (average from 2015 to 2019) by CA and SA country to add to Table 12.5]

Brazil, French Guyana, Suriname, Guyana and Venezuela are significantly dependent on fish protein (Oxenford and Monnereau, 2018). However, although CA and SA countries are among the major net exporter of fish and a large producer of aquaculture, they have the lowest per capita consumption worldwide (10.9 Kg yr^{-1} in 2013) (FAO, 2017). This is predicted to change by 2030 as the total fish consumption is expected to increase in Latin America more than 33% (FAO, 2018).

12.5.2.2.2 Coastal protection and tourism

Coastal ecosystems provide important ecosystem services such as protection to storms, sea level rise and erosion by reducing the wave energy, increasing sedimentation and reducing erosion. Healthy and functioning coastal ecosystems are important in order to reduce the human vulnerability, but also to support adaptation (and mitigation) in coastal areas to climate change impacts (Spalding et al., 2014; Muñoz Sevilla and Le Bail, 2017).

Habitat destruction, biodiversity loss, rising invasive species, increasing SLR, extreme events, coastal erosion, flooding, and salinization of groundwater in CA and SA will put coastal areas and their services, and the infrastructure under significant risk with important consequences at several levels (Villamizar et al., 2017). Recent studies warn that coastal areas of Ecuador, Peru, Nicaragua, El Salvador, Guatemala, Guyana, Suriname and Honduras show the most populated areas and the highest risks of the region to more frequent and high-intensity storms, higher sea-levels, and more severe floods (Calil et al., 2017; Villamizar et al., 2017; Nagy et al., 2019). Models predict that under higher population growth rates, more than 9 million people living in low elevation coastal zones (LECZ) of CA and SA will be affected by SLR and coastal flooding (Neumann et al., 2015). Without the implementation of adaptation strategies and adequate conservation measures more than 4 million people living in CA and SA coastal areas of low elevation will be exposed to more intense and frequent events such as flooding as a consequence of SLR, storms and extreme El Niño events (Reguero et al., 2015).

Coastal tourism, one of the most important sectors in the global economy offering a variety of natural and cultural activities (Carvache-Franco et al., 2019) will be impacted by climate change with significant implications in major sectors such as the economy, transportation, fishing or agriculture. In addition, substantial effects are expected on the tourist preferences, tourism operator strategies and transportation market (Weatherdon et al., 2016; Bindoff et al., 2019). Tourism in Ecuador is the third economic sector after bananas and shrimp with more than 1.5 million visitors during 2017. The coastal tourism (i.e., Santa Elena province) is a touristic spot for recreational activities related mainly with water sports such as surfing, ecotourism, and beach (Carvache-Franco et al., 2019). In Chile, coastline (over 4000 km in length from north to south) with beaches and resorts of various features, provide a significant space for tourism and recreation primarily because of the suitable beaches for relaxing, swimming and water sports (González and Holtmann-Ahumada, 2017). In Argentina, recreational fishing (shore- and boat-based), netting, and spearfishing of bony fishes, sharks, rays and chimaeras are very popular pastime activities alongshore with an important economic relevance (Venerus and Cedrola, 2017).

[PLACEHOLDER FOR SECOND ORDER DRAFT: More information about the economic importance of tourism (GDP), employment, etc. (information to be added to figure)]

12.5.2.3 Adaptation, policy and management framework, governance and financing

Current adaptation initiatives to climate change impacts for ocean and coastal ecosystems have increased since AR5, however yet only a few CA and SA governments have specific adaptation efforts on ocean and coastal areas. Many of these adaptation initiatives are located in the recent Framework Laws in Climate Change (i.e., Guatemala, Peru), National Adaptation Plans (NAPs), Sectoral Adaptation Plans, and in the recent Nationally Determined Contributions (NDCs). Countries such as Argentina, Belize, Costa Rica, El Salvador, Panama, Peru, Suriname, Venezuela show the lowest number of adaptation strategies or actions in their national or international compromises on ocean and coastal ecosystems, while Brazil, Chile, Colombia, Ecuador, Guyana, Nicaragua have an intermediate level. At the moment, Guatemala shows the highest level of commitments in relation to ocean and coastal ecosystems and their services which have been incorporated in the different Framework Laws of Climate Change, national (NAP) and sectoral adaptation plans, and iNDCs.

To date, the management of marine and coastal resources in CA and SA countries are mainly based on partial restrictions, temporal and spatial closures, quotas, total allowable catch, closed areas and seasons, limited access, fishing reserves, marine reserves, non-take zones, MPAs, and nursery closures (Singh-Renton and McIvor, 2015; Bertrand et al., 2018; Lluch-Cota et al., 2018). Fishery governance in CA and SA is mainly established from the Central Governments with some important inputs from non-governmental conservationist entities in the case of Ecuador and Costa Rica (Lluch-Cota et al., 2018), or the scientific advisory of governmental and research institutions in the case of Peru, Chile and Venezuela (Singh-Renton and McIvor, 2015; Bertrand et al., 2018).

In Brazil, the management of the major fisheries do not follow an ecosystem approach, although for some small-scale fisheries applies the precautionary approach (Singh-Renton and McIvor, 2015). In Peru, the industrial fishery governance follows an adaptive management approach and include several measures (i.e., stock assessments, catch limits). On the contrary, the management and control of artisanal medium and small-scale fisheries in Peru is minimal with an important lack of regulations, control, and management actions (Bertrand et al., 2018). In Chile, the small-scale fishery of benthic-demersal resources is managed through the granting of exclusive territorial use rights (called TURFS) with established quotas defined by the central authority (Bertrand et al., 2018). In Argentina, marine recreational fisheries have been largely unregulated with non-official fisheries monitoring programs at the national level encouraging to recreational fishers to develop small-scale commercial fishing operations, non-controlled and contributing to the overexploitation of some key coastal stocks (Venerus and Cedrola, 2017).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Complement with more information about management strategies (specifically related to MPAs) in CA and SA]

Additionally, other instances have been implemented in order to control illegal fishing activities such as the 'Comprehensive policy for the development of sustainable fisheries' or the National Plan for Sustainable Aquaculture (PlaNDAS) in Colombia (OECD, 2017). In Chile, MPAs play a key role in adaptation to climate change (Gelcich et al., 2015), and an increasing funding has been inverted in initiatives to reduce the vulnerability of fisheries and aquaculture to climate change (OECD, 2017). Since 2016, Argentina is developing a strategy to implement and ecosystem-based fisheries management with support from the Global Environment Facility program (GEF). Additionally, Argentina has strengthened the regulation and control system on fishery sectors by implementing a system of on-board cameras and better regulations. South America countries, such as Argentina and Chile, are promoting the local consumption of seafood and the certification of its fishery products (OECD, 2017). Brazil and Chile have developed actions on climate change adaptation in the coastal zone thought of new studies, methodologies, told and research institutions, while Uruguay is incorporating stakeholders in their adaptation strategies (Nagy et al., 2015).

In CA and SA, the participation of women fishers is not equally considered, being excluded during the decision-making processes. This is mainly due to the management and policy decisions have been traditionally focused on direct, formal, and paid fishing activities (Harper et al., 2017). However, some examples in SA (Chile) show how when laws and policies contribute to incorporate women and gender equality and increase the woman visibility, not only increase the women empowerment as improved fishers, workers but also enhance the management of the marine resources (Frangoudes and Gerrard, 2018; Gallardo-Fernández and Saunders, 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: More information about gender]

In CA and SA, coastal protection has relied heavily on hard and costly measures such as the construction of seawalls and bulkheads which sometimes fails by exacerbating problems of coastal erosion or damage in adjacent areas (Spalding et al., 2014; Lins-de-Barros and Parente-Ribeiro, 2018).

Coastal ecosystems such as beaches show low conservation values mainly due to the loss of dunes and the construction of infrastructures focused on tourism and residential building which generate an interruption of the natural dynamic of beaches and a consequent low protective index to tides, waves, extreme events or tsunamis (González and Holtmann-Ahumada, 2017). Some of the strategies implemented to conserve and protect the biodiversity and dune ridges have failed due to the lack of guidelines for monitoring and management (González and Holtmann-Ahumada, 2017).

The adaptive capacity of tourism to extreme and abrupt environmental conditions is relatively highly, however still remains unknown how this industry will respond to long-term changes driven by climate change (Weatherdon et al., 2016). [PLACEHOLDER FOR SECOND ORDER DRAFT: More information about tourism]

12.5.2.4 Priorities, improving the knowledge base, policy needs and recommendations

In CA and SA, adaptation strategies must be developed in order to eradicate poverty and the increment of food security in fishing and aquaculture communities by building resilience in a multi-dimensional and multi-sectoral context in agreement with the different international (Paris Agreement, SDGs, NDCs) and national instruments (NAPs). Adaptation strategies should create opportunities for fishers, fish-farmers and fish-workers in order to increase their flexibility and sustainability (target species, emerging fisheries), empower the local stakeholders (multilateral fisheries agreements), improving public awareness and simplifying regulations (Weatherdon et al., 2016; Kalikoski et al., 2019).

Ecosystem Based Fishery Management (EBFM) arises a suitable tool to be implemented that should incorporate the complexity and the relationships among species within ecological systems (Long et al., 2015) with the aim of avoiding the degradation of the ecosystems and its services, minimizing the risk of irreversible changes and maintaining the long-term socioeconomic benefits (Gullestad et al., 2017).Climate change coastal adaptation in CA and SA should necessarily consider the level of vulnerability to disasters as a consequence of the rapid urbanization, environmental degradation, or the loss of biodiversity. Coastal adaptation should not only based on the construction of coastal defences, but on the contrary should incorporate the protection, management and restoration of environmental and natural resources, as well as, ecosystems such as beaches, dunes, wetlands and coral reefs on coastal planning strategies through an ecosystem-based management (EBA) (Spalding et al., 2014; González and Holtmann-Ahumada, 2017; Scarano, 2017).

Finally, adaptation measures for ocean and coastal ecosystems should be accompanied by climate data, early warning systems, improved local institutions, the construction of adequate infrastructure, major funding for capacity building, and an enhanced engagement of women (Leal Filho, 2018).

Incorporate Women need to be incorporated in the public policies thought improving the current information databases, visibility, encouraging the participation of new women generations, promoting the participation of women in trade organizations, increasing the access to education and technical training and access to productive assets (FAO, 2016b; Harper et al., 2017; Gallardo-Fernández and Saunders, 2018).

Table 12.6: [PLACEHOLDER FOR SECOND ORDER DRAFT: Current Framework Laws on Climate Change that address climate adaptation in relation to ocean and coastal ecosystems and their services]

[PLACEHOLDER FOR SECOND ORDER DRAFT: Specific adaptation contributions and strategies for ocean and coastal ecosystems found in the National Determined Contributions in CA and SA (NDC) comparing iNDCs (2015-2020) and NDCs (2020-2025)]

[PLACEHOLDER FOR SECOND ORDER DRAFT: objectives, strategies and actions found in CA and SA National and Sectoral Adaptation Plans for climate change in relation to ocean and coastal ecosystems and their services]

Table 12.7: [PLACEHOLDER FOR SECOND ORDER DRAFT**:** Relative importance of marine fisheries and marine aquaculture (in %) in relation to the total fisheries and aquaculture developed in CA and SA region in terms of capture (fishery), production (aquaculture), GDP, employment and women participation (considering direct and indirect activities) (capture data are means (%) from 2014 - 2017, from CEPAL). Include updated information from 2015 - 2020]

Figure 12.10: [PLACEHOLDER FOR SECOND ORDER DRAFT: Indicators of ocean and coastal services for CA and SA]

12.5.3 Water

In CA and SA, as in many other regions of the world, water plays a fundamental role for ecosystems, societies and economies. The water cycle is profoundly affected by climate change and increasing water use and allocation. While impacts related to water vary widely across locations, the region is recognized as one of the most affected by current and particularly future severe hydrological changes with an increasing number of lowland populations depending on streamflow from mountain areas (*high confidence*) (Heinke et al., 2019; Viviroli et al., 2019).

In most countries of the region, adaptation to changing water resources is therefore a priority in climate change policies which benefited from the definition and incipient implementation of Nationally Determined Contributions (NDCs) through a stronger focus on e.g., improving water efficiency, water quality and governance at multisectoral level in the UNFCCC framework (UNEP, 2015). Since IPCC AR5 substantially more adaptation experiences in the field of water have accumulated but only a small fraction thereof is documented in the scientific literature (*high confidence*). As for most aspects of adaptation more information is available in the grey literature (e.g., government documents, project reports with variable standards of quality and evidence). The following sections assess the state of knowledge in different fields of water related adaptation to changing hydrology.

12.5.3.1 Water supply and demand

In some parts of CA and SA, (seasonal) water scarcity is already a serious challenge to local livelihoods (*high confidence*) (Mekonnen and Hoekstra, 2016). In the context of strong impacts from climate change, changing water allocation systems and increasing water use, a growing number of people potentially affected by water scarcity can be observed (Kummu et al., 2016) and this number is projected to increase in the region (*high confidence*) (Veldkamp et al., 2017). However, most of the available and cited studies have a global or continental scale, and model results may bear considerable uncertainties at local level. Enhanced water scarcity levels can result not only from reduced water quantity but also from poor water quality. A subsequent decline of water quality can be observed in different glacier-fed river systems in the NWS and SWS regions, as sulphides and heavy metals are leached out from recently exposed sediments due to glacier retreat (*medium to high confidence*) (Milner et al., 2017; Vuille et al., 2018). These processes represent important adaptation challenges for human water provision systems and occur across multiple sectors of water use, including agriculture (Mekonnen et al., 2015), domestic use (Buytaert and De Bièvre, 2012), energy (Van Vliet et al., 2016; Zhang et al., 2018), health (Aparicio-Effens et al., 2016; Lizarralde Oliver and Ribeiro, 2016), tourism (Milner et al., 2017), housing (Milner et al., 2017), employment (Desbureaux and Rodella, 2019) and others (*high confidence*).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Data (in table or figure) on population growth and increasing human water use/consumption, and agricultural development]

Many regions in CA and SA highly rely on hydroelectric energy and due to considerable rising energy demand, hydropower capacity is constantly extended (IHA, 2018; Schoolmeester et al., 2018). Worldwide, SA features the second-fastest growth with about 4.1 GW additional annual capacity installed (2017–2018). Most of that growth is concentrated in Brazil (3.4 GW) which has by far the highest capacity installed with around 100.3 GW in the entire Americas. More than 20 GW are currently installed in the NWS region (including Colombia with 11.7 GW), about 7.3 GW can be found in the SWS (Chile), >21 GW in the SES (including Argentina with 11.2 GW) and around 15.4 GW in the NSA regions (Venezuela), and also CA has a high dependency on hydropower (up to 90%). Although much smaller in total capacity (7.1 GW) the CA region also has a high dependency on hydropower (up to 90% for countries like Costa Rica) with investment plans for further extensions in (IHA, 2018).

While current and planned future hydropower capacity extensions support carbon emission reductions, several large projects in the region have important social implications, lack local acceptance and are, thus, highly controversial, e.g., in the NWS region (Carey et al., 2012; Duarte-Abadía et al., 2015; Hommes and Boelens, 2018). Additionally, hydropower is increasingly affected by declines or shortage of water supply, including in the NWS and SWS where glaciers continue to shrink and alter river runoff regimes (*high confidence*) (Carey et al., 2014; Drenkhan et al., 2015; Buytaert et al., 2017; Drenkhan et al., 2019) and in CA where increasing evaporation rates and drought risks are expected to occur (*medium confidence*) (CEPAL, 2017). Depending on the social, economic and political context, such environmental changes can entail conflict potentials with competing water sectors and high economic losses (*medium to high confidence*) (Vergara et al., 2007; Vuille et al., 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of adaptation with respect to water quality and groundwater, and interconnections between water quality, water scarcity, floods and social justice issues]

12.5.3.2 Water security, management and governance

In CA and SA, about 65% of the population have access to safely managed drinking water while another 31% only dispose of basic water services. Hence, around 222 million people have no connection to safe drinking water services and an additional 25 million do not have access to a basic water system in the CA and SA region (*medium to high confidence*) (WHO and UNICEF, 2017). Strong disparities prevail regionally and the most vulnerable (without access to water supply and sanitation service) belong to low-income groups and rural or informal settlements (WWAP, 2019). While in the SWS nearly the entire population has access to safely managed drinking water services in 2017 (Chile: 99%), this share strongly decreases for the NWS (Peru: 50%, Colombia: 73%) and even more towards the CA (Nicaragua: 52%, Guatemala: 56%) (*medium to high confidence*) (UNICEF and WHO, 2019). Additionally, many major cities (e.g., Lima, Peru) in the region have very strong disparities in water provision, access and consumption (varying by an order of magnitude or more per capita), typically reflecting differences in economic wealth, and social injustice more generally, within the cities (*high confidence*) (Miranda Sara and Jameson, 2016; WWAP, 2019).

Some studies conclude that flood risk is increasing more strongly in the region than water stress (Arnell et al., 2016). However, considerable uncertainties exist concerning future water related risks because both flood and water scarcity risks are strongly influenced by human intervention, management and adaptation as well as socio-economic development. Both climatic and non-climatic (human) drivers of water related risks have strong spatio-temporal effects on catchment scale. Flood risk is expected to increase primarily in the NWS, SAM, SES regions during the 21st century (*medium-high confidence*) (Arnell and Gosling, 2016; Alfieri et al., 2017).

In many regions of CA and SA, past water management addressing water scarcity has been dominated by extending the water supply system including large infrastructure projects. Reasons include low local water availability in dry regions, such as the Pacific coast (SWS and parts of NWS region) and growing cities, industries, and large-scale agriculture with a dependence on upstream mountain water sources. At present, these 'hard path' interventions are highly contested due to negative effects exacerbating local water conflicts (Carey et al., 2012; Drenkhan et al., 2019), potentially leading to increasing water demand, vulnerabilities and water shortage risks (Di Baldassarre et al., 2018), and, hence, limiting adaptive capacity (*high confidence*) (Ochoa-Tocachi et al., 2019). There is medium evidence of such upstream-downstream effects in the region, for instance, in relation with reservoirs, land-use practices or ecological restoration and conservation (*medium confidence*) (Veldkamp et al., 2017). In particular reservoir construction and management (for hydropower and/or agricultural use) is a highly contested issue in many parts of Central and South America, which has led to numerous social conflicts (*high confidence*) (Isch et al., 2012; Drenkhan et al., 2019). Integrated water scarcity and flood management via multi-purpose project planning seems promising as it allows for interlinking different types of water use potentially reducing conflict potentials among stakeholders and enhancing long-term project feasibility. However, only a few experiences exist in e.g., the NWS, most of them without long-term implementation (*medium confidence*) (Barriga et al., 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of water conflicts in the context of climate change adaptation]

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of existing flood management efforts and future response]

Many countries in the region promote a move towards more integrative and sustainable management of water resources through new legislations. For instance, new water laws including principles of Integrated Water Resources Management (IWRM) have entered into force, among other, in the CA, NWS and SWS regions (Nicaragua: 2007, Peru: 2009, Ecuador: 2014, Costa Rica: 2014) or are underway (Colombia: since 2009). However, two remaining countries in the CA region (Guatemala and El Salvador) have not been able to adopt new water laws yet, and current realities in all regions show that there are major challenges on this transformative pathway, related to aspects such as political and institutional instabilities, governance structures, fragmented service provision, lack of economies of scale and scope, corruption, or social conflicts (*high confidence*) (Castro, 2019; WWAP, 2019).

Several water management solutions with evidence or expectancy of success have been adopted over the past years. Examples include green infrastructure development, water sowing and harvesting (Bustamante et al., 2012; Ochoa-Tocachi et al., 2019), ecological restoration and ecosystem based adaptation (Harvey et al., 2017; Reid et al., 2018). In addition to improve water security and clean drinking water for downstream population centres watershed conservation can generate co-benefits for flood mitigation (Tellman et al., 2018). Furthermore, demand-oriented approaches focus on incentives for the reduction of water use through changes in people's habits, efficiency increase and smart water management (Gleick, 2002) which are promoted in some regions, such as the NWS (e.g., Ecuador, Peru) through 'water culture' policies. Recent years have also seen a stronger consideration of knowledge from indigenous and farmer communities for adaptation baseline and planning processes, in particular in regions with high percentage of indigenous population (such as the NWS and SWS regions). In some cases, such practices have facilitated additional water storage capacities (*high confidence*) (Bustamante et al., 2012; Ochoa-Tocachi et al., 2019) but in general the effectiveness is not yet well assessed. However, there is evidence for the NWS (Peru) and NSA regions (Guyana) that important co-benefits are generated for sustainable development (*medium to high confidence*) (Bremer et al., 2016; Spencer et al., 2017).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of successful water management and adaptation solutions and synergies/trade-offs with sustainable development]

Figure 12.11: [PLACEHOLDER FOR SECOND ORDER DRAFT: interwoven drivers and measures of adaptation to water resource changes]

Table 12.8: Synthesis of observed and projected glacier changes, and associated impacts and adaptation efforts across the Andes. [PLACEHOLDER FOR SECOND ORDER DRAFT: void cells in table to be completed].

Country Total glacier area Total glacier area trend	Latitude Longitude (in DD) (glacier type)	Mountain range Glacier *Area* *Volume* *Max. altitude*	Climate trends	Glacier change	Impact	Adaptation	Legal frameworks
VE 0.1 km^2[1] (*NA*)	8.6°N 71.0°W (IT)	Sierra Nevada de Mérida Pico Humboldt 0.1 km^2[1] *4942 m asl.*		**OBSERVED:** A: entire mountain range: -97% (1952–2011), Pico Humboldt: -85% (1985–2011)[1] **PROJECTED:** **A/V:** the last remaining glacier in Venezuela under risk of extinction within the next years[1]			No glacier law. Initial framework for Climate Change Law (Proyecto de Ley del Cambio Climático) was set in 2018 (?).
CO 37 km^2 (2018) -56% (1986–2017)[2]	4.8°N 75.4°W (IT)	Cordillera Central Nevado Santa Isabel 0.6 km^2 [2] *4965 m asl.*		**OBSERVED:** **A:** -36% (1990s–2016)[3] MB: -2.5 m w.e. yr^{-1} (1987–2005)[3] -0.4 to -1.0 m w.e. yr^{-1} (2000–2018)[4] **PROJECTED:** **A/V:** under risk of extinction within the next years[3]			Framework Law no. 1931 on Climate Change underway (regulation by 2021)
	4.9°N 75.3°W (IT)	Cordillera Central Nevado del Ruiz 8 km^2 *5321 m asl.*		**OBSERVED:** **A:** -51% (1986–2017)[2] **PROJECTED:** **A/V:** will likely remain after 2050[3]			

EC 44 km^2 (2018) -54% (1980s– 2017)[5]	0.5°S 78.1°W (IT)	Cordillera Real Antisana 16 km^2[6] ~0.6 km^3[6] *5760 m asl.*		**OBSERVED:** **A:** -33% (1979–2007)[7] **MB:** -0.2 to -0.4 m w.e. yr^{-1} (2000–2018)[4] **V:** **PROJECTED:** **A:** -71% for RCP2.6 and -80% for RCP8.5 (2016–2050); -72% for RCP2.6 and -98% for RCP8.5 (2016–2100)[6] **MB:** **V:** -70% for RCP2.6 and -78% for RCP8.5 (2016–2050); -71% for RCP2.6 and -97% for RCP8.5 (2016–2100)[6]	**OBSERVED:** **PROJECTED:** **GD:** would be completely lost under RCP8.5[6,8] **LF:** 10 new glacier lakes (before 2050)[6] **EC:** potential for new ecosystem formations		No Glacier or Climate Change laws
		Cordillera Real Cotopaxi km^2 *m asl.*		**OBSERVED:** **A:** -52% (1976–2016) **MB:** **V:** **PROJECTED:** **A:**			
PE 1114 km^2 (2018) -54% (1962– 2016)[9]	9.2°S 77.5°W (OT)	Cordillera Blanca *449 km^2* [9] *6768 m asl.*	**T:** +0.31°C/decade (1969–1998)[10] **P:** +60 mm/decade (1983–2012)[10]	**OBSERVED:** **A:** -38% (1962–2016)[9] **MB:** 0 to -2.0 m w.e. yr^{-1} (2000–2018)[4] **V:** **PROJECTED:** **A:** -58% for RCP2.6 and -99% for RCP8.5 (2003–2100)[11] **MB:** **V:**	**OBSERVED:** **RC:** 'peak water' passed in 7 deglaciating catchments[12] **WQ:** acid rock drainages and decreasing pH **LF:** 16 new glacier lakes (2003–2018)[13] **EC:** **SC:** **PROJECTED:** **GD:** only highest summits (over ~5800 m asl.) would remain under RCP8.5[8]	**IMPLEMENTED** **GI:** green infrastructure with reactivation of ancient infiltration systems for water harvesting in the Central Andes[14] **MP:** GLOF early warning system at lake Palcacocha [not fully implemented] (Palcaraju glacier, Quilcay)[15] **MS:** GLOF early warning system at Lake 513 (Hualcán glacier,	Framework Law on Climate Change (2018)[16] explicitly including glaciers

					LF: 19 new glacier lakes: 3 (imminent), 6 (before 2050), 10 (after 2050)[13]	Carhuaz); lake monitoring at Palcacocha (Huaraz) **FI:** public investment projects	
	13.5°S 71.1°W (OT)	Cordillera Vilcanota *255 km² [9]* *4.65 km³* *(catchment scale)* *6384 m asl.*	**OBSERVED:** **T:** +0.1 to 0.4°C/decade (1965–2009)[17] **P:** slightly decreasing, not significant[17] **PROJECTED** **P:** -19% to -33% (DJF, wet season) until 2100[18]	**OBSERVED:** **A:** -48% (1962–2016)[9] **MB:** **V:** -18% (1988–2016)[19] *(catchment scale)* **PROJECTED:** **A:** -59% for RCP2.6 and –97% for RCP8.5 (2009-2100)[11] **MB:** **V:** -19% for RCP2.6 and -20% for RCP8.5 (2016–2050); -24% for RCP2.6 and -91% for RCP8.5 (2016–2100)[19] *(catchment scale)*	**OBSERVED:** **RC:** **WQ:** **LF:** 14 new glacier lakes (2009–2018)[13], partial drying-out of non-glacial lakes (1985–2012)[20] **EC:** drying of several cushion bogs, three amphibian species have colonized deglaciated areas[21] **SC:** using stones instead of ice blocks at pilgrim procession Qollur Rit'i[22] **PROJECTED:** **GD:** only highest summits (over 6000 m asl.) would remain under RCP8.5[19] **LF:** 27 new glacier lakes: 7 (imminent), 10 (before 2050), 10 (after 2050)[13]	**IMPLEMENTED** **GRI:** traditional micro-reservoirs (Pitumarca)[23] **MP:** multi-purpose project design (flood retention and water availability for agriculture)[24,25] **MS:** GLOF early warning system at lake Riticocha (Chicón glacier, Urubamba)[24] **PLANNED** **GRI:** river valley damming for hydropower supply (Pitumarca); (re)construction of qochas (microreservoirs) for local water supply[25] **MS:** debris flow early warning system at Santa Teresa (Sacsara, Vilcabamba)[26] **PO:**	
BO 346 km² (2014) -~35% (1980s–2013)[27]	16.2°S 68.2°W (OT)	Cordillera Real Zongo 2 km²[28] 8.6 km³ [28] *6000 m asl.*		**OBSERVED:** **A:** **MB:** -0.4 to -0.6 m w.e. yr^{-1} (2000–2018)[4] **V:** -12% (1997–2010)[28] **PROJECTED:** **A:** **MB:**	**OBSERVED:** **PROJECTED:** **GD:** only highest summits (over ~5800 m asl.) would remain under RCP8.5[8]		No glacier law. Initial draft on Climate Change law Framework Law of Mother Earth and Holistic Development for

				V: -40% for RCP2.6 and -89% for RCP8.5 (2016–2100)[28] **PF:** -95% (2050) and -99% (2099) for SRES A1B compared to present-day (1950–2000)			Living Well (2012)
CH 23,641 km^2	33.5°S 70.1°W (ST)	Cordillera Central Echaurren Norte 0.2 km^2[29] *3880 m asl.*		**OBSERVED:** **A:** **MB:** -0.35 (1977–2012)[29] to -0.59 m w.e. yr^{-1} (1955–2015)[30] **V:** **PROJECTED** **A:** **MB:** **V:**			National Inventory of Glaciers (2012), currently being updated No Glacier Law underway
		Monte Fitz Roy xx km^2 *3375 m asl.*					
AR 8484 km^2 [31]	41.1°S 71.8°W (ET)	Patagonia Norte Monte Trovador Mistral 1.3 km^2[32] *~2300 m asl.*		**OBSERVED:** **A:** -10% (2000–2012)[32] **MB:** -0.54 m w.e. yr^{-1} (2000–2012)[32] **V:** -0.015 km^3 (2000–2012)[32] **PROJECTED:** **A:** **MB:** **V:**			National Inventory of Glaciers (2018)[31] Glacier Protection Law No. 26.639 (2010-2019)

Notes:
Country: VE = Venezuela, CO = Colombia, EC = Ecuador, PE = Peru, BO = Bolivia, CH = Chile, AR = Argentina
Total glacier area: according to publication of last National Glacier Inventory.
Glacier type: IT = inner tropical (Tropical Andes), OT = outer tropical (Tropical Andes), ST = subtropical (Dry Andes), ET = extratropical (Patagonian Andes)
Climate trends: T = temperature trend, P = precipitation trend
Glacier change: A = area trend, MB = mass balance trend, V = volume trend, PF = permafrost trend
Impact: GD = glacier development, RC = river runoff change, WQ = water quality impact, LF = glacier lake formation, EC = ecosystem impact, SC = socio-cultural impact
Adaptation: GRI = grey infrastructure, GI = green infrastructure, MP = management and planning, MS = monitoring system, FI = financing, PO = policy
[1] Braun and Bezada (2013); [2] IDEAM (2018); [3] Rabatel et al. (2018); [4] Dussaillant et al. (2019); [5] Cáceres (2018); [6] Cuesta et al. (2019); [7] Rabatel et al. (2013); [8] Vuille et al. (2018); [9] INAIGEM (2018); [10] Schauwecker et al. (2014); [11] Schauwecker et al. (2017); [12] Baraer et al. (2012); [13] Guardamino et al. (2019); [14] Ochoa-Tocachi et al. (2019); [15] Frey et al. (2018); [16] MINAM (2018); [17] Salzmann et al. (2013); [18] Neukom et al. (2015); [19] Drenkhan et al. (2018); [20] Hanshaw and Bookhagen (2014); [21] Seimon et al. (2017); [22] Bolin (2009); [23] IMA (2016); [24] Barriga et al. (2018); [25] Drenkhan et al. (2019); [26] Frey et al. (2016); [27] Ramírez (2014); [28] Réveillet et al. (2015); [29] Masiokas et al. (2016); [30] Farías et al. (2017); [31] IANIGLA (2018); [32] Ruiz et al. (2017)

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of national adaptation policies related to water in national communications to the UNFCCC, NDC's]

12.5.3.3 Limits to adaptation

Limits to adaptation have been framed under dimensions of physical, ecological, technological, economic, and social and institutional limits, and been defined as the points where adaptation actions fail to protect things that are assigned a value (Adger et al., 2009). More recently, limits to adaptation have also been framed in the context of Loss and Damage (Mechler and Schinko, 2016), with a distinction of soft and hard limits (Leal Filho and Nalau, 2018). In CA and SA, only limited explicit information on limits of adaptation in relation to water is available, such as for instance on possible path dependency of institutions and associated resistance to change (Barnett et al., 2015). This information, however, would be strongly needed for specific national policy planning on Loss and Damage (Roberts and Pelling, 2018). Examples of soft adaptation limits (options to avoid intolerable risks currently not available) in the context of CA and SA include for example water scarcity levels reduced through improved water harvesting technologies using traditional knowledge as a complementary measure to limited grey infrastructure (Ochoa-Tocachi et al., 2019). An example for hard adaptation limits (intolerable risks cannot be avoided) in the region is loss of livelihoods and cultural values due to glacier shrinkage e.g., in the NWS (Peru) (Jurt et al., 2015).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Evidence of existing, emerging or projected water related limits to adaptation]

In many parts of CA and SA the level of success of respective adaptation measures depends much on the governance of projects and stakeholder-based processes and is closely related to their effectiveness, efficiency, social equity and political legitimacy (*medium to high confidence*) (Adger et al., 2005). Recent reviews have found that water related processes are among the most frequent climatic stimuli of adaptation in mountains, with the Andes mountain countries featuring high with the number of adaptation actions worldwide in mountains countries (McDowell et al., 2019) (*robust evidence*).

12.5.3.4 Research and policy needs and financing

Availability of data (environmental, social, economic, etc.) in the region is improving but still limited and a barrier to the assessment and to evidence based adaptation. In the water sector there is a need for improvement of available data on water supply and demand in terms of quality and spatio-temporal observations. Therefore, novel mechanisms, such as local citizen science based initiatives (Buytaert et al., 2014; Tellman et al., 2016) which engage the public e.g., via crowdsourced monitoring (Njue et al., 2019), can support the robust estimation and flexible data collection and, additionally, generate a broader acceptance of local communities in a context of distrust towards scientists and local conflicts.

There are indications that blended bottom-up and top-down adaptation approaches are most effective as their integration enables the possibility to include scientific knowledge with impact and scenario assessments into local adaptation agendas (Huggel et al., 2015b) using a risk-based approach with a range of future scenarios in order to develop robust adaptation measures and strategies (*medium confidence*) (Ludwig et al., 2014). National policies on climate change, water protection, regulation and management laws as well as clearly defined adaptation efforts (e.g., NDCs) are important focal areas of adaptation in the water sector. Notable in the jurisdiction field is a first worldwide Glacier Protection Law in Argentina (2010–2019), a Framework Law on Climate Change, promulgated in Peru (2018) and underway in Colombia and Venezuela. Additionally, important measures relating to benefit sharing mechanisms for ecosystem services and headwater conservation have been implemented in the last years (e.g., in Peru) (Ochoa-Tocachi et al., 2019).

In terms of financing water management and adaptation, only about 50–70% of required financial resources are currently annually allocated to meet the national targets in the water, sanitation and hygiene (WASH) sector for the Sustainable Development agenda 2020–2030 (SDG 6) in CA and SA, except for the NSA (Venezuela) and the SES regions (Argentina, Uruguay, Paraguay) which are below the 50% level. In the CA region, this share drops down to less than 50% only with Panama allocating more than 75% of required financial resources. These funding deficits pose important limitations for future water provision and adaptation to changing water resources. To close this WASH coverage gap towards the achievement of the

SDGs by 2030, additional financial and implementation efforts are needed (*medium to high confidence*) (WHO, 2017). Furthermore, financing mechanisms have been promoted, tested or implemented (in some countries based on new legislation, e.g., Peru) that produce incentives for sustainable water management such as redistribution of ecosystem service payments for water provision (cf. section 12.3.1).

Currently, at least 21 different water funds are being implemented across CA (4), the NWS (13), NES (1) and SES (3) engaging public, private and civil society stakeholders in order to contribute to improved governance and conservation of watershed systems. These financing mechanisms are important for increasing water security levels through nature-based solutions, however, only a few experiences have been evaluated as successful due to insufficient implementation, low decision-making of some stakeholder groups and poor evidence-based approaches (*medium confidence*) (Bremer et al., 2016; Leisher et al., 2019).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Assessment of water pricing/valorisation, cost-benefit aspects to be inserted]

12.5.4 Food, fiber and other ecosystem products

The effects of climate change on humans, via natural systems, characterize the impact related to depletion of ecosystem services (Scholes, 2016; IPBES, 2018). These services are reflected on food options and production, and products for the biodiversity (e.g., fibre, seeds, fruits, roots, wood). The predicted impact in a world with mean temperature higher than 2C, are patchy in ecosystems around the globe, although, in most cases, a net negative effect (IPBES, 2018).

The rate of change in climate might impact ecosystems in CA and SA in different levels affecting the adaptive capacity of social and ecological systems, and the capacity provision of food, fibre and ecosystem services.

Considering the food systems vulnerability and adaptation, the indicators tend to be associated with production, agricultural incomes, food availability, consumption or rural subsistence, with a low direct cause-effect on ecosystems impacts. The social component of this question refers basically to the producer (traditional communities, rural households, agricultural enterprises, agribusiness) or the consumer (rural subsistence households or urban consumers). Management ecosystem resources, to increase resilience of food production, are expressed as positive effect on pollination (IPBES, 2017; Scarano et al., 2018), water availability, soil conservation.

However, assessing the vulnerability of the food system and adaptation measures, requires a deeper understand of the vulnerability of specific elements of the system, and the links between production and consumption (Nolasco et al., 2017).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Information on different tropical ecosystems and temperate forests, dry forests, savannah, grasslands, mountains, mangroves will be provided]

While global agriculture is responsible for around 25% of greenhouse gases (GHG) emissions from human activities, and the global trend since the 70s is a reduction of GHG per unit of agricultural product (Bennetzen et al., 2016), the sector faces big challenges related to climate change adaptation.

Food production systems in CA and SA face enormous challenges derived from climate change, leading to higher uncertainty in yields for major nutritional crops (d'Amour et al., 2016). Adaptation capacity in the area is limited and, therefore, it is highly probable to observe that climate change affects availability and access to food, and an increased price variability (Feldman and Cortes, 2016). Major issues refers to landscape planning considering natural and production (food, fiber, biofuel) systems, considering the distribution of rural population and increasing urbanization in CA and SA regions (Smith, 2013).

12.5.4.1 Food security

While small-scale farmers have a big contribution to food production and food security, especially in developing economies, they face global policies oriented towards global commodity markets and the entrepreneurship capacity with the capital required investments, despite a growing consensus on its ecological incompatibilities, social limitations and economic risks (Pokorny et al., 2013; Fernandez et al., 2019) describe a dualistic agricultural Chilean sector in which subsistence farms coexists with commercial exploitations, in this scenario, the most affected by climate change are small-scale producers.

Even though there are large improvements in food availability in several regions, including Latin America, there is also a tendency of a dramatic decline in food self-sufficiency in many countries, such as Peru (Porkka et al., 2013). According to Rolando et al. (2017) the increasing food demands in a region with limited agricultural land and affected by climate changes calls for agricultural intensification in the tropical high-Andean Puna.

CA presents a growing dependence on food imports (Porkka et al., 2013; d'Amour et al., 2016) found a teleconnection between food supply shocks and vulnerability, this link is especially relevant for countries with a high share of poverty, in this analysis Central America is recognized as one of the world's area most sensitive to supply shocks in maize. In the same area, for the case of production of dry bean, an important crop for subsistence, the analysis of simulations of Eitzinger et al. (2017) shows a future corridor of reduced yields from Lake Nicaragua to central Honduras, indicating a decrease of 10 to 38%. Based on surveys, Niles and Salerno (2018) found that a 23% and a 53% of farmers in Costa Rica and Nicaragua, respectively, declare experienced a climate shock. In the same study, they found that households that reported experiencing a climate shock were 1.73 times more likely to be food insecure.

12.5.4.2 Crop production

Crop production is related to natural and human factors. Climate is one of the main determinants of agricultural production. Climate change effects in the region are heterogeneous (among and inside countries) and can be extremely large (Feldman and Cortes, 2016).

Rodríguez De Luque et al. (2016) implemented an analysis of projections by crop and countries in the region, they found that the growth of corn and bean production would fall mainly in Nicaragua, El Salvador, Guatemala, Honduras, Colombia, Venezuela and Brazil. Rice production is the one that would present the greatest drops compared to the scenario in which climate change does not occur. The production of soybeans would be significantly affected in Ecuador and Uruguay, while for other producers a neutral or positive impact would be experienced. Wheat production would be adversely affected, especially in large producers and trader countries such as Argentina, Brazil and Uruguay.

Table 12.9: [PLACEHOLDER FOR SECOND ORDER DRAFT: Showing the main crops in the region and by country with observed and projected impacts of CC]

12.5.4.3 Adaptation mechanisms

Strategies for climate change adaptation in the region must considerate the development of varieties whose characteristics allow resistance to drought and high temperatures, as well as the implementation of climate projection programs that offer information allusive to the possible changes that may occur in rainfall patterns and temperatures in the short and medium term (Roco et al., 2014; Rodríguez De Luque et al., 2016; McMartin et al., 2018). McMartin et al. (2018) reported that in Argentine, agricultural technologies are not necessarily changing, but the economic activity is shifting to accommodate climate and adapting to recognize changes in water availability and ideal growing conditions. The same study found that in Colombia, coffee plantations are moving farther up the mountain with land at lower elevations converted other uses.

Bacon et al. (2017) found protective benefits of agroforestry to obtain less severe yield losses due to climate damages in coffee production in Nicaragua. Integrated food-energy approaches are also important for climate change adaptation on the area given that they provide multiple benefits, including the coproduction of food, animal feed, and organic fertilizers (Sharma et al., 2016).

For the case of maize production in northeast Brazil, an important staple food, the results of Martins et al. (2019), using crop simulations, suggest that it is necessary to develop cultivars with a longer crop cycle and tolerant to high temperatures; additionally, irrigation becomes essential to sustaining productivity in adverse climate change scenarios.

Parraguez-Vergara et al. (2018) using case analyses, found that indigenous and traditional agriculture in six countries of the region present and interesting potential to food sovereignty, based in existing traditional knowledge and practices. Bocco et al. (2019) studying indigenous adaptation practices in Mexico indicate that terrace flat terrains, with deeper soils and higher residual soil moisture than sloping soils, allowed for coping with climatic changes and water scarcity.

In the same line, de Sousa et al. (2018) claim that capacity development interventions are important to promote climate change adaptation practices among farmers in Central America. For the same subregion, Son et al. (2018) develop an agricultural drought assessment method to inform the risk. Mulligan (2015) encourage for a more focused intensification of agriculture to minimize negative projected effects of climate change on tropical areas of the region.

Information has an important role in climate change adaptation and there is a recognized gap between climate science and farmers. An interesting case facing this gap is the implementation of local technical agro-climatic committees in Colombia (Loboguerrero et al., 2018), these committees allow to share and validate climatic and weather forecast; and crop modelling results.

Table 12.10: [PLACEHOLDER FOR SECOND ORDER DRAFT: Summarizing studies related to climate change adaptation determinants in CSA Region]

Imbach et al. (2017) report about potential coupled impacts of climate change on pollinators and crops considering the coffee production of Latin America, highlighting the urgent need of implementation of a responsive management strategy.

[PLACEHOLDER FOR SECOND ORDER DRAFT: More information of “Adaptation in livestock and fisheries sectors”]

12.5.4.4 Advances and gaps in adaptation

Since AR5, important advances at institutional level are observed based on the development and implementation of national adaptation plans for agriculture and forestry sector among countries. Nevertheless, it is important to evaluate actions described in plans and contrast with situation in absence of those plans.

[PLACEHOLDER FOR SECOND ORDER DRAFT: Table or figure showing the development and implementation of national adaptation plans for agriculture and forestry sector by country]

It is necessary to advance in the identification of relevant elements for public policies to promote the development of a climate-smart agriculture, in a frame where factors associated with climate have become important for producers, given recent experiences of drought and lack of water (Clarvis and Allan, 2014; Roco et al., 2016).

Limited information regarding cost-benefit analyses of adaptation is available in the region. Roco et al. (2017) in a study in Mediterranean Chile found that the use of a set of adaptive practices presented a significant and positive effect on agricultural productivity; practices related to irrigation and water management showed the biggest effect. Despite of that, it is important to advance in a better understanding of adaptation effects to avoid maladaptation and promote site-specific and dynamic adaptation options considering available technologies.

McMartin et al. (2018) indicate evidence of limitations of infrastructure to reduce vulnerability in face floods and droughts. In the same line, Porras et al. (2019) found that water mismanagement, overexploitation, and conflicts over access are linked to the lack of application and neglect of formal rules in Rio del Carmen watershed in Mexico. Additionally, Calliari et al. (2019) based on a case in the same country, give evidence of the importance of the quality and strength of relationships among formal public organizations to promote climate adaptation.

[PLACEHOLDER FOR SECOND ORDER DRAFT: More information of "Barriers for adaptation"]

12.5.5 Cities, settlements and key infrastructure

CA and SA are the second most urbanized region of the world (UNDESA, 2019). It is still on an upward growth curve, although it is slowing down. Some of the most populous urban areas in the world (with more than 10 million inhabitants) are in CSA. São Paulo, located in Brazilian inland, is the largest one in the region and the fourth worldwide, with almost 22 million inhabitants. It is followed by three coastal metropolises: Buenos Aires and Rio de Janeiro, with over 13 million inhabitants on the Atlantic Coast; and Lima, with more than 10 million on the Pacific Coast.

The number of inhabitants of large, medium and small sized cities (under 10 million inhabitants) are expected to keep growing up to 2030 (UNDESA, 2019). The concentration of huge metropolitan areas on coastal zones, as much as the increasing number of small cities by the sea (Barragán and de Andrés, 2016), escalates the exposure on the region.

Urban vulnerability to extreme events (droughts, storms, increasing temperatures and sea level rise) varies greatly in a region that has a very varied physiology. However, the main determinants of this urban vulnerability are recurrently: poor infrastructure and construction standards, high densities, overload of local ecosystems and sources of water and energy, poverty concentration and weak governance, which dangerously increases the risks to human health, ecosystems and its services, water supply, food and energy in the urban areas (see Section 12.3).

A recent study shows that cities adaptation in the region, considering all Latin America, have been reported by most of capitals, having Quito stood out, although 87% of cities over 1 million inhabitants in the region have not reported any adaptation initiative up to the date of the study (Araos et al., 2016).

12.5.5.1 Challenges and opportunities

City adaptation in CA and SA face major social, economic and political challenges, shared by its many countries and deeply reflected in the urban territory, increasing vulnerability and demanding priority on political and economic investments (Filho et al., 2019). Those challenges are related to inequality, poverty and informality (Romero-Lankao et al., 2014). These features are boosted in the combination with weak and unstable political and governmental institutions, which recurrently suffer with corruption, fragile governance structure and reduced capacity to fund adaptation measures (Rasch, 2016).

As a result, it shapes the urban territory with socio-spatial segregation and the growth of population in informal settlements combined with the occupation of different risk areas, as we may see in the many slums still growing from medium to megacities on the region (Rasch, 2017; Government of Ecuador, 2019). This may also be seen in the persisting housing deficit and the iterant deficiency distribution of infrastructure and urban services – including sanitation and transportation (Filho et al., 2019). Such significant deficits, on the other hand, are also a huge opportunity to build new infrastructure and housing within a climate-adapted standard (Creutzig et al., 2016; Government of Brazil, 2016).

Poverty and inequality can be particularly relevant to affect adaptive capacity in cities, although the interference of poverty in local vulnerability is complex and may vary in different contexts (Nelson et al., 2016). Nowadays 21% of urban population in Latin America and Caribbean lives in slums (UN-Habitat, 2018).

A strong governance of climate change has been a major challenge and is affected by many factors. For some countries, as reported by Ecuador (Government of Ecuador, 2019), lack of coordination between sectoral and vertical government instances are still delaying adaptation, while including diverse stakeholders and improving adaptation strategies through encouraging and learning from community-based adaptation experiences has been an opportunity (Archer et al., 2014) in examples such as in the Regional Climate Change Adaptation Plan of Santiago implementation (Krellenberg and Katrin, 2014).

[PLACEHOLDER FOR SECOND ORDER DRAFT: RCP scenarios and its opportunities/requested adaptation measures in the region]

12.5.5.2 Main Goals

[PLACEHOLDER FOR SECOND ORDER DRAFT: goals for cities pointed in SDGs and New Urban Agenda concerning the Region]

All countries in the region have published their Nationally Determined Contribution (NDC) except French Guyana. Many, such as Brazil, Colombia and Peru, emphasize the central role of cities, local governments and urbanization investments in achieving their goals and Ecuador emphasize the importance of local government participation on the NDC design. Although the guidance towards integrating actions with local governments is not so robust, many of them, such as Argentina, Brazil and Chile`s, define that this guidance will be developed in their National Adaptation Plans (NAP). Uruguay`s NDC is one of the exceptions that includes specific guidance for local adaptation, such as implementing a Guide for the Preparation of Land-use Planning Instruments by 2020 and adaptation measures in at least 30% of the cities with over 5,000 by 2025. Guatemala, Uruguay, Brazil and Venezuela`s NDC highlight the importance of adapting urban transportation systems. Costa Rica, Venezuela and Uruguay`s documents also highlights the importance of local land-use plans, incorporating risk variables, for adaptation (Government of Argentina, 2016; Government of Brazil, 2016; Government of Costa Rica, 2016; Government of Guatemala, 2017; Government of Uruguay, 2017; Government of Colombia, 2018; Government of Venezuela, 2018; Government of Ecuador, 2019).

Six out of nineteen countries in the region has already published its NAP (Brazil, Chile, Colombia, Ecuador, Guatemala and Peru), being recurrent the sectoral approach in transport, agriculture, housing and energy (Government of Brazil, 2007; Government of Peru, 2010; Government of Chile, 2014b; Government of Ecuador, 2015; Government of Colombia, 2016; Government of Guatemala, 2016). Only Chile, however, have developed their sectoral NAP for Cities emphasizing its participatory process of design, the importance of the land planning and local management. Its action lines include improving urban mobility, creating green infrastructure projects, increasing public spaces, developing sustainable public buildings and encouraging the use of clean energy in housing (Government of Chile, 2018).

12.5.5.3 Governance and Financing

Adaptation in CA and SA faces the need for improving governance and long-term support, considering the importance of the engagement of sectoral government instances and a multiplicity of stakeholders in all levels in adaptation initiatives (Anguelovski et al., 2014; Chu et al., 2016). A survey including 736 cities around the word and 41 in Latin America has highlighted the tendency of LA cities to have above average large planning teams dedicated to Climate Change, although they reported being not clear the liability for climate change response planning (Aylett, 2015).

Experiences such as the one in metropolitan Lima, the Climate Action Strategy for the Lima Province (Metropolitan Municipality of Lima, 2014; Miranda Sara and Baud, 2014; Miranda Sara et al., 2017), show that the ability to enrol as many stakeholders as possible in the adaptation process can be determinant for their success (Anguelovski et al., 2014; Rosenzweig et al., 2018). This is of paramount importance to the region, where cities are shaped by social inequality and a fragmented urban fabric, especially when considering its impact in the decision-making arena (Chu et al., 2017). The engagement of local players in this example has been central to spreading and mobilizing different types of knowledge and creating the metropolitan and local networks able to support adaptation measures. The inclusive process is also a goal on the example from the Chile Municipal Network facing Climate Change (RedMuniCC) engaged on developing participatory strategic plans for climate adaptation and mitigation (https://www.redmunicc.cl/).

Water, transportation and sanitation are some of the central systems to creating a resilient urban environment whose limits do not match cities' administrative boundaries. Therefore, adapting urban systems will require a high multi-level governance capacity with strong multi-players horizontal and vertical coordination (Bai et al., 2016; Miranda Sara et al., 2017) and with an inter-sectoral systemic approach. In the adaptation of metropolitan areas, the many administrative units that compose it intensify the need of coordination capacity. Examples dealing with those governance challenges are the metropolitan plan of Rio de Janeiro and Lima.

The Rio de Janeiro Metropolitan Strategic Plan engaged thousands of people, including representatives of the many municipalities of the Metropolis, sectoral representatives of the state government, academia, non-governmental organizations, class entities, the private sector, experts and members of the social movement, in different instances of participation in each stage of its elaboration (Government of Rio de Janeiro, 2018). Lima approves its Climate Action Strategy after a participatory and conservative process with the technical group on climate change from the Metropolitan Environmental Commission, focusing on the reduction of water vulnerabilities related to drought and heavy rain scenarios, on the basis of which 21 (out of 52) Lima districts municipalities are developing and implementing their adaptation measures.

[PLACEHOLDER FOR SECOND ORDER DRAFT: information access; transparency; knowledge improvement]

Financing climate adaptation in the CSA has been another major challenge. New forms of financing and leadership focused on community-based solutions have been developed in the region to overcome that issue (Castán Broto and Bulkeley, 2013; Archer et al., 2014; Paterson and Charles, 2019). Also national legislation can help financing adaptation measures. Peruvian Law on the Retribution Mechanism of Eco-Systemic Services and Code (Supreme Decree Nº 009-2016-MINAM) (Miranda Sara and Baud, 2014) in addition to the Ley Marco de la Gestión y Prestación de los Servicios de Saneamiento (Supreme Decree 019-2017-Vivienda), defines that every Potable Water Company is capable to add 1% to the urban water tariff to guarantee ecosystem services related to water generation, treatment and reuse with natural and green infrastructure projects, decides they are due to develop Adaptation Climate Plans into their cities and invest another 4% extra into the water tariffs on their adaptation measures.

[PLACEHOLDER FOR SECOND ORDER DRAFT: Financing agencies and priorities]

12.5.5.4 Urban Design and Planning

Both the shape of the city and the activities held in it have an impact on carbon emissions and on adaptation and mitigation opportunities (Creutzig et al., 2016; Satterthwaite et al., 2018). Combining urgent measures, strategic action (Chu et al., 2017) to long term planning is central for cities adaptation (Filho et al., 2019). Planning is also crucial to avoid maladaptation (Filho et al., 2019). Lack of information at local scale, human resources and financing, as well as the concreteness of non-resilient urban structures (Creutzig et al., 2016), can hinder both short- and long-term adaptation in CSA, demanding multilevel and transversal governance, but also the adoption of not so costly urgent measures by local governments (Rosenzweig et al., 2018).

In this sense, some initiatives have focused on developing urban and building codes to establish new adapted standards of sustainable construction, taking advantage of the dynamics of private investments in order to discontinue the reproduction of non-resilient city growth and the increases in vulnerability. The Chile's Sustainable Housing Construction Code (Government of Chile, 2014a), for example, require a risk assessment at the design stage of the project by a qualified professional and the definition of mitigation measures if necessary.

Strategic adaptation initiatives have been an approach adopted by many cities in dealing with the multilevel complexity and the inter-sectoral nature of the urban system, with gains in fostering leadership and facing the prevalence of neoliberal oriented urban development in the global south (Chu et al., 2017). An example is the Medellin's metropolitan green belt that, focusing on problems such as irregular settlements, inequality and poor governance, have articulated programs and projects of the Municipality of Medellin and the municipalities of the Vale do Aburra in a strategic long-term planning. Places with informal and vulnerable settlements were aimed to be transformed with the belt's integration areas: ecoparks and eco-gardens (Municipality de Medellín, 2012).

[PLACEHOLDER FOR SECOND ORDER DRAFT: data providing/downscaling; monitoring]

12.5.5.4.1 Land use

[PLACEHOLDER FOR SECOND ORDER DRAFT: assessment on adaptation concerning land use]

12.5.5.4.2 Housing, informality and risk areas

Informality and precariousness in housing is one of the most sensitive issues for adaptation in CA and SA cities, as in Asia and Africa (Satterthwaite et al., 2018; UN-Habitat, 2018). It is an important factor of vulnerability and risk and reaches worrying dimensions, particularly in large cities. Lack of sanitation, transportation, environmental health, in addition to violence, exclusion and, along with this, property insecurity, have been faced by urban governments, accumulating good practices in the last decades (Satterthwaite et al., 2018).

However, most of these government housing programs and initiatives do not focus on adaptation, although they often focus on risk reduction. An example is the Integral Resettlement in Medellin (Municipality of Medellín, 2013), a program that offers housing to families who have had to leave their homes because of a natural event or being under risk (http://isvimed.gov.co/). The demand for qualified housing (both in relation to the construction of new housing units and settlements, and in relation to the requalification of existing precarious settlements), and all the necessary infrastructure and services, is also an opportunity to adapt the city that still it was not taken advantage of properly.

[PLACEHOLDER FOR SECOND ORDER DRAFT: housing deficit (quantity and quality); segregation and suburbanization; (in)justice in access and deficiency in urban basic services; violence; occupation of risk areas]

12.5.5.4.3 Green and grey infrastructure

The necessary investment in urban infrastructure to meet basic needs has been seen as an opportunity in many places, if one can guarantee that all the new infrastructure built will help adapt the urban system (Creutzig et al., 2016), as with the introduction of nature-based solutions (NBS). The concept of NBS gained featured while embraces other well-known concepts such as UGI (urban green infrastructure), EbA (ecosystem-based adaptation), and ESS (ecosystem services) (Pauleit et al., 2017). In Petrópolis, a medium-sized city in the hills of Rio de Janeiro state, for example, the water company Águas do Imperador, which has the public concession to provide wastewater service, has implemented 10 nature-based micro wastewater treatment plants, using biologic systems with biodigestors and wetlands for final phytoremediation. They don't demand energy to operate, and generate biogas used locally (Herzog and Rozado, 2019).

Hybrid solutions (mixing green and grey infrastructure) have also been adopted in order to achieve better efficiency in flooding and landslide prevention, coastal protection, and sanitation. (Rosenzweig et al., 2018) This is particularly important in urban environments where the hard infrastructure has frequently been alone in facing those impacts (Castán Broto and Bulkeley, 2013). Although the literature points out the importance of hybrid solutions, in CA and SA these examples are still scarce. The most noticeable is a supplement to the existing infrastructure with newly built first rain garden in a public space in São Paulo city, that was implemented by a collective bottom-up effort. The rain garden of the 'Araucária' Square collects stormwater from about 900 m^2 paved area, into the gardens that infiltrate most of the water, only in heavy climatic episodes the overflow goes to the urban drainage system (Herzog and Rozado, 2019).

Green infrastructure has important initiatives focused on improving city-nature metabolism balance and restoring the connection between the city and its rural and natural surroundings, recognizing thus their interdependency. In these initiatives, coastal vegetation and riverbank preservation and recovering can provide protection to urban areas while also benefiting the environment (Rojas et al., 2019). Besides, enhancing green functional areas and streets in the city may reduce heat bubbles, provide beautiful landscapes, and even capture CO_2 (Rosenzweig et al., 2018). In Costanera Sur, Buenos Aires, public initiative and scientific knowledge were articulated to protect an auto-regenerated Plata riverbank, which had received demolition material to create land for future real estate development close to the central area. Nowadays the reserve offers numerous ecosystem services for residents and attract visitors activating the tourist industry (http://oics.cgee.org.br/).

Managing the water in the city in an adaptive way, using both grey and green solutions, has been central to reducing impacts such as flooding, while also contributing to water quality and supply. A recent study covering 70 cities in Latin America calculated that 91 million people in these cities would benefit from improving main watersheds with green infrastructure (Tellman et al., 2018). A project in Sao Paulo City for the Jaguaré river, a tributary of Pinheiros river (one of the most important in the city), proposes to enable a huge landscape transformation with the renaturalization of the riverbanks, creation of built wetlands in multifunctional public parks along the waterway. This project is an opportunity to shift the paradigm of large, expensive and monofunctional engineered-solution of concrete-built water storage to manage stormwater with nature-based solutions (Herzog and Rozado, 2019).

12.5.5.4.4 Mobility and Transportation

[PLACEHOLDER FOR SECOND ORDER DRAFT: assessment on adaptation concerning mobility and transportation systems]

12.5.5.5 Advances and Gaps

A comprehensive review of more than 4000 case studies on mitigation in world literature shows a huge deficit of studies on South and Central American cities (Lamb et al., 2019) Studies on urban adaptation initiatives in CA and SA are still incipient in the literature, while also being underreported by municipalities. In addition, several practical results have not yet been demonstrated (Araos et al., 2016). Few examples are available, especially when considering the ratio of urban intensity of the Region. A study on Colombian cities, however, has shown an increase in the attention to adaptation in political agendas (Schaller et al., 2016; Koch, 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Gaps in adaptation by sector, enhancing opportunities]

12.5.6 Health, wellbeing and the changing structure of communities

12.5.6.1 Climate change adaptation and human health

Climate change impacts multiple dimensions of human health. Table 12.11 summarizes the environmental effects of climate change and the resulting health effects. Adaptation measures to mitigate the impact of climate change and weather variability are usually measures that are justified even in the absence of climate change. See IPCC AR5 WGII Chapter 12 (Adger et al., 2014) for a characterization of adaptation measures for climate change. The topic is even more complex when considering human health. Adaptation can be direct (e.g., by staying indoors and using cooling system during heat waves), or indirect (e.g., any measure that reduces *Aedes aegypti* larval habitat will also diminish the risk of dengue infections that would increase because of increasing suitable habitat associated with climate change). Hence, in some way, any measure that decreases the risk of a climate-sensitive disease is an adaptation measure for climate change.

Adaptation can be implemented through public policies, private households' responses, or communal management. See Table 12.11 for examples of these types of measures in relation to different health outcomes. The group of measures that are most directly linked to diminishing health risk because of climate change and weather variability are those related with climate services for health. Climate services have been developed traditionally to provide sector-specific short and mid-term weather forecast to inform decisions about agricultural production and natural disasters prevention (WHO and WMO, 2016). Climate services are tailored products for a specific sector that allow decision makers and practitioners to plan interventions (Mahon et al., 2019). More recently, climate services, such as early warning systems and forecast models, have been promoted for the health sector (WHO and WMO, 2012; WMO, 2014; WHO and WMO, 2016;

Thomson and Mason, 2018). To guide this process, the Global Framework for Climate Services (GFCS) was developed as the policy mechanism to support the development of climate services for the health sector and other key sectors (Lowe et al., 2014). The GFCS aims for stakeholder engagement between health and climate actors at all levels to promote the effective use of climate information within health research, policy and practice.

An integrated health-climate surveillance system for dengue control has been implemented in Machala, Ecuador (Lowe et al., 2017). The system produces high-resolution risk information as the basis for improving seasonal dengue early warnings (Lowe and Stewart-Ibarra, 2013; Lowe et al., 2017). Additionally, an interdisciplinary multinational team with researcher in the border of Ecuador and Peru created a cooperation network for climate-informed dengue surveillance (Quichi et al., 2016) and successful binational collaboration resulted in the local elimination of malaria (Krisher et al., 2016). Similar is the innovative community-based data collection to understand and find solutions to rainfall-related diarrheal diseases in Ecuador (Palacios et al., 2016). The Brazilian Observatory of Climate and Health is a similar case, where climate and health information for the region of Manaus, Amazon region is organized and disseminated (Barcellos et al., 2016). It has been shown in Brazil how information on drought conditions can be used to reduced health impacts of drought using a national disaster risk reduction framework (Sena et al., 2016). The institutional setting of these initiatives is heterogeneous. But usually, there are linkages between the academia and local or national authorities.

There exist many examples where climate services have been used to construct vulnerability maps for health outcomes. The institutional setting is also heterogeneous. But in this case, there exist isolated cases promoted by academia, while partnership with the public sector also does exist. Dengue, malaria, and Zika vulnerability maps using climate, social, and environmental information has been developed in Brazil and Colombia (Cunha et al., 2016; López-Álvarez, 2016; Pereda, 2016; IDEAM, 2017). Weather monitoring to alert the population about heat and cold waves is a common services that the national meteorological institutes in the region supply, as in the case of Uruguay (Bidegain, 2014). Argentina is focused on improving the health system by using Climate Change Risk Map System as a tool that identifies the risks and allows assessing their management (OPS and WHO, 2018).

Vulnerability and risk maps have turned into an important tool for disasters risk management. These maps have been developed at city level, in places such as Bogotá, and Cartagena de Indias, and the municipality of Mocoa, in Colombia (Yamin et al., 2013; Guzman Torres and Barrera Arciniegas, 2014; Tehelen and Pacha, 2017; Zamora, 2018) and for the metropolitan district of Quito in Ecuador (Tehelen and Pacha, 2017). In addition, vulnerability maps has been done for the primary roads network of Colombia (Tehelen and Pacha, 2017). At the regional level, vulnerability maps using climate change probability, disaster risk and food insecurity variables has been produced for the Andean region, which includes different municipalities of Bolivia, Colombia, Ecuador and Peru (WFP, 2014). Further, it was developed a vulnerability map for countries of Latin American and Caribbean region using a Climate Change Vulnerability Index (CAF, 2014).

Some countries have developed National plans on health including the role of climate. Chile has a Climate Change Adaptation Plan of the Health Sector that proposes several actions to enhance monitoring, institutions and citizens information and education (Ministry of Health of Chile and Ministry of Environment of Chile, 2016). Based on the identification of vulnerability to climate change, Colombia has developed eleven regional adaptation plans to strengthen institutional capacities; climate change education for behavioral changes; and cost estimation to promote health resilience (WHO and UNFCCC, 2015). In addition, El Salvador implemented actions to strengthen health infrastructure through high latrines for housing in flood communities, as well as other measures focused on water supply and quality based on an education and awareness program (Ministry of Environment and Natural Resources of El Salvador, 2013).

Early warning systems using climate services has been also implemented to predict and take sudden action regarding health outbreaks. In Brazil and Panama is currently operational an early warning system for the surveillance of dengue fever transmission (Codeço et al., 2016; McDonald et al., 2016). In Colombia and Uruguay the national meteorological institutes and national emergencies systems manages early warning system for hydrological disasters (Bidegain, 2014; IDEAM, 2017).

National Risk Management Plans or National Disaster Response Plans are a tools for adapting to climatic-disasters effects, that can help to diminish death and injuries because of disasters. These Plans are generally promoted by governments as national instruments that guide the processes of estimating, preventing and reducing disaster risk. An updated national risk management plans has been found for Guatemala (CONRED, 2014), Honduras (COPECO, 2014), El Salvador (Ministry of Health of El Salvador, 2017), Costa Rica (CNE, 2016), Ecuador (SGR, 2018), Peru (SGRD et al., 2014), Argentina (Ministerio de Seguridad de Argentina, 2018), Bolivia (VIDECI, 2017), Chile (ONEMI, 2015) and Colombia (UNGRD, 2015).

Early warning systems are one of the main measures of adaptation regarding to climate-related disasters. There exist at least 24 early warning systems in South America to avoid deaths and injuries from floods in the countries such as Argentina, Colombia, Bolivia, Brazil, Peru, Uruguay and Venezuela (Bidegain, 2014; Moreno et al., 2014; Dávila, 2016; López-García et al., 2017; Carrizo Sineiro et al., 2018). A total of 149 emergency prevention and response systems are reported in Central America (UNESCO, 2012). However, only 37 can be considered early warning systems properly working, because many of these systems are not operating, are not functioning properly, or do not meet the requirements to be considered early warning systems (UNESCO, 2012). In both regions most of the systems were implemented to prevent disasters caused by floods, given that these are the weather events more linked to population risk (UNESCO, 2012; Dávila, 2016). Although there are other systems designed to prevent landslides or multiple threats, but to a lesser extent (UNESCO, 2012; Dávila, 2016). Likewise, many of these systems have a centralized mode, that is, the monitoring is managed by a single responsible organization (UNESCO, 2012; Dávila, 2016). There is a proposal by the Andean Regional Program for the Strengthening of Meteorological, Hydrological, Climate and Development Services to implement an early warning system for flooding in local communities in Suches-Titicaca (Bolivia) and Catamayo-Chira (Ecuador-Peru) basins (CIIFEN, 2017). One of the strengths of this proposal was the interaction of authorities, institutions responsible for the response and community members who worked in an articulated and consensual way. In addition, some countries implement programs for the relocation of families who are in risk condition, like in Bogotá and Medellin, Colombia (World Bank, 2014; Watanabe, 2015).

The information produced by the integrated health-climate surveillance systems and early warning systems many times is systematized to be communicated to the authorities and general public. In the case of the dengue early warning system in Panamá, the information is distributed in a monthly bulletin that is used by health authorities to inform vector control activities (McDonald et al., 2016). A guide for dengue prevention and risk communication ahead of Brazil's 2014 FIFA World Cup was produced using climate knowledge as an input (Lowe et al., 2014; Lowe et al., 2016). In Colombia, the Intersectoral National Technical Commission for Environmental Health monthly publishes a bulletin with regional weather forecast and potential effects on health (CONASA, 2019). Paraguay improves epidemiological surveillance and train first level health staff with information campaigns and prevention of diseases sensitive to climate change, and promote health networks with the participation of civil society (Environmental Secretariat of Paraguay, 2011).

Public information based in climate services can be considered directly as an adaptation measure to climate change and weather variability to diminish health risk. More indirect is the linkage with private and communal management measures, because they might be adopted even in the absence of climate change. That is, if the population lives in a hot city, they will find a way of cooling regardless of future projections of mean temperatures or the predicted future frequency of heat waves.

Building grey infrastructure has been a popular adaptation measure to reduce deaths and injuries because of floods and. For example, infrastructure has been improved at schools, public buildings and drainage systems in cities such as Bogota, Colombia (World Bank, 2014) and La Paz, Bolivia (Fernández and Buss, 2016). In Brazil, channel works to reduce the flooding of the Tiete River, which crosses the metropolitan area of Sao Pablo, has been done based on simulated flood scenarios (Hori et al., 2017).

A study in Santiago, México City, Buenos Aires, and Bogotá show that households' responses to deal with weather variability are mainly air conditioner / swamp coolers, open windows, wet the floors, shade trees, and seat stoves (gas, electric, oil, or other) (Romero-Lankao et al., 2014). Responses vary depending on the geography, social conditions and weather of the city. In addition, technological transfer to improve health, and thermal comfort and safety in homes in cold weather areas programs has been applied to the high Andean habitat of the Puno Region (Ministry of Environment of Perú, 2016).

Communal management can be considered other group of adaptation measures to diminish health risk. Communal management is a way of collective action between individuals, where they cooperate to jointly produce a good, or avoid a bad, that do not only bring benefits to them but also to other individuals (Ostrom, 2015). Communal management can be relevant, e.g., when thinking about mosquito breeding sites control regarding vector borne diseases, like malaria or dengue. Communal management can be relevant when thinking of new places where vector-borne diseases can emerge because of changes in climate. Andersson et al. (2015) test whether community mobilization adds effectiveness to conventional dengue control in Nicaragua using a cluster randomized controlled trial. They found that community mobilization showed a lower risk of infection with dengue virus in children (relative risk reduction 29.5%, 95% confidence interval 3.8% to 55.3%), fewer reports of dengue illness (24.7%, 1.8% to 51.2%), fewer houses with larvae or pupae among houses visited (house index) (44.1%, 13.6% to 74.7%), fewer containers with larvae or pupae among containers examined (container index) (36.7%, 24.5% to 44.8%), fewer containers with larvae or pupae among houses visited (Breteau index) (35.1%, 16.7% to 55.5%), and fewer pupae per person (51.7%, 36.2% to 76.1%).

Finally, there is evidence that forest conservation and natural protected areas can diminish the cases of malaria and diarrhea in Brazil, and the cases of children diarrhea in Colombia (Bauch et al., 2015; Herrera et al., 2017; Chaves et al., 2018). Forest conservation can improve hydrological cycle control and soil erosion that can help to improve water quality and water-borne diseases. In addition, forest cover can help to diminish mosquitos' breeding sites, who are vector that transmit malaria and dengue. Again, these are measures that might be taken independently of climate change. However, they can help to design policies in sites where these problems do not currently exist but can emerge as a consequence of climate change and the increase in the frequency of weather extreme events.

Table 12.11: Environmental effects from climate change, health effects, and adaptation.

Environmental effects from climate change	**Health effects**	**Adaptation**		
		Public	**Private**	**Communal**
Extreme weather events Frequency Severity Geography	Thermal stress deaths / illness	Vulnerability mapping Nature restoration to avoid 'urban heat islands' (urban and landscape) Early warning systems Training of health staff to recognize and treat heat strain.	Air Conditioner / Swamp coolers Open windows Wet the floors Shade trees Heat stoves (gas, electric, oil, or other) Building isolation	Training of community health volunteers to recognize and treat heat strain
	Injury / deaths from floods, storms, cyclones, bushfires	Vulnerability mapping Infrastructure (e.g., berms) to reduce flood impacts Early Warning systems Relocations programs National Risk Management Plans	Defensive infrastructure Nature restoration to diminish runoff because of soil erosion and hydrological cycle control	
	Poor nutrition due to reduced food yields and dehydration due to limited fresh water supply			
Effects on ecosystems (land and sea) and on particular species	Microbial proliferation: food poisoning (e.g., *Salmonella spp*), and unsafe drinking water	Vulnerability mapping Nature restoration to increase hydrological cycle control		Integrated health-climate surveillance systems

		Integrated health-climate surveillance systems Expanded health insurance arrangements Enhance interaction between health and agricultural authorities to have integrated monitoring of food-borne and animal diseases		
	Changes in vector-pathogen host relations and in infectious disease geography/seasonality:	Vulnerability mapping Nature restoration to avoid temperature increase, and hence increase vector breeding sites (mosquitoes) Fumigation of large areas Integrated health-climate surveillance systems Operational early warnings Pilot early warning systems Vaccination programs (e.g., rotavirus in the USA) Expanded health insurance arrangements Public information	Bed nets Screens in doors and windows Repellent and insecticides Mosquito breeding sites control Avoid outdoor activities during peak vector activity	Training of community health volunteers to collect blood smears for malaria diagnosis. Community led campaigns to eliminate vector habitat
	Impaired crop, livestock and fisheries yields leading to impaired nutrition, health, survival			
Sea-level rise: salinization of coastal land and freshwater; storm surges				
Environmental degradation: land, coastal ecosystems, fisheries	Loss of livelihoods, displacement, leading to poverty and adverse health: mental health, infectious diseases, malnutrition, physical risks			Communal actions to help families rebuild after natural disasters
	Sexual abuse and violence associated with crowding in shelters following natural disasters			
	General	Public information Integrated health-climate surveillance systems		
Based in [IPCC, 2014];(McMichael et al., 2006) and own elaboration				

12.5.6.2 Migration and trapped populations

Migration is an impact of climate change, but also an adaptation strategy when people continue with their lives in a different but stable situation, or when members of the families send remittances to those that remain in the affected areas (*robust evidence, high agreement*). The remittances create opportunities for adaptive capacity building, for infrastructures, agricultural supplies, food, education, health.

It is hard to differentiate when climatic hazards are the main cause of migration, or when they interact with other political, social or economic reasons (Brandt et al., 2016), for example with monoculture-propelled deforestation in the Southern Cone (Heslin et al., 2019). Surveys to migrants usually find that the main reported reason for migration is to find a job or to increase the household income (OIM, 2017) but the underlying reason for the lack of job or income is rarely examined. At times, positive climatic conditions also appear to facilitate migration (Gray and Bilsborrow, 2013).

Some areas are more sensitive to generate climatic migration: the Andes, the dry areas of the Amazonia, northern Brazil, or Northern CA. Drier areas face the highest rates. In a study of 8 countries around the world, including Guatemala and Perú, Warner and Afifi (2014) found a link between rainfall variability and food insecurity which could lead to migration in areas of high prevalence of rainfed agriculture and low diversification. In CA, younger individuals are more likely to migrate in response to hurricanes and especially to droughts (Baez et al., 2017). But migration is often the last resort for rural communities facing water stress problems in CA and SA (Magrin et al., 2014; Ruano and Milan, 2014). In Bolivia, glacial retreat has not triggered new migration flows and had a limited impact on the existing migratory patterns (Kaenzig, 2015).

Some rural and urban threatened or affected populations may be immobilized because of floods or droughts, and are called "trapped populations" (Thiede et al., 2016). These populations are supposed to face a double set of risks: they are unable to move away from environmental threats, and their lack of capital makes them especially vulnerable to environmental changes (Black et al., 2011). In Guatemala, migrating to the U.S. is becoming dangerous and expensive; these trends expose local populations to the risk of becoming trapped in the near future in a place where they are extremely vulnerable to climate change (Ruano and Milan, 2014). A recent survey by OXFAM in Guatemala (Aguilar et al., 2019) found no correlation between migration to the U.S. and severe food insecurity in households, but the correlation became significant if the level of food insecurity was moderate, suggesting that families in extreme hardship did not have the resources to migrate.

But some populations just have chosen not to move, as in Peru, where immobility in dissatisfied people is more likely to be caused by attachment to place than resource constraints (Adams, 2016; Correia and Ojima, 2017). Some populations have chosen to adapt relying in their indigenous and traditional knowledge (Boillat and Berkes, 2013). In SA, climatic variability increases the likelihood of inter-province migration, rather than trapping populations. In a study of interprovincial migration motivated by temperature, an exception arose in Bolivia, and even if that could suggest a trapped population (Thiede et al., 2016), it is not clear if they want to stay and adapt.

12.5.6.3 Gender and climate change

According to Casas Varez (2017), there is ample empirical evidence that the ravages of climate change are not of equal scope for men and women since women, due to socially constructed gender norms and associated structural gaps, suffer the consequences of global warming more severely (*high evidence, high agreement*). In issues such as drinking water, energy, natural disasters, impacts on health and agriculture, women are affected in greater proportion by the consequences of the global warming, further widening structural gender gaps underlying Latin American societies (Casas Varez, 2017). For example, in a rural community in Nicaragua vulnerable to drought, short-term coping was more common among the women, especially among female heads of household, while adaptive actions were more common among the men; there are gendered inequalities in access to and control over different forms of capital that lead to a gender-differentiated capacity to adapt, where men are better able to adapt and women experience a downward spiral in their capacity to adapt and increasing vulnerability to drought (Segnestam, 2017).

However, women are not always the more vulnerable group (*medium evidence, high agreement*). In Brazil, Peru and Mexico, female headed households tend to be slightly less vulnerable and more resilient than male headed households, even though some exceptions can be found in rural Peru and urban Brazil when looking at sub-groups (Andersen et al., 2017). In turn, some studies suggest that women establish a more friendly relationship with the environment and towards natural resources than men (*low evidence, low agreement*); studies on masculinities and environment confirm this tendency (Brough et al., 2016).

The relationship between gender and adaptation to climate change demands an analytical framework that connects environmental problems with social inequalities in a more complex way (Godfrey, 2012). An intersectional approach contributes to better capture the diversity of adaptive strategies that men and women adopt vis-à-vis climate change. Particular constellations of race, gender, class, age or nationality reveal more complex realities (*high evidence, high agreement*). Recent studies emphasize that a gender approach to social inequalities ought to move beyond just looking at men and women as experiencing the impacts in a differentiated manner; rather, an intersectional analysis illuminates how different individuals and groups relate differently to climate change, due to their situatedness in power structures based on context-specific and dynamic social categorizations (Kaijser and Kronsell, 2014).

12.5.6.4 Perceptions, knowledge, participation and adaptation policies

Perception and understanding of CC can be seen as an adaptive feature. In CA and SA, the consciousness of CC as a threat is growing, a situation that could be related to a growth in climate justice activism, as well as to the occurrence of extreme weather events of all kind (Forero et al., 2014; Magrin et al., 2014; Capstick et al., 2015). Wealth and education are positively and significantly related to climate change concerns (Evans and Zechmeister, 2018). Anyhow, some communities do not associate their problems with the scientific concept of CC, so discussions as if it is human induced, the causes, or relations with other problems, can become irrelevant (Sapiains Arrué and Ugarte Caviedes, 2017). It is necessary then to consider the amplification of the understanding, but also the complex and diverse ways to cope with CC. Even communities affected by the same changes do not necessarily perceive them in the same way, as seen in Southern Brazil (Bonatti et al., 2016). In some places of Peru, rather than adapting to climate change, people adapt to climate change to their social worlds (Rasmussen, 2016). It is also noticeable that the interpretations of change, its causes and effects, can widely vary (Paerregaard, 2018; Scoville-Simonds, 2018).

Perceptions tend to be different in rural and urban areas (Sherman et al., 2015). In the rural areas, it is highly related with agriculture, but also with biodiversity loss, greater solar radiation or changes in the oceans (Infante and Infante, 2013; Postigo, 2014; Jacobi et al., 2015; Barrucand et al., 2017; Martins and Gasalla, 2018; Meldrum et al., 2018; Córdoba Vargas et al., 2019; Viguera et al., 2019). In places as the Amazonia, there is an increased perception with age (Funatsu et al., 2019). In Mediterranean Chile, younger, more educated producers and those who own their land tend to have a clearer perception than older, less educated, or tenant farmers, but they do not have a clear perception or how it may affect their yields and farming operation (Roco et al., 2015). In some dry and humid Ecuadorian montane forests, peasants perceive CC in the same way as scientific data, but they are at odds to predict the changes and consider that they may not be prepared and only can be reactive (Herrador-Valencia and Paredes, 2016).

There is a wide range of social movements that address adaptation. Some engage and participate in policy and planning, often having good results at the local level (Krellenberg and Katrin, 2014; Nagy et al., 2014; Waylen et al., 2015; Bizikova et al., 2016; Chelleri et al., 2016; Merlinsky, 2016; Villamizar et al., 2017). On the contrary, top-down approaches without participation have shown to be less effective (*robust evidence, high agreement*) (Stein and Moser, 2014; Ruiz-Mallén et al., 2015; Sherman et al., 2015). Effective adaptation and mitigation responses depend on policies and measures at multiple scales, but especially on the involvement of the more exposed and vulnerable people. In Quito (Ecuador), the participation of experts, communities and citizens has been effective; more inclusive planning processes correspond to higher climate equity and justice outcomes in the short term, but also an emphasis on building dedicated multi-sector governance institutions may enhance long-term programs stability, while ensuring civil society voice in adaptation planning and implementation (Chu et al., 2016). Several experiences of local adaptation have been successful with local participation (FAO and Fundación Futuro Latinoamericano, 2019). Local and indigenous knowledge participation and agency, expressed through diverse demands, is thought to be more addressed and deepened (Nagy et al., 2014; Jurt et al., 2015; Stensrud, 2016). Anyhow, some people do not always participate in CC adaptation, in cases because they rather point to areas as education, sanitation or social assistance (Bonatti et al., 2019).

Representations of CC can also emerge as critiques and resistances, that expose that CC labelled politics or interventions have posed even bigger risks, or do not address poverty issues (Lampis, 2013; Pokorny et al., 2013; Ojeda, 2014).

Although there are many public policies, most of them they have become symbolic, in conflict with prevailing economic policies and practices. In some cities of Colombia, adaptation plans are in conflict with other policies (Koch, 2018). In coastal areas of Venezuela and Uruguay, there exists insufficient support in multiple areas such as social attitudes and behaviour, knowledge, education and human capital, finance, governance, institutions and policy (Villamizar et al., 2017).

12.5.7 Poverty, livelihood and sustainable development

This section contributes with a synthesis on poverty, livelihood and sustainable development in CA and SA. These concepts are refined in AR6 and described in more detail by chapter 8 in this WGII report. Climate change impacts are increasing and exacerbating poverty, inequalities meanwhile it is bringing up new risks affecting both poor to rich populations, challenging sustainable development pathways. At the same time poverty, high levels of inequalities and existing vulnerabilities also worsen climate change impacts, where those already suffering are losing their livelihoods and reducing their development options (*high evidence, high agreement*).

Opportunities for a climate proof governance configuration may emerge based on socially supported agreements which can contribute to increase transparency, negotiate and balance power relationships allowing the views of the poor, vulnerable and excluded to be heard (Miranda Sara and Baud, 2014). But existing unbalances on power relations, corruption secular problems and high levels of risk tolerance (Miranda Sara and Jameson, 2016) constitute climate proof governance barriers in putting more effective adaptation and/or preventive measures in place. Corruption, particularly in the city construction and infrastructure, have proven to be a barrier for the development in the region even reproducing and reconstructing risks (French and Mechler, 2017). Critical infrastructure and valuable assets continue to be placed in vulnerable areas (Calil et al., 2017) evidencing the persistence of maladaptation and adaptation deficit to current climate variability (Villamizar et al., 2017). Governments, being national to subnational, must guarantee the rights to all, but technical, financial resources and political constraints for climate resilience at their disposal still need to be enhanced (Satterthwaite et al., 2018). Good quality infrastructure and services and better housing quality are at the centre of reducing risks from extreme climate events (Satterthwaite et al., 2018; Dodman et al., 2019).

Poorer and most vulnerable groups evidence limited political influence, fewer capacities and opportunities to participate in decision and policy making, are less able to leverage government support to invest on adaptation measures linked with poverty, inequality and vulnerability reduction (Miranda Sara et al., 2017; Reyer et al., 2017; Dodman et al., 2019), where climate knowledge construction and risk perceptions affect decision-making on defining implementation priorities being less able to cope and to adapt avoiding "adaptation injustices" (Miranda Sara et al., 2017). Climate change challenges those already vulnerable and disfavored, moving societies in the region in the opposite direction in their search for resilience, equity and sustainable development, where the CA and SA high levels of inequality and informality (Bartlett et al. 2016) remain the biggest challenges considering current and expected risks attributed to climate change for adaptation measures being effective (Rosenzweig et al., 2018; Dodman et al., 2019).

The interaction of biophysical projected impacts with the existing vulnerabilities in the region, as hunger, malnutrition and health inequalities, arising from its social, economic and demographic profile, may affect development and well-being in the region in a variety of ways (Reyer et al., 2017) such as increasing poverty and inequality putting at risk the paths for sustainable development (Reckien et al., 2017). Impacts are likely to occur simultaneously in different sectors being uncertain and not uniform, exacerbating those of the poorer but also creating new vulnerable groups (Miranda Sara and Jameson, 2016; Rosenzweig et al., 2018; Dodman et al., 2019).

According to ECLAC report (ECLAC, 2019), the Gini index average for 18 Latin American countries (including Mexico) decreased from 0.543 to 0.466 (from 2002 to 2017), but it is slowing down: from 1.3% (2002 to 2008) to 0.8% (2008 to 2014) and to 0.3% (2014 to 2017). Inequality is a historic and structural characteristic of the region, despite important improvements over the last 15 years it continues to be highly unequal (ECLAC, 2019). Meanwhile poverty is increasing, after 12 years in which poverty and extreme poverty rates in the region decreased. In 2018, 29.6% of Latin America's population was in situation of poverty (182 million people), a percentage that increased from 27.8% in 2014 (164 million). Extreme poverty, meanwhile, increased to 10.2% in 2018 (63 million people) from 7.8% in 2016 (46 million people). Showing a modest positive tendency because poverty levels shrank by 1 percentage points between 2002 and 2018. Poverty and extreme poverty rates are higher among children, adolescents, young people, women (Reckien et al., 2017) migrant groups (Dodman et al., 2019) and the population residing in rural areas. The first Sustainable Development Goal aims to eradicate extreme poverty - measured as people living on less than $1.90 per day - everywhere, by 2030 (SDG, 2015), although researchers argue that poverty is frequently misunderstood (Castán Broto and Bulkeley, 2013), mischaracterized and that the income-based poverty lines are spurious indicators of poverty which do not reflect the multiple dimensions of poverty. The second SDG aims to reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions by that same year and only 11 of Latin American countries would be able to reach it following the current tendencies (ECLAC, 2019).

Unpaid care and household work condition and support the labor market, local economies and overall social and individual wellbeing; the low participation of women in income earning opportunities contrasts with their role in unpaid activities (ECLAC, 2019). Most of these work goes unregistered in official statistics and has no influence in decision making processes, country surveys highlight that the economic contribution of unpaid work could represent between 15% and 24% of the GDP depending on country (ECLAC, 2019). Despite important progress in women's human capital and sociodemographic changes in society, gender differences in labor markets continue to be an unjustifiable form of inequality; for every hour worked women receive work income that is, on average, 17% lower than those of men of the same age, education, presence of children in their home, condition of rurality and type of work (OIT, 2019). Women have gained greater presence in the paid labor market but it is usually in those segments that present the greatest flexibility: part time, informal, self-employment, casual (OIT, 2019). CA and SA are highly urbanized with increasing urban population accounted for 78.8% in 2010 of the totals (ECLAC, 2016). The vast majority of the poor are living in urban areas while urban extreme poor is increasingly becoming more relevant (Rosenzweig et al., 2018; Dodman et al., 2019), with those who live in informal settlements and work in the informal economy form a critical part of each city's economy (Satterthwaite et al., 2018). One of the greatest challenges for climate change adaptation is how to build resilience for the billion urban dwellers who are estimated to live and work in what are termed informal settlements (Satterthwaite et al., 2018). IPCC's Third, Fourth and Fifth Assessments from Working Group II recognized the higher risks associated with poor living conditions, substandard housing, inadequate services, location in hazardous sites due to lack of options and the need to work more strongly on strengthening governance structures involving residents, community organizations amongst others (Wilbanks et al., 2007; Revi et al., 2014).

Poverty and disaster risk reduction interlinked with climate change adaptation avoiding competition for resources and leakages to corruption processes may share a focus on identifying and acting on local risks and their root causes, even if they have different lenses through which to view risk (Satterthwaite et al., 2018). Cities are more propense to act on climate change when they identify co-benefits to solve local problems (Margulis, 2016).

Extreme temperatures are a major health concern around the globe, especially in the face of climate change [IPCC, 2014; IPCC, 2018]. Studies have found that most heat related mortality occurs indoors and in permanent homes, while indoor exposure to heat waves results from non- functioning AC in situations where mechanical cooling is needed, pointing to the critical need of resilient power infrastructure to avoid power outages due to system overheating and equipment failure during heatwaves (Sailor et al., 2019). With 1.5ºC warming a significant increase in heat wave magnitude and frequency is expected in Africa, South America and Southeast Asia, and under 2ºC scenario heat waves will become more intense and more frequent (Dosio et al., 2018). Studies on urban climate in metropolitan area of Rio de Janeiro point to the role of vegetation coverage as temperature regulator, and contrary to suburban development in Europe and Northern America, show that in countries such as Brazil suburban areas usually have little or no tree coverage, lack of adequate housing and sanitation, therefore often registering temperatures as high or even higher than in central city areas (Peres et al., 2018).

[PLACEHOLDER FOR SECOND ORDER DRAFT: Hazards such as Tsunami, storm surges coastal flooding, coastal erosion and ocean acidification with loss of fisheries which can be devastating for the coasts and their communities (particularly Central America, Peru, Colombia, Equator and Chile); Marine ecology and oceans (including degradation) vis-a-vis poverty and sustainable development]

In the delta of the Amazonas, most small urban centres are expanding over land highly vulnerable to sea level rise and exposed to floods, landslides, and climate and socioeconomic shocks that combine with lack of basic service and infrastructure provision, and adequate income levels (Mansur et al., 2016; Almeida et al., 2018).

For the Amazon region, poverty levels and inequality are higher than in any other parts of the region due to marginalization and isolation, it is considered a hotspot for climate change impacts and the uncertainty is increasing leading to high risks in the future (Filho et al., 2016; Lapola et al., 2018; Zavaleta et al., 2018). Amazon livelihoods are been affected during extreme events (Pinho et al., 2014). The projections at 2°C shows that major river levels will be reduced by ~25%. Water availability and security, health, infrastructure, energy access, education, and food production with implications for nutrition security affecting by consecutive years of extreme droughts and flooding and reducing the ability of local population to cope (Brondízio et al., 2016; Zavaleta et al., 2018). Climate change is affecting the health of human populations where indigenous Amazonian populations have a high reported prevalence of nutritional deficiencies (Zavaleta et al., 2018). Nelson et al. (2016) analyzed the response of communities in Northern Brazil to drought events in 1997–1998 and 2011–2012 during those period, poverty levels declined due to an aggressive government-implemented poverty reduction program. However, the population risk management strategies did not change, and their food insecurity increased during the more recent drought event. Authors concluded that poverty reduction alone is not enough to improve population adaptive capacity, as people will not necessarily invest in risk management activities when added income is available.

The region relies heavily on the agricultural sector. An important percentage of Argentina's exports are related to primary products and manufactures of agricultural origin, given projected impacts of climate change, the vulnerability of its productive system is potentially high (CEPAL, 2014). [PLACEHOLDER FOR SECOND ORDER DRAFT: Other cases, El Orinoco]. This will affect country development opportunities directly or indirectly impacting social vulnerability levels.

An increase in land surface affected by periodic floods associated with the phenomena of 'Sudestadas" in the estuary of the Río de la Plata is expected. Without greater efforts in adopting structural and non-structural measures to counter the negative impacts of floods, it is expected that social vulnerability will increase in north and south Metropolitan Buenos Aires, specifically along the margins of the Matanza – Riachuelo and Reconquista rivers ((CEPAL, 2014; Magrin et al., 2014). Increases in average Rio de la Plata water levels will also likely affect different components of public services (sewerage, thermic plants), transportation system, electricity distribution, road and railroad networks along the coast (CEPAL, 2014). In the region of Comahue, reduced flows in the Neuquen and Limay rivers in the periods 2020, 2030, 2050 and 2070 will affect electric energy provision, water provision for urban areas and irrigation (CEPAL, 2014).

Social policies, poverty reduction and inequality are articulated with difficulty to the reduction of daily or chronic risk (Allen et al., 2017) being exacerbated by climate risks with the most vulnerable populations suffering. Chronic and every day risks associated with poor access to infrastructure, services, incomes, housing, tenure, education, security, location and poor-quality environment, networks and having a voice are often exacerbated and generate new unknown risks by climate change (Satterthwaite et al., 2018)). Risks are seldom distributed equally and highlight socioeconomic inequalities and governance failures (Rasch, 2016). Existing inequalities in the provision and consumption of services such as drinking water and sanitation are bound to be exacerbated by future risks and uncertainties around water availability and vulnerabilities associated with climate change scenarios (Miranda Sara et al., 2017), more so in cities located in low elevation coastal zones where climate related extreme weather patterns are expected to impact more strongly (McGranahan et al., 2007; Villamizar et al., 2017). In Lima, with only 9 to 12 mm of rain per year, regional vulnerability to climate change is compounded with socio economic inequalities in water distribution reinforcing negative loops (Miranda Sara and Jameson, 2016). Those with no water connection pay ten times more than those with connections but consume on average less than 25 lpcd. (Miranda Sara et al., 2017). Residents are often caught in 'risk traps', accumulated cycles of everyday risks and small-scale disasters, sometimes in spite of or as a result of investments undertaken to improve substandard living conditions. These processes and conditions perpetuate risk accumulation cycles, calling for a more detailed understanding of what constitutes and causes risk (Allen et al., 2017).

Although Peru was declared an upper middle-income country in 2015, the 2017 Coastal Niño affected it severely, even 2019 provides better results, the government recognized that poverty increased again in 2018 with 380 thousand new poor, 1,3% GDP loss and only 2% economic growth, without considering the deaths and destruction (French and Mechler, 2017). The costly reconstruction is still undergoing, and it is calculated near 7 thousand million US dollars. It was after the disaster, in 2018 that the Peruvian government passed the Legal Framework for Climate Change, Law No. 30754, a unanimous decision in Congress, to assure the integration and transversally of the climatic component in public policies and investment projects, being its Code in the process of approval.

There is urgency for social systems to better respond to climate related risks and increase their adaptive capacity (Lemos et al., 2016). In responding to climate change, communities deploy a combination of assets that define their overall adaptive capacity, this involves both interventions that address climate related risks (specific capacity) and structural deficits (generic capacity), synergies and a strategic balance between both is necessary (Eakin et al., 2014; Lemos et al., 2016). The institutional context of adaptation can undermine one form of capacity with repercussions on the other compromising overall adaptation and sustainable development (Eakin et al., 2014).

Tensions and conflicts may result from differing perceptions and knowledge on vulnerabilities and risk which can hinder the acceptance of adaptation measures and implementing stronger adaptive or preventive actions (Miranda Sara and Jameson, 2016). Adaptation measures have tradeoffs that need to be acknowledged and acted upon, most importantly by developing the capacity to convene discussions that draw in all key stakeholders and commit them to do things differently, starting by co-producing knowledge that aids decision making (Almeida et al., 2018; Hardoy et al., 2019). Adaptation measures by themselves cannot address the causes of increasing climate related risks (Dodman et al., 2019).

Adaptation initiatives to reduce poverty, improve livelihood and achieve sustainable development in the region range in scale and scope, from planned and collective interventions to autonomous and individual actions, they include structural measures and grey infrastructure to non-structural and use of nature-based solutions. Some are community led and bottom up, others are government led or a combination of both. There is experience in a great variety of measures, including developing financial instruments to strengthen and protect livelihoods and assets such as collective insurance schemes and payments for Ecosystem Services (PES) as well as early warning systems (Dávila, 2016; Hardoy and Velásquez, 2016; Lemos et al., 2016; Porras et al., 2016); working directly on improving production aspects such as seed distribution programs, promoting access to irrigation, implementing better agricultural practices, combining livelihoods strategies, accessing and using climate forecasts, using appropriate construction techniques and small scale interventions on neighborhood layout, circulation and safety, integral urban upgrading initiatives that involve housing, secure tenure, risk reducing infrastructure and services, territorial and urban planning, developing regulatory frameworks, and incorporating nature based solutions (Stein and Moser, 2014; Hardoy and Mastrangelo, 2016; Almeida et al., 2018; Desmaison et al., 2018; Satterthwaite et al., 2018).

An analysis of three mitigation (avoided deforestation) projects in Belize by Kongsager (Kongsager and Corbera, 2015) showed that adaptation outcomes were rarely reported by project beneficiaries particularly with respect to improved livelihoods and adopted adaptation practices in agriculture. The authors conclude that the current carbon market approach is unlikely to deliver both mitigation and adaptation as implementers do not have enough resources, capacities or incentives to deal with both dimensions. Carbon standards will have to make adaptation mandatory as part of the approval process for mitigation initiatives.

12.6 Integration of Impacts and Adaptation

[PLACEHOLDER FOR SECOND ORDER DRAFT]

12.7 Case Studies

12.7.1 Nature-Based Solutions in Quito, Ecuador

Nature-Based Solutions- NbS is related to the maintenance, enhancement, and restoration of biodiversity and ecosystems as a means to address multiple concerns simultaneously (Kabisch et al., 2016). Ecosystem-based Adaptation- EbA can be seen as a type of NbS deployed in response to climate change vulnerability and risk (Greenwalt et al., 2018), combining the objectives of reducing the vulnerability of human and natural systems (IPCC, 2014).

Cities are among the locations most vulnerable to climate change (Hardoy and Pandiella, 2009; CAF, 2014; IPCC, 2014) facing a challenge to promote resilience (Allen et al., 2018), especially in Latin America, with around 81% of urban population and high rates of poverty (30%) (ECLAC, 2019) (see Section 12.5.5).

The Municipal Quito District- MQD, in Ecuador, also follow this pattern of urbanization reaching over 3 million inhabitants by 2020, of which 80% will live in urban areas (DMQ, 2016). Its population has increased four-fold over the last 30 years and a combination of problems (economic crisis, debt, lack of government services, lack of planning) has led to legal and illegal occupation of slopes (Hardoy and Pandiella, 2009; Cuvi, 2015). Over 670,000 people live in high-risk areas, and overall, 43.5% of its inhabitants live below the national poverty line (Obermaier, 2013).

Climate change has increased mean temperatures in Quito between 1.2°C and 1.4°C over the last 100 years (PNUMA and FLACSO, 2011), while the precipitation presented a general tendency to decrease in 8% of the annual rates (DMQ, 2016). The possible increase in the average temperature, as well as the greater presence of periods of drought, can increase ongoing glacier retreat and affects the water storage capacity of the *páramos* ecosystems (Bradley et al., 2006), reducing clean water supply and irrigation capacity due to high reliance on glacier basins. These extreme weather events combined with intensive land use change in recent years have affected infrastructure, human settlements, agriculture, and forests, while the loss of glaciers and *páramos*, threatens food security and water and hydropower supplies over MQD (Municipio del Distrito Metropolitano de Quito, 2012; Grafakos et al., 2018). Another important climate change effect is about solar radiation and heat island effects since Quito is one of the highest cities in the world and its population is exposed to extreme levels of UV radiation almost half of the year (DMQ, 2016).

Given the great climatic vulnerability, the city of Quito was one of the pioneers in Latin America regarding the adaptation to climate at the municipal level (Obermaier, 2013). In 1993, the municipal law of the city was reformed to create responsible authorities for adaptation, including protected areas and landscape management as an important action. As a result, Quito gradually implemented new risk management and adaptation strategies, including hillside management (1997), intensification of glacier monitoring (1998), flood control (1999), watershed protection fund- FONAG (2000), urban agriculture- AGRUPAR (2002), *quebradas* restoration (Obermaier, 2013).

The commitment of local actors to adaptation, including public, private and civil society, is high; problems have been identified jointly, and innovative actions are being implemented in risk management, recuperating degraded urban forests, climate-resilient agriculture, managing water resources and institutional capacity building (Obermaier, 2013). Two of them will be described here as an important baseline and succeed EbA project. The first one is FONAG, that was established to facilitate, in alliance with local institutions and actors, the protection of the water basins that supply 75% of the MDQ water consumed. It was established through a financial mechanism that executes programs and projects of conservation, climate vulnerability, ecological restoration and environmental education, for a new water culture and integrated water resources management (Krchnak, 2007; Kauffman, 2014; DMQ, 2016). It was an innovative in pioneered the use of trust funds in a voluntary, decentralized mechanism for financing watershed conservation. FONAG lessons learned in community-level watershed management has inspired several other water funds (Kauffman, 2014).

The second one is AGRUPAR, that was launched as a public initiative to provide training, marketing support, technical, agricultural and organizational assistance, among others (Thomas, 2014; Cuvi, 2015; Rodríguez-Dueñas and Rivera, 2016). AGRUPAR is focused poorer communities that live further away from the city's basic service net and provides seeds and seedlings, as well as free spaces within the city where local farmers can sell their products (Rodríguez-Dueñas and Rivera, 2016; Clavijo Palacios and Cuvi, 2017). Quito's goal is to produce 30–40 percent of its food locally over the next few decades, mostly in the region's rural areas (UCCRN, 2018). Currently, AGRUPAR has more than 1,300 gardens in operation that have had direct impacts on nutrition, on the generation of work for women, on the production of healthy food for the population and the promotion of agrobiodiversity, the reduction of runoff and carbon footprint, the recycling of organic waste, among others (Cuvi, 2015).

These programmes moved away from emergency response to integrated disaster prevention and more recently to Climate Change Strategy (2009) and Climate Action Plan (2012, 2015), that included mitigation and adaptation in an integrated way and participatory governance. This gradual implementation of EbA programmes was crucial to build capacity before engaging in a more strategic approach (Obermaier, 2013). Protected Areas, Natural vegetation restoration, Landscape and watershed management, Urban agriculture and Green Urban areas are key components to address the most important effects of climate change in the urban and peri-urban area of DMQ.

[PLACEHOLDER FOR SECOND ORDER DRAFT: Address additional challenges, such as air pollution, solar radiation, greenspace, urban heat island]

12.7.2 Anthropogenic soils, an option for mitigation and adaptation to climate change in Central and South America. Learning from the "Terras Pretas de Índio" in the Amazon

Amazon Dark Earths (ADEs), also known as "Terras Pretas de Índio", are anthropogenic soils derived from the activities associated to settlements and agricultural practices of pre-hispanic societies in the Amazon (Woods and McCann, 1999; Lehmann et al., 2003; Sombroek et al., 2003). Most of the ADEs identified so far are 500 to 2500 years old (de Souza et al., 2019). According to Maezumi et al. (2018a) polyculture agroforestry allowed the development of complex societies in the eastern Amazon around 4500 years ago. Agroforestry was combined with the cultivation of multiple crops and the active and progressive increase in the proportion of edible plant species in the forest, along with hunting and fishing. The formation of ADEs, as a result of these activities, provided the basis for a food production system that supported a growing human populations in the area (Maezumi et al., 2018a).

Amazon Dark Earths are the result of the accumulation and incomplete combustion of waste materials such as ceramic artifacts and organic residues from harvest, weeding, food processing (including cooking) and other activities (Lima et al., 2002; Hecht, 2003; Kämpf et al., 2003). ADEs are characterized by their increased fertility in relation to adjacent soils; with high contents of organic carbon (C) (mainly as charcoal) as well as inorganic nutrients, especially phosphorus (P) and calcium (Ca); and high Carbon/Nitrogen ratios. They also exhibit high cation exchange capacity (CEC) and moisture retention among others properties (Hecht, 2003; Kämpf et al., 2003; Falcão et al., 2009). Charcoal content is a key indicator of pre-hispanic fire activity and sedentary occupation, which is evidence of the anthropic origin of these soils (Hecht, 2017; Maezumi et al., 2018b).

Accumulation of organic residues and low intensity fires management are recognized as key elements for ADEs formation. ADEs originated around settlements show a relatively high density of ceramic artifacts and are named *Terras pretas*. They present a higher content of Ca and P than those originated from agriculture activities which are known as *Terras mulatas* (Hecht, 2003).

There is a robust and growing body of research from different disciplines that gives high relevance to ADEs in the region. It has been shown through archaeological and paleoclimatic data that Amazonian societies which based their agricultural management on "Terras Pretas de Índio", were more resilient to the changing climate due to increased soil fertility and water retention capacity (de Souza et al., 2019). Additionally, low organic carbon degradability over long time periods, associated with high contents of charcoal or pyrogenic carbon, makes these soils an important C sink (*high confidence*) (Lehmann et al., 2003), which is particularly relevant in an area like the Amazon, that could change from a net carbon sinks to a net carbon source as a consequence of anthropogenic climate change (Maezumi et al., 2018b).

The indigenous agricultural practices which originated ADEs are thought to be associated with a more sedentary agricultural model than the current slash and burn and shifting cultivation practices. Although this is a controversial topic, as the precise definition of slash and burn and shifting cultivation is presently under discussion (Hecht, 2003); several present-day local and indigenous agricultural practices, including in-field burning and nutrient additions from food processing and residue management, have been recognized as promoting high organic carbon and nutrient soil contents similar to the ones found in ADEs (Hecht, 2003; Winklerprins, 2009).

At present, ADEs are estimated to cover up to 3.2% of the Amazon basin and are highly valued for their persistent fertility, becoming a key resource for sustainable agriculture for Amazon communities in a climate change context) (Altieri and Nicholls, 2013; Maezumi et al., 2018a; de Souza et al., 2019). Based on the lessons learned from the Terras Pretas de Índio, some researches have proposed the development of technologies to promote a new generation of anthropogenic soils (e.g., Kern et al. (2009); Lehmann (2009); Schmidt et al. (2014); Bezerra et al. (2016); Kern et al. (2019)). [PLACEHOLDER FOR SECOND ORDER DRAFT: More information about the different technologies proposed to the development of new generation of anthropogenic soils will be provided]

To preserve the knowledge and practices associated with these soils is vital for sustainable agriculture in a climate change scenario in the Amazon. It will contribute to the preservation of valuable indigenous knowledge and has the potential to contribute to the development of new adaptation and mitigation technologies and solutions.

12.7.3 Towards a Metropolitan Water-related Climate Proof Governance (re)configuration? The case of Lima, Perú

The Lima-Callao Metropolitan City (from now on, just Lima) capital of Perú shows interesting lessons learned on water related climate proof governance reconfiguration facing recurrent water related climate disaster: a) when disasters affect poorest and richest populations, actors can prioritize the integral city´s resilience and development and work in a coordinated and *concertated* manner across institutional levels and geographical scales even having different ideas, discourses and power, recognizing that no one single actor has enough power; b) water related climate change scenarios require comprehensive, transverse, multi-sectoral, multi-scalar and to include multiple types of actor´s knowledge (expert, tacit, codified and contextual embedded) managing the tensions and even conflicts when some knowledge is not being shared, exchanged, restricted, not recognized nor included and lower risk perception and higher risk tolerance are present (Miranda Sara and Baud, 2014; Siña et al., 2016; Miranda Sara et al., 2017).

Lima is the second driest city in the world is highly vulnerable to drought and heavy rain peak events in the surrounding Andean highlands associated to climate change. Located in the Pacific Coast, has more than 10 million inhabitants, suffering both flooding, mudslides disasters and water stress, together with unequal distribution and water pricing, where one million inhabitants suffer lack of water connections (Miranda Sara, 2017; Montero and García, 2017; Vázquez-Rowe et al., 2017; Schütze et al., 2019). This is related to a lack of city long-term planning integrated with water and risk management, climate risks and climate change scenarios being ignored by residents, private sector and decision-makers alike, particularly when budget allocation for preventive actions was necessary (Miranda Sara and Jameson, 2016; Allen et al., 2017).

Lima is being affected by ENSO, Coastal Niño and heavy rain peak events being held more frequently since 1970, 1987, 1998, 2012, 2014, 2015 and 2017, see (Miranda Sara and Jameson, 2016; Vázquez-Rowe et al., 2017). In 2014, the Water Company (SEDAPAL) together with the Lima Metropolitan Municipality (LMM), ANA and other organizations agreed a Lima Action Plan for Water (Schütze et al., 2019) on basis of which SEDAPAL approved the Water Master Plan (SEDAPAL, 2014) and right after the COP20 held in Lima the same year, the Lima Municipality (LMM) approved the Climate Change Strategy developing adaptation and mitigation measures (Miranda Sara and Baud, 2014). Those instruments were based on technical and scientific action research within inclusive, interactive and iterative *concertation*[1] multi actor processes.

The 2015 municipal elections followed by national government elections shifted Lima's and Peru's political power to parties associated to climate deniers at a high cost to the city infrastructure and housing, along rivers, ravines and slopes, flooded by heavy rain peak events, *huaycos*[2], potable water cuts (Vázquez-Rowe et al., 2017) and vector-borne diseases affecting particularly the poorer but also richer inhabitants [PLACEHOLDER FOR SECOND ORDER DRAFT: to include table with data]

In 2017, the Peruvian government passed the Framework Law for Climate Change, Law No. 30754, a unique unanimous political decision in Congress, to assure the integration of climate change concerns in public policies and investment projects, being its Code in the process of approval after an inclusive process of prior consultation with Peruvian indigenous communities. This Law defines Local Government mandates on investing in adaptation and mitigation measures, localizing climate change Action Plans [PLACEHOLDER FOR SECOND ORDER DRAFT: To include the District Local Governments developing their Climate Action Plans and the Reconstruction with Changes Law with sanctions of jail to those putting in risk people or infrastructure]

2019 elections have brought new local authorities to Lima whom half of them are already developing their Climate Action Plans to implement the Metropolitan Strategy and LMM major has recently become the South American Major´s representative at the Global Covenant of Majors Steering Committee. It is after the 2017 disaster and meanwhile dealing with corruption scandals associated to Lava Jato, agreements are being built up for collaborative action, even within those actors with diverse water and development discourses as well as different territorial and city visions, whom are contributing from different territorial scales and government levels.

[PLACEHOLDER FOR SECOND ORDER DRAFT: To prepare a table or map]

Peru is both a highly centralized and institutionally fragmented country, Lima Local Governments do not manage strategic services such as potable water nor river basin management (in hands of Housing and Agriculture Ministries respectively); and, the Metropolitan City is managed by 2 Regional Governments, 2 Provincial Local Governments and 51 District Local Governments each one with democratically elected authorities.

Peruvian institutionalized culture of participation has led to a broader concept of *concertation*, with practices of collaborative planning which allow actors to build up socially supported agreements, decisions and take actions without losing their principles. These processes are being applied again after 2017 disaster to reduce risks, to adapt and anticipate uncertain and unknown futures, introducing climate change concerns under a complex political and institutional environment surrounded by corruption scandals. *Concertation* provides inclusive spaces in which more and more types of actors could meet to debate issues, negotiate solutions and develop proposals for action. "*Concertation*"[3] can be seen as another form of social learning.

The Climate Action Plan has reopened the Climate Change Technical Group of the Environmental Commission lead by the LMM; the recently formalized River Basin Council Management integrated by multiple actors across the three river basins in the Metropolitan City (Chillon, Rimac and Lurin) is developing the River Basin Management Plan lead by the National Authority of Water (ANA) and the new Lima Urban Development Plan (LMM) is being developed with the support of a High level Consultation Group. These processes are contributing to build trust and motivate actor´s to undertake joint planning to pursue a climate proof water related governance (re)configuration.

[1] FOOTNOTE: *Concertación* processes involve a variety of actors and have become mandatory in various contexts.

[2] FOOTNOTE: Peruvian word for land and mudslides

[3] FOOTNOTE: "Key characteristics are learning-by-doing, combined with the construction of knowledge through various social networks. The latter implies the validation (or contestation) of the knowledge of a variety of participating actors, and a highly sensitive and complex process of dialogue–negotiation– *concertación*–conflict management and consensus building (or not). Such processes are constantly evolving cycles" (Miranda Sara and Baud, 2014).

Actors can build up agreements, even having different discourses, bringing in knowledge from different territorial scales and institutional levels and being able to initiate collaborative actions over time, providing inputs for scenario building on cities towards future forms of water and climate change adaptations; however, such processes may include strong discussions, conflicts and the recognition of other's discourses, and require to include all types of knowledges (expert, codified, tacit and contextual) in order to build up scenarios capable of "visualizing" and anticipating what might happen when conditions change towards the future. Those processes require democratic, transparent and decentralized institutions, providing strong mandates and political will supporting them, so the views of the poor, vulnerable are included and can make themselves heard.

Opportunities for a socio-political and technological water governance reconfiguration are emerging based on socially supported agreements (Miranda Sara and Baud, 2014). Even one dominant actor network configuration sets discourses, rules and implementation, power relations are being renegotiated and emerging new networks including socio-ecological concerns and wider communities, attempting that the views of the poor, vulnerable and excluded can make themselves heard, even their power remains limited. The water governance configuration faces the paradox that current water demands of all users combined may no longer be feasible within ecological limits and future climate change consequences.

12.7.4 Strengthening water governance for adaptation to climate change: managing scarcity and excess of water in the Pacific Coastal area of Guatemala

Guatemala experiences high climate inter-annual variability now increased from the effect of climate change (INSIVUMEH, 2018; Bardales et al., 2019). Impacts on human settlements, agriculture and ecosystems result from both excess and reduced precipitation. Guerra (2016) argues that deficient integrated water resource management in the country is the main reason for those impacts. A case in point is that of rivers Madre Vieja and Achiguate where an intense El Niño event triggered dryer conditions and, in turn, a crisis and conflict that reached national proportions. Progress in local water governance helped to solve that crisis and contributed to tackle challenges posed by reduced precipitation and flood risk in southern Guatemala.

The ENSO event that started in November 2014 and ended in July 2016 (CIIFEN, 2016) has been the most intense since records commenced in 1950 (NOAA, 2019). Its effects were felt in different parts of the world and, Guatemala and the rest of Central America experienced an intense water scarcity due to a significant reduction in rainfall (IICA, 2015; Scientific American, 2015). River flow in the dry months is related to precipitation levels in the previous rainy season and thus, ENSO has an effect on river flow rates. Two of the main rivers in the Pacific coast of Guatemala, Madre Vieja and Achiguate, dried out completely at the beginning of 2016, triggering a nearly violent local conflict that caught attention at the national level (Guerra, 2016; Gobernación de Escuintla et al., 2017). In addition to the severe drought, the rivers dried because of over-extraction by multiple users (60 in the case of Madre Vieja). This had happened before to a lesser extent in the last 20 years during the critical months of the dry season. Lack of regulation, coordination mechanisms, information, and other elements of water governance was the root cause of the problem, exacerbated by the drier conditions during the intense El Niño resulting in the intensification of an existing conflict (Guerra, 2016).

Roundtables were set up to foster dialogue between numerous stakeholders including communities, agro-export companies, governmental organisations, municipalities, all led by the local governor (Gobernación de Escuintla et al., 2017). Agreements included: to keep a minimum of the rivers flowing all the way to the sea; to set up a monitoring and verification system for levels of river flow; and to restore riparian forests. A system was set up to monitor river flow in different points along the rivers on a daily basis in the dry season using a simple Whatsapp-based system to communicate the warnings and monitor compliance. Four years on, the rivers had not dried out and conflict was kept to a minimum. Rural communities can use rivers for recreational purposes and for fishing all year round, whilst plantations (large and small) can use water for irrigation (rationally) and keep producing. Similar schemes and interactions started happening in other rivers in the Pacific coast of Guatemala, with positive results, particularly keeping the rivers flowing all through the dry season as can be seen in the report of river flows for years 2017, 2018 and 2019 (ICC, 2019b).

A key actor in the improvement of water governance has been the Private Institute for Climate Change Research (ICC). This is a unique initiative that was created in 2010 and is funded primarily by the private sector of Guatemala to help the country advance in climate change mitigation and adaptation (Guerra, 2014). The institute works alongside local governments, communities and private companies in several topics apart from integrated water management. Its role is merely technical-scientific being in charge of the water monitoring system, generating data on weather and hydrology, and providing support to other stakeholders.

Local governance was also essential for the implementation of flood risk management actions. Guerra et al. (2017) explained how impacts were significantly reduced in the Coyolate river watershed, also in the Pacific coast of Guatemala, thanks to flood protection that was designed and implemented in a technical and integrated manner. This was a result of strong and active participation of local communities, companies and the local municipality who demanded the central government to invest effectively. The stakeholders provided some resources (financial and in-kind) and inspected the works. Some flat areas of the lower Coyolate watershed used to flood annually causing economic damage for communities. The areas covered by flood risk measures have not flooded which has avoided losses as well as created conditions for investment to come and create jobs, improving life conditions for locals. Other processes of participation and interaction between the authorities, the private sector and communities have taken place in other watersheds for planning, action and investment for flood risk management. The ICC has played a role by studying flood-prone areas, building capacities in communities, fostering public-private coordination mechanisms, and providing much-needed technical assistance to local governments (ICC, 2019a).

Although some may argue that water governance is in the realm of development, it has made contributions in reducing direct and indirect impacts of climate events and therefore, it can be seen as a key element for climate adaptation.

12.8 Knowledge Gaps

[PLACEHOLDER FOR SECOND ORDER DRAFT: text to be developed on the knowledge gaps assessed and listed below:

- In CA, gaps are identified in further research on modelling ecological responses to climate change, regional climate models and simulating climate influences on crop yields and diseases.
- There is a significant lack of climate data for CA and remote regions of SA, making decision making as well as the adoption of adaptation measures very difficult.
- There is limited literature on severe risks and limited specific and explicit consideration of risk drivers in the region.
- There is an information gap to assess climate change-influenced migration processes in CA and SA.
- There is still significant gap on identifying limits to adaptation in CA and SA.
- There is a knowledge gap about the likely impact of climate change on NES biodiversity.]

12.9 Conclusion

Central and South America is a broadly heterogeneous region in terms of topography, ecology, demography, economy, culture and climate. Its landscapes range from ice in Patagonia to the Amazon forest (one of the major carbon sinks in the planet and determinant of precipitation balance in the region), through the Atacama Desert and the tropical lowland and mountain forests of CA. The topography includes extensive areas of low elevation coastal zones (highlighting Venezuela Guiana and Suriname), vast plains in contrast to the Andean mountains. It includes indigenous populations (some still isolated as in the Brazilian Amazon) and global megacities, such as São Paulo, Buenos Aires and Rio de Janeiro. The region relies on a strong agrarian economy in which small producers and large industries participate, but also large industrialized urban centres, oil production and mining. It is the region with the largest social inequality on the planet.

This heterogeneity has important consequences on delaying the assessment of exposure, vulnerability and climate change impacts, particularly when combined with scientific knowledge gaps, in process climate scenarios downscaling and still insufficient reporting of climate change adaptation action in the region. The adaptive capacity of the region is equally complex to assess and deeply impacted by weak governance systems, consistent deficits of infrastructure to sustain economic activities and basic needs, plus the financing challenge.

Many extreme events are already impacting the region and projected to intensify, such as warming temperatures and dryness in NES, sea level rise, coastal erosion and ocean and lake acidification in SWS and NWS, or increasing rainfall in SES. The region in also threatened by a general warming and increasing frequency and severity of droughts, with associated decrease in water supply, increase in landslides and flash floods.

Both CA and many regions of SA were recently exposed to severe droughts. In CA 10.5 million people are living in a so-called dry corridor, a region with an extended dry season. A water crisis in Brazil affected the major cities of the country between 2014 e 2016, becoming more frequent since then. Severe droughts have also been reported in Paraguay and Argentina. In contrast, the urbanised areas of NSA are highly exposed to extreme floods (41% of urban population in the Amazon Delta and Estuaries). The coral reefs in coastal CA and NES are extremely valuable for ecological and also touristic reasons and are exposed to increased sea temperatures. The Cerrado and the Atlantic Forest (two important biodiversity hotspots where ca. 72% of Brazil's threatened species can be found) are exposed to different hazards due to climate change. Many coastal areas and its huge urban population and assets are exposed to sea level rise.

The region is highly vulnerable. The Amazon, Cerrado and Pantanal biomes have documented several highly or moderately vulnerable protected areas. CSA region is one of the most urbanized in the planet and its cities are vulnerable for many reason, being inequality a basic driver of it. In the central area of Brazil, for example, Cuiabá and Brasília are expected to experience an expressive increasing on the number of deaths related to heatwaves up to 2080. The high poverty levels in NES booster its increasing vulnerability to droughts, both in cities and rural areas, where people already suffers from natural scarcity of water. NEB's coral reefs have its vulnerability to warming oceans increased by anthropogenic disturbances (urban development, marine tourism, overexploitation of reef organisms, and oil extraction).

Many impacts in the economy are expected from climate change. CA may experience 5.4% economic losses in the agricultural sector by the years 2050 and 19.1% of the GDP by 2100 under A2 climate change scenario. Subsistence farmers and urban poor are expected to be the most impacted by droughts and variable rainfall in the region. Brazil is also expected to experience GDP losses, particularly due to desertification in NES, but also by severe drought periods in SES. The increasing on water scarcity in NES will impact on food security, human health and well-being. The biomes of the region are already being impacted and expected to suffer great harmful changes. The conversion of Eastern Amazonian rainforest to savannah due to extended periods of droughts is expected to increase in RCP 8.5 scenario, which brings severe biodiversity loss and dysfunctional ecosystem services, affecting food production, climate regulation, carbon sink and hydroelectric capacity. The extinction of 657 plant species in the Cerrado has been documented due to the combination of land cover change and concurrent climate change. The impacts of the many landslides and floods in SES affect mainly the urban poor neighbourhoods and are responsible for the majority of the deaths related to natural disasters in the country. Sea-level rise and intense storms surges in the large coast of SES is expected to impact the tourism and industry in general.

Climate change is so threatening several systems (glaciers in the Andes, coral reefs in Central America, loss of the Amazon forest) in CA and SA, which are already approaching critical conditions under risk of irreversible damage. Many CA countries are already ranked as the highest risk level worldwide due to its exposure combined to high vulnerability and low adaptive capacity. Facing those risks will require good knowledge of its drivers and adaptation options. However, there is still limited literature focusing on severe risks and risk drivers for the region. In CA and many areas of SA the lack of climate data and proper downscaling are clearly challenging the adaptation process.

In the way of facing those many challenges, all countries in the region but French Guyana have submitted its Nationally Determined Contributions (NDC) and many have published its National Adaptation Plans (NAP), establishing priorities and formulating its own policies to cope with climate change.

Many adaptation initiatives have been assessed in the many sectors. Action focused on reducing poverty and improving livelihood has increased in planned and autonomous initiatives, community led or government led or the combination of both, engineering or nature based solutions (NDS). The focus on inclusion and enrolling of the full range of stakeholders in adaptation processes has shown good results in the region. NBS are becoming more frequent. There are still only few reported experiences of specific adaptation efforts on ocean and coastal areas in the region comparing to the increase of those initiatives world. Adaptation focused on water management has been adopted, being the most frequent in mountains, as in the Andes. Disasters reduction solutions have been also applied, such as Early warning systems, with many examples in South America and some in Central America. Those so many and diverse initiatives are still poorly reported and evaluated in scientific literature, what challenges its assessment and improvement.

Frequently Asked Questions

[START FAQ 12.1 HERE]

FAQ 12.1: How is food production being affected by climate change in Central and South America?

[PLACEHOLDER FOR SECOND ORDER DRAFT]

[END FAQ 12.1 HERE]

[START FAQ 12.2 HERE]

FAQ 12.2: What are the main impacts expected in Central and South America urban areas due to climate change?

Cities are vulnerability hotspots to climate change as they concentrate people, assets, and economic activities that are particularly vulnerable to climate change. They are particularly vulnerable due to their geographic location, and socio economic, demographic and environmental characteristics which make them highly sensitive to the impacts of climate change, even though the regions contribution to climate contributors at a global scale.

Latin America is the second most urbanized region in the world after North America (81% of its population). The number of cities has multiplied by six in fifty years, there are 2000 cities with more than 20,000 inhabitants and 129 with more than 500,000 inhabitants concentrating half of the region's urban population (222 million), while 65 million reside in megacities. The region has practically concluded its urban transition and urban population is growing at a lower rate, however new trends emerge such as migration between cities, increase in the number of secondary cities, and creation of megaregions and urban corridors. Unsolved urban pressures have resulted in an increase of the urban residential informal sector, going from 6% to 26% from 1990 to 2015, with importance in cities with less than 500,000 inhabitants.

Coastal areas increasingly concentrate a greater number of urban centres. Researchers indicate that between 3 to 4 million inhabitants will be exposed to coastal flooding from sea-level rise by the end of the century for either of the RCPs considered in SA alone. A study on population in cities with more than 100,000 people between 1945 and 2014 shows the concentration of small cities close to fragile ecosystems such as bays, estuaries and mangrove forests, reinforcing a historic trend, thus resulting in higher concentrations of population and economic activities along the coast. The same study highlights that the number of coastal cities (less than 100 km from the coast) went from 42 to 420, a tenfold increase between 1945 and 2014. The degradation of coastal ecosystems limits their ability to mitigate risks and provide essential ecosystem services, port and tourism as well as income opportunities.

Impacts in cities are associated with an increase of extreme events (heavy rain, drought, storms), increases in temperatures and heat waves, glacier retreat and sea level rise, all with important variations across the region and a strong influence of ENSO. Cities are increasingly dealing with floods, landslides, storms, tropical cyclones, water stress, fires, vectors, etc. with consequences over infrastructure, activities, built and natural environments and the populations overall wellbeing.

Glacier retreat will affect water run off towards the Pacific and water provision in Metropolitan cities that rely on rivers that originate in the high Andes such as Lima, La Paz, Quito and Santiago. Being Lima the second driest city in the world is highly vulnerable to drought and heavy rain peak events associated to climate change.

Metropolitan Santiago has to prepare for a more arid and hotter climate and higher temporal variability; the area is crossed by the Maipo and Mapocho rivers that rely on regional precipitation and glacial waters, together these two rivers cover 70% of the demand of potable water and approximately 90% of the demand of water for irrigation.

In Bogota (Colombia) lower precipitations are expected in the coming decades and a tendency of increased extreme events, the protection of fragile ecosystems such as 'paramo' will be crucial for water supply to the city. Water provision for urban populations is a particularly fragile sector to climate change in a context where there are still important water provision and sanitation coverage needs and quite unequal distribution in urban and peri-urban areas aggravated by contamination of water sources as in the case of Pedernales, Ecuador.

Sea level rise impacts cities located in low elevation coastal zones, not only because of direct coastal flooding and coastal erosion but also aggravates the impact of storm surge and heat wave energy and saltwater intrusion. Sixty-eight percent of the population of Surinam and thirty-one percent in Guyana live below 5 meters the sea level, many sectors of Georgetown, the capital of Guyana are below sea level, while most economic activities and infrastructure in Caribbean islands such as airports, tourism infrastructure, roads, and others are located in vulnerable coastal areas. The estuary of Rio de la Plata and lower delta of the Parana River where Metropolitan Buenos Aires is located, will be affected by floods with varying recurrence and increased frequency and intensity of storm surges. Climate projections estimate increases in temperature and precipitation for the pampean region where an important number of cities are located.

Over 80% of losses associated with climate related risks concentrate in urban areas, and between 40 and 70% in cities with less than 100,000 inhabitants, most probably as a result of limited capacities to manage disaster risk together with climate change scenarios and low level of investments.

[END FAQ 12.2 HERE]

[START FAQ 12.3 HERE]

FAQ 12.3: How is climate change impacting and expected to impact migration in Central and South America?

In CA and SA, people migrate temporarily or permanently to contiguous areas within their country or abroad, most likely because of economic, security and/or political reasons. The decision to migrate can also be prompted or even motivated by sudden or slow climatic hazards, such as cyclones and storms, droughts, heat waves, heavy rainfalls that result in floods or landslides, among other impacts. There is a lot of heterogeneity across demographic groups and countries that determine the causes for migration. In the case of interprovincial migration, exposure to monthly temperature shocks has the most consistent effects, relative to monthly rainfall shocks and gradual changes in climate.

Much of the climate-related migration is directed toward urban areas. Poor people are usually the most affected, but they are not always capable of migrating: in poor countries, higher temperatures reduce the probability of migration to cities and to other countries, consistently with the presence of severe liquidity constraints. These people are said to be "trapped", but there are also many populations that make the decision to stay in their home areas in spite of potential severe risks. They need institutional, political and economic support to enhance their adaptive capacity, considering their limits to adaptation, as in areas that will be affected by sea level rise or severe water scarcity.

When some members of the family migrate, especially abroad to places such as the US, the role of remittances is important, as they help to build adaptive capacity in the places of origin. One of the main groups migrating abroad, to Mexico and the US, consists of families from rural areas in Guatemala, Honduras and El Salvador. They are mostly families living under poverty levels that rely on rain fed, subsistence agriculture for their livelihood, and are being affected by observed droughts during the last decade.

Migration can be closely related with food security, when people move to areas where agricultural livelihoods and food sources are more secure. The settlement and relocation of large populations may lead to food insecurity and social risk. So migration can be adaptive, but can also pose threats, increasing the vulnerability of the people that migrate and of the families who stay behind, particularly if loans were taken to pay for the expenses of the trip.

Urban areas in CA and SA are often unable to absorb those populations, having high rates of unemployment, overcrowding and increasing the spread of infectious diseases. Receiving countries, particularly in North America, are now increasing the controls to stop the flux of people, but the socio-economic pressure exacerbated by higher climate variability on the sending countries is also increasing.

Some factors that determine the propensity of a population to migrate or stay are education, land tenure, gender, age, access to information, access to rents, other forms of land degradation, power structures, among others. For example, if the extreme variations in climate threats the access to land and water, the coping possibilities for smallholders can be undermined, forcing migration, as seen in flooded villages of Honduras and communities enduring glacier recession and shifting hydrologic regimes in Peru. In both cases, stress motivated new forms of migration that reinforce dominant power structures. In northern Brazil, the absence of programs for rent transferences is thought to have a more important role in migration than droughts. In Brazil, migration rates are increasing and climate change will likely exacerbate the country's regional inequalities, as the most developed regions gain population and welfare, while the least developed regions lose it.

Despite assumptions and evidence from some contexts that show that male labour migration is most responsive to shocks, there is evidence that women are sometimes more likely to act in an insurance role. Migration does not necessarily conduce to feminization, as when all the family migrate. Age also influences the possibility of migration, as in Central America, where younger individuals are more likely to migrate in response to disasters, especially droughts.

Migration as a consequence of climate change is expected to increase from coastal areas, as activities such as tourism and fishing will be affected. Other places facing severe situations are the dryer areas in the Andes, northern Brazil and Central America. Latin America could see more than 17 million climate migrants by 2050, moving from areas with lower water availability and crop productivity, and from areas affected by rising sea level and storm surges.

[END FAQ 12.3 HERE]

[START FAQ 12.4 HERE]

FAQ 12.4: How are indigenous knowledge (IK) and practices being incorporated in the design and implementation of adaptation practices and policies in Central and South America?

Indigenous populations in CA and SA have been estimated at about 30 million people, belonging to more than 800 indigenous groups that are mostly settled the Amazon forest, the Andean ecosystems and the CA highlands. They are often categorized as vulnerable to climate change, but this assessment has been defined as simplistic. Indigenous peoples, although frequently affected by socioeconomic inequalities and power asymmetries with negative impacts in their vulnerability, have particular and diverse knowledge systems that allow them to cope with, or prevent, many climate change impacts. Indigenous knowledge systems can be traced in agricultural practices, agrobiodiversity, art, rituals, ceremonies, customary laws, foods, materials, social structures and medicine, among others.

There is increasing evidence of the role that indigenous knowledge and practices can play to inform and guide adaptation practices and policies. These processes are intrinsically associated with the long-term interactions between communities and their environments, the customary ways for knowledge transfer and productive skills that are passed from one generation to the next, including the diversity of languages, which is critical for a detailed conceptualization and description of socio ecological processes and environmental changes, among others factors.

Some examples in the region comprise a wide range of adaptation practices like increasing species diversity in agricultural systems by community seed exchanges and adapting varieties to cope with climatic change in Peru; promoting ancient and highly diverse crop systems such as the chakra in Ecuador and the milpa in Guatemala; and monitoring changes in communal ecological-agricultural calendar cycles by observation, systematization and knowledge exchange among communities in Brazil. Other activities include establishing a dialog between indigenous and non-indigenous knowledge; recognizing changes in ecological indicators like birds' migration patterns, insect behaviour and phenology of fruit species; and monitoring variations in humid and dry periods which affect slash-and-burn agricultural practices. Other examples include community owned solutions for fire management based on indigenous and local governance and fined-tuned understanding of environmental indicators that are found in Venezuelan, Brazilian and Guyanese Indigenous communities.

Many of these adaptations practices of indigenous and small farmers are related with agriculture, livestock and fisheries; and are exposed to climatic stressors like heat waves, increasing temperatures, changes in rainfall patterns, droughts, floods, plant diseases, winds, hail and extreme climatic events. There is a robust and increasing literature on the resilience of indigenous and local livelihoods, and also on the ongoing struggles for access to key resources, land rights and resource degradation, which reveals the impacts of power relations involved in resource management, determining the dynamic of non-climatic stressors that affect indigenous and local communities and their knowledge systems, and in consequence, their adaptation practices and capacity.

Participation of local populations in adaptation planning has proved to deliver satisfactory results; while, on the contrary, top-down approaches are often problem ridden. Maladaptation can occur when local perceptions and knowledge is not considered in policy and decision-making processes. In Costa Rica, for example, there are conflicts between farmers, government agents and outside researchers, who offer solutions based on their own perceptions of the situation rather than local farmer perceptions. Farmer resistance to proposals which do not solve problems that they see as important frustrates government representatives, who perceive farmers' antagonism as arbitrary. In the Peruvian Andes, peasant and government responses to glacial receding are not articulated in some cases and even conflicting in others.

Despite the increasing recognition and integration of IK in adaptation practices and policies, important barriers for a more effective and transformative integration remain. Some of the most relevant ones are limited participation of indigenous and local communities in adaptation planning and lack of sufficient consideration of non-climatic socioeconomic drivers of vulnerability. These include "westernization" of traditional educational systems which disrupt intergenerational knowledge transmission and affects indigenous languages and worldviews vitality; inadequate partnerships for research and planning which undervalue indigenous and local actors and their contribution to adaptation; conflicts over indigenous land tenure and uses resulting in the loss of indigenous rights over their ancestral lands; counterproductive economic or market incentives that prioritize mitigation over adaptation negatively impacting food sovereignty and livelihoods; and weak governance mechanisms to enforce indigenous rights.

In general, these barriers show that the visions and interests of indigenous and local communities are not always incorporated in negotiations and decisions, which constitute an epistemological marginalization; despite the international commitment to incorporate IK in multilateral discussions. Scientific knowledge remains prioritized over traditional, indigenous, and local knowledge; but some transformative efforts are emerging. In the multilateral scenario, for instance, Bolivian indigenous organizations are proposing to include in the climate change debate, contributions based on indigenous philosophy and worldviews to contest normative conceptions of development with impact on the global community.

Several strategies have been proposed to overcome these barriers, including building effective and respectful partnerships among indigenous and non-indigenous researchers to co-produce climate change research; strengthening the governance with indigenous peoples protecting the integrity and vitality of indigenous knowledge systems; recognizing indigenous peoples as active actors who are continually developing autonomous strategies to preserve their practices, beliefs and knowledge; and considering all programs and strategies in the light of local, regional and global historical, socioeconomic and geopolitical contexts that can modulate their fairness and integral performance. The implementation of these and other strategies could significantly enhance the impact, effectiveness and sustainability of vulnerability assessments and adaptation planning and action.

[END FAQ 12.4 HERE]

[START FAQ 12.5 HERE]

FAQ 12.5: How are inequality and poverty limiting adapting capacity to climate change of vulnerable communities in Central and South America?

The relationship between poverty, social inequality, and climate change is complex. Here we understand poverty as a multi-dimensional concept including health, education, and standard of living, i.e indicators of drinking water, sanitation, energy, housing and assets. In general, poverty and particularly extreme poverty, decreases population adaptive capacity through many channels, like lack of information, education, or access to financial capacity. Therefore, poor populations have higher vulnerability, and usually are geographically located in regions with higher exposure, making them suffer larger damage because of weather extreme events. This situation feedbacks the poverty situation, making it hard for the population to improve their wellbeing, even to end in a chronic poverty trap. These feedbacks are reinforced by social inequality, decreasing the population ability to cope and recover from the damage suffered. By social inequality, we refer mainly to income inequality, despite the fact that we recognize that other types of inequalities exist (both within the social and natural systems).

Poverty, inequality and climate change are related in many dimensions. Developing countries in CA and SA show heterogeneous agricultural production systems and farm typologies. The impact of climate change in agriculture in CA and SA depend on many determinants, like water availability, irrigation infrastructure, as well as socioeconomic characteristics. While large scale farming is vulnerable to weather shocks, small scale farmers are more vulnerable to the impacts of climate change due to heavy dependence on agriculture for livelihood. Larger and poorer households are associated with higher odds of food insecurity. However, while CA and SA can have an important role for advances in global food security, trade liberalization processes would also lead to more environmental pressures in some regions across the region.

Income reductions translate into incomplete insurance; i.e. vulnerable households manage to protect their consumption only partially from the negative income shocks caused by major hazards. Governments in the region has implemented several poverty reductions programs. However, a study in Northern Brazil shows that population risk management strategies to droughts and food insecurity has not change between 1997 and 1998 and 2011–2012. This indicates that income redistributive and policy alleviation policies alone do not necessarily improve climate risk management, and complementary policies are needed.

Moreover, impacts on livelihoods can become chronic. Major shocks, like weather extreme events, reduce and destroy public assets like schools, hospitals, and transport infrastructure, as well as private property like housing, machinery, crops and livestock. For example, Hurricane Mitch (1998) decimated over 80,000 hectares of agricultural land, the large majority of which was used by small farm holders for subsistence agriculture in CA. The lack of financial capacity of poorer households and regions increases their vulnerability and diminishes their resilience to recover from the shock, and may push households into a vicious circle.

Poverty strengths several mechanisms that relates climate change and health risk in CA and SA. Social marginality constraints healthcare access, making poor people or regions often to go undiagnosed or untreated. Some studies showed that marginalized people lack key registrations necessary to access public services in Buenos Aires (Argentina), Mexico City (Mexico) and Santiago de Chile (Chile), while under-reporting, diagnosis, and treatment has been shown regarding women with leishmaniasis in Colombia, and risk of diarrheal diseases because of floods in Amazonian riverine communities in Acre State (Brazil). Biases on health access and diseases reporting by marginalized population usually are reflected in spatial differences. For example, in Brazil's Amazonian North there were 2.2 medical doctors/1,000 inhabitants in 2018, compared to São Paulo (4.95), Rio de Janeiro (5.08), Rio Grande do Sul (9.02) and Santa Catarina (9.52) in the richer South-East and South regions. Also, pregnant women in remote or road-less Amazonian municipalities received less antenatal care. These social inequities underlie systematic biases in health data-quality and hinder reliable estimation of disease burdens, including socially differentiated distribution of disease or birth/death registrations; diagnosis bias due to the effects of differential access to quality healthcare; and diagnosis avoidance due to stigmatized conditions or socioeconomic or cultural differences in health-seeking. This mechanism again reinforces the vicious circle between the impacts of climate change, vulnerability, adaptive capacity and poverty. In addition, as a response to the poor healthcare by formal institutions, other forms of healthcare systems can be operating and remain unseen. This is the case of Mayan-speaking indigenous communities in Guatemala the existence of health institutions based on indigenous knowledge may reinforce the lack of universal coverage by formal state healthcare, increasing mis-estimations of morbidity, mortality and cause-of-death among disadvantaged groups.

[END FAQ 12.5 HERE]

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